A construction of four-dimensional Yang–Mills theory

First sections for review; the detailed proofs are incomplete.

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Part I
The main theorem

1 The result

How can a model on a lattice give a quantum field theory on continuous space and time? We study this question for four-dimensional Yang–Mills theory, starting with Wilson's lattice model. On a finite lattice the variables are finitely many unitary matrices, and their law is an ordinary probability measure. The coupling is the parameter controlling the weight assigned to the lattice energy; a local observable is a function of matrices near one point. The problem is to choose them so that their correlations—the averages of their products—have a limit as the lattice becomes finer and the box becomes larger.

We construct such limits for matrix groups with simple nonabelian Lie algebra (Definition 2.1). The spacing tends to zero first, at each fixed box size. From any increasing sequence of box sizes, one can then select a subsequence along which all the required correlations converge. The limiting correlations may be averages of correlations from several component states; each component is called a phase. One selected phase gives quantum fields on a Hilbert space: translations have nonnegative energy, the fields transform with spacetime symmetries, and fields at spacelike separated points commute (Definition 2.16). For nearby fields, one common finite sum of fields at their centre approximates their effect inside any correlation with distant fields, with a stated error bound (Definition 2.20). The two-point function of a distinguished field has a logarithmic correction at short distances, which separates it from the finite sums and products of free-field kernels defined in Definition 2.22. The result does not establish a mass gap, a positive lower bound on energies above the vacuum.

2 Definitions

We begin with the finite lattice model and the observables whose averages will converge. The limiting correlations may be averages of correlations from component states; we call the components phases. We then define the fields reconstructed from one selected phase. The last definitions state the two short-distance tests: one compares nearby products with fields at one point, and the other specifies a two-point limit and the free-field kernels it excludes. Throughout, $\mathbb{N} = \{0, 1, \ldots\}$; empty products equal 1, and correlation functions of order zero are included.

2.1 The lattice and its measure

Definition 2.1 (Gauge group)**.** A gauge group consists of an integer $m \geq 1$, a closed subgroup G of the group $U(m)$ of unitary $m \times m$ matrices, and a real linear subspace $\mathfrak{g}$ of the complex matrix space $M_m(\mathbb{C})$ such that

$$X \in \mathfrak{g} \quad \Longleftrightarrow \quad \exp(sX) \in G \text{ for every } s \in \mathbb{R}\,. \tag{2.1}$$

The elements of $\mathfrak{g}$ are the infinitesimal directions in which a matrix in G can move away from the identity; this set of directions is the Lie algebra of G. It is simple if $[X, Y] = XY - YX \neq 0$ for some $X, Y \in \mathfrak{g}$, and every real subspace $J \subset \mathfrak{g}$ with $[\mathfrak{g}, J] \subset J$ equals 0 or $\mathfrak{g}$. The condition says that these directions do not contain a smaller nonzero subspace preserved by commutation with every direction in $\mathfrak{g}$.

The matrices in G will be the variables of the lattice model. The specified embedding in $U(m)$ fixes the trace used in its energy. We do not assume that G is connected.

Definition 2.2 (Periodic lattice)**.** For G as in Definition 2.1 and a positive integer ℓ, let $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$ and write e_μ, $1 \le \mu \le 4$, for its coordinate unit vectors. The sites are the points of Λ_ℓ. A positive link (y, μ) joins y to $y + e_\mu$. A link configuration is a map

$$U : \Lambda_\ell \times \{1, 2, 3, 4\} \longrightarrow G\,. \tag{2.2}$$

The value assigned to the reversed link is $U(y, \mu)^{-1}$.

A link matrix describes parallel transport along one edge. Multiplying four of these matrices around an elementary square compares transport along its two coordinate paths. This product, the plaquette matrix, is the local quantity from which the Wilson energy is formed.

Definition 2.3 (Plaquettes and energy)**.** For a configuration as in Definition 2.2, set

$$\begin{aligned} U_{y;\mu\nu} &= U(y, \mu)U(y + e_\mu, \nu)U(y + e_\nu, \mu)^{-1}U(y, \nu)^{-1}, \\ E_\ell(U) &= \sum_{y \in \Lambda_\ell} \sum_{\mu < \nu} \left(1 - \frac{1}{m} \operatorname{Re} \operatorname{tr} U_{y;\mu\nu}\right)\,. \end{aligned} \tag{2.3}$$

The energy counts each unordered pair of coordinate directions once. Each summand lies in $[0, 2]$, so $0 \le E_\ell \le 12\ell^4$. The probability measure weights configurations according to this energy.

Definition 2.4 (Wilson measure)**.** Let G be as in Definition 2.1 and let h be normalised Haar measure on G, the left-translation-invariant probability measure. For $\beta \in \mathbb{R}$, the Wilson measure on the configurations of Definition 2.2 is

$$\begin{aligned} Z_{\ell,\beta} &= \int e^{-\beta E_\ell(U)} \prod_{y,\mu} h(dU(y, \mu)), \\ \mu_{\ell,\beta}(dU) &= \frac{e^{-\beta E_\ell(U)}}{Z_{\ell,\beta}} \prod_{y,\mu} h(dU(y, \mu))\,. \end{aligned} \tag{2.4}$$

The bound on E_ℓ makes $Z_{\ell,\beta}$ finite and strictly positive for every real β. Increasing β gives greater relative weight to configurations with smaller energy. This parameter is called the bare inverse coupling: it is specified in the original lattice measure, before any variables are integrated out. Changing coordinates in the internal matrix space independently at each site acts as follows.

Definition 2.5 (Gauge transformations)**.** For configurations as in Definition 2.2 and $\gamma : \Lambda_\ell \to G$, define

$$U^\gamma(y, \mu) = \gamma(y)U(y, \mu)\gamma(y + e_\mu)^{-1}\,. \tag{2.5}$$

An observable F on the configurations of Definition 2.2 is gauge invariant if $F(U^\gamma) = F(U)$ for every U and γ.

2.2 Local observables and the continuum limit

An observable is a function of the link matrices; its expectation is a measurement in the finite lattice model. To pass to a field at a point, we use observables that depend on links near one site. Their coefficients may change as the lattice is refined. To follow one observable through these changes, we give it a fixed index, separate from its formula on a particular lattice.

Definition 2.6 (Observable indices and insertions)**.** Let $\mathcal{T}$ be a set of observable indices. An insertion with index $v \in \mathcal{T}$ at a site $y \in \Lambda_\ell$ is a function on link configurations (Definition 2.2):

$$I(v, y; \cdot) : G^{\Lambda_\ell \times \{1,2,3,4\}} \longrightarrow \mathbb{C}, \qquad U \longmapsto I(v, y; U)\,. \tag{2.6}$$

For a finite set $\Sigma \subset \mathbb{Z}^4$, write $\Sigma \bmod \ell$ for its image in Λ_ℓ. The insertion is supported on the links inside $y + (\Sigma \bmod \ell)$ if it takes the same value on any two configurations that agree on all links whose two endpoints belong to this set.

The index set is arbitrary at this stage; no linear structure or degree is required. Our construction uses pairs consisting of a degree and a gauge-invariant polynomial in the Taylor coefficients of a classical connection (Definition 6.4). A connection is a smooth $\mathfrak{g}$-valued one-form that specifies infinitesimal parallel transport; its Taylor coefficients record its local behaviour near a point. That polynomial specifies the classical observable to be represented, while the insertion is its actual function of link matrices. The continuum field obtained from an observable index will be part of the hierarchy below. We next choose the physical sizes of the lattices. The index N refines the spacing and p increases the volume.

Definition 2.7 (Wilson family)**.** For a gauge group (Definition 2.1), a Wilson family $\mathcal{W}$ consists of strictly increasing positive integer sequences $(D_N)_{N\in\mathbb{N}}$ and $(L_p)_{p\in\mathbb{N}}$, real numbers $\beta_{N,p}$, normalised Haar measure h on G, a set $\mathcal{T}$ of observable indices, and insertions $I_{N,p}(v, y; U)$ as in Definition 2.6 on $\Lambda_{L_pD_N}$. We write

$$a_N = \frac{1}{D_N}, \qquad \ell_{N,p} = L_pD_N, \qquad \mu_{N,p} = \mu_{\ell_{N,p},\beta_{N,p}}\,. \tag{2.7}$$

The lattice has L_pD_N sites per side, separated by physical distance a_N; thus its physical side is L_p. Taking $N \to \infty$ at fixed p refines the lattice without enlarging the box. Taking $p \to \infty$ enlarges the box. The spacing a_N is a short-distance cutoff; its limit to zero is called the ultraviolet limit. The family also specifies the observables whose averages are to have limits.

Definition 2.8 (Regularity)**.** A Wilson family (Definition 2.7) is regular if:

Reg1. There exists $b_* > 0$ such that, for every $\varepsilon > 0$, there exists N_0 with

$$\left| \frac{\beta_{N,p}}{\log\left(\frac{1}{a_N}\right)} - b_* \right| < \varepsilon \qquad (N \geq N_0,\ p \in \mathbb{N})\,. \tag{2.8}$$

Reg2. Each $U \mapsto I_{N,p}(v, y; U)$ is continuous.

Reg3. Each insertion is gauge invariant in Definition 2.5.

Reg4. For every $v \in \mathcal{T}$, there exists a finite set $\Sigma_v \subset \mathbb{Z}^4$ such that every $I_{N,p}(v, y; \cdot)$ is supported on the links inside $y + (\Sigma_v \bmod \ell_{N,p})$ in Definition 2.6.

The same finite set in Reg4 works at every spacing, volume and site. Its diameter in physical units tends to zero with a_N. Condition Reg1 is a logarithmic growth law for the bare inverse coupling, uniform in the volume; its positive slope is left unspecified here.

Definition 2.9 (Physical positions and moments)**.** Let $r_\ell(j)$ be the representative of $j \in \mathbb{Z}/\ell\mathbb{Z}$ in $\left(-\frac{\ell}{2}, \frac{\ell}{2}\right]$. The physical position of $y \in \Lambda_{\ell_{N,p}}$ is

$$\hat{y} = a_N \big(r_{\ell_{N,p}}(y^\mu)\big)_{\mu=1}^4\,. \tag{2.9}$$

For each tuple of sites, first take the expectation of the product of insertions. The moment $M_{N,p}$ is this expectation; the smeared quantity $C_{N,p}$ then weights these moments by a test function and sums over physical positions. For a Wilson family (Definition 2.7) and $v = (v_1, \dots, v_n) \in \mathcal{T}^n$, define the moment and its smearing by a function f by

$$\begin{aligned} M_{N,p}(v; y) &= \int \prod_{i=1}^{n} I_{N,p}(v_i, y_i; U)\, \mu_{N,p}(dU), \\ C_{N,p}(v; f) &= a_N^{4n} \sum_{y \in \Lambda^n_{\ell_{N,p}}} f(\hat{y}_1, \dots, \hat{y}_n) M_{N,p}(v; y)\,. \end{aligned} \tag{2.10}$$

A correlation is the expected product of the insertions at the chosen sites. The factor a_N^{4n} is the lattice-volume factor in the sum, which is compared with a continuum integral. We test only configurations of distinct points.

Definition 2.10 (Separated tests)**.** For $n \in \mathbb{N}$, put

$$\mathrm{Conf}_n = \{x \in (\mathbb{R}^4)^n : x_i \neq x_j \text{ whenever } i \neq j\}\,. \tag{2.11}$$

A separated test is a smooth function $f : (\mathbb{R}^4)^n \to \mathbb{C}$ whose closed support is compact and contained in Conf_n. We write $\operatorname{supp} f$ for the closed support, the closure of the set on which f is nonzero.

Such a test vanishes near every coincidence of two points. It can depend jointly on all n points and need not be a product of one-point tests. To compare a torus with Euclidean space, its support must also stay inside the physical box.

Definition 2.11 (Tests supported inside a volume)**.** A separated test f (Definition 2.10) is supported inside volume p if

$$\operatorname{supp} f \subset \left(\left(-\frac{L_p}{2}, \frac{L_p}{2} \right)^4 \right)^n . \tag{2.12}$$

Definition 2.12 (Ordered limit)**.** For a Wilson family (Definition 2.7), $P : \mathbb{N} \to \mathbb{N}$ and kernels $K_n(v; x)$, convergence along P means the following. For each fixed n, list $v \in \mathcal{T}^n$ and separated test f (Definition 2.10), the test is supported inside volume $P(k)$ eventually in k, and there are numbers $R_p = R_p(n, v, f) \in \mathbb{C}$ such that

$$\begin{aligned} \lim_{N \to \infty} C_{N,p}(v; f) &= R_p \qquad\qquad (f \text{ supported in volume } p), \\ \lim_{k \to \infty} R_{P(k)} &= \int_{(\mathbb{R}^4)^n} f(x) K_n(v; x)\, dx\,. \end{aligned} \tag{2.13}$$

Here the moments are those of Definition 2.9 and the support condition is Definition 2.11. The kernel is indexed by n and v; the numbers R_p may also depend on the test f. The first limit uses the entire spacing sequence at every volume containing the test. The second limit enlarges the volume along P. At volumes not containing the support, R_p is unconstrained. A strictly increasing P contains each fixed compact test support eventually, since $L_{P(k)} \to \infty$.

Definition 2.13 (Periodic separation)**.** For a Wilson family (Definition 2.7) and volume p, the periodically separated configurations are

$$\{x \in (\mathbb{R}^4)^n : x_i - x_j \notin L_p \mathbb{Z}^4 \text{ whenever } i \neq j\}\,. \tag{2.14}$$

Distinct points can represent the same point of a torus if their difference is a period. This definition excludes those coincidences as well, and is used for fixed-volume correlation kernels.

2.3 The continuum objects

Definition 2.14 (Spacetime and Schwartz tests)**.** Write $x = (\vec{x}, t(x)) \in \mathbb{R}^3 \times \mathbb{R}$ for spacetime, with $t(x) = x^4$, Euclidean norm $|x|$ and

$$\mathsf{m}(x) = t(x)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{x : |\vec{x}| \le t(x)\}\,. \tag{2.15}$$

A vector is spacelike if $\mathsf{m}(x) < 0$. Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^4; \mathbb{C})$, the smooth complex functions whose derivatives decay faster than every inverse power of $|x|$, with the corresponding Schwartz topology. This topology is generated by the seminorms

$$p_{M,\alpha}(f) = \sup_{x \in \mathbb{R}^4} (1 + |x|)^M |\partial^\alpha f(x)| \qquad (M \in \mathbb{N},\ \alpha \in \mathbb{N}^4).$$

Hilbert inner products are antilinear in the first variable. The next definition lists the Hilbert space and fields before imposing their physical properties. A field is smeared against a Schwartz function because its value at a single point need not be an operator.

Definition 2.15 (Relativistic data)**.** A relativistic datum consists of a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of field labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$, a vector $\Omega \in \mathcal{D}$ called the vacuum, and fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ linear over $\mathbb{C}$ in $\ell \in \mathcal{L}$ and $f \in \mathcal{S}$ (Definition 2.14). The field label ℓ specifies which quantum field is smeared; the test function f specifies how it is tested in space and time. It includes an antilinear map $\ell \mapsto \ell^*$, a unitary translation representation $T(b)$ that varies continuously when applied to any vector, and a projection-valued Borel measure E on $\mathbb{R}^4$. The measure E records the joint momentum spectrum of translations: its value on a Borel set is the projection onto states with momenta in that set. For each vector ψ, its diagonal measure $\mu_\psi(A) = \langle \psi, E(A)\psi \rangle$ is finite, with total mass $\|\psi\|^2$.

There is also a unitary representation U of a group $\mathcal{P}$ of spacetime translations and orientation- and time-preserving Lorentz transformations; it varies continuously when applied to any vector. An identification κ gives $\mathcal{P}$ the same transformations and topology as the proper orthochronous Poincaré group. Writing $\kappa(g) = (\Lambda_g, b_g)$, this identification respects identity and the multiplication

$$(\Lambda, b)(\Lambda', b') = (\Lambda\Lambda', \Lambda b' + b)\,. \tag{2.16}$$

Here Λ is invertible, has determinant 1, preserves m and maps $\overline{V}_+$ onto itself. Pure translations $\tau(b) = \kappa^{-1}(1, b)$ satisfy $U(\tau(b)) = T(b)$. For each $g \in \mathcal{P}$ there is a complex linear bijection $\rho(g)$ of $\mathcal{L}$. The action on tests is $(g \cdot f)(x) = f(\Lambda_g^{-1}(x - b_g))$.

A projection-valued Borel measure here is normalised by $E(\mathbb{R}^4) = 1$, has $E(\varnothing) = 0$, satisfies $E(A)E(B) = E(A \cap B)$ and $E(A)^* = E(A)$ on Borel sets, and is countably additive: for disjoint Borel sets, finite sums of their projections converge on every vector to the projection of their union. One can assign arbitrary operators to non-Borel sets, which none of the assertions uses. The map * on field labels is not required to be an involution. No continuity or finite-dimensionality condition on ρ is included in the data.

Definition 2.16 (Wightman conditions)**.** The relativistic datum of Definition 2.15 satisfies the following conditions, with spacetime and test functions as in Definition 2.14 and $p \cdot b$ the Euclidean dot product.

W0. For every $\psi \in \mathcal{H}$ and $b \in \mathbb{R}^4$,

$$\langle \psi, T(b)\psi \rangle = \int e^{ip\cdot b}\, \mu_\psi(dp), \qquad E(\mathbb{R}^4 \setminus \overline{V}_+) = 0\,. \tag{2.17}$$

Every vector fixed by all $T(b)$ belongs to $\mathbb{C}\Omega$.

W1. The vacuum has norm one, $\mathcal{D}$ is dense, and

$$\mathcal{D} = \operatorname{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_j, f_j)\Omega : j \in \mathbb{N},\ \ell_i \in \mathcal{L},\ f_i \in \mathcal{S}\}\,. \tag{2.18}$$

For every $\ell \in \mathcal{L}$ and $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ into $\mathcal{H}$ in norm. For every ℓ, f and $\psi, \chi \in \mathcal{D}$,

$$\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle\,. \tag{2.19}$$

W2. The maps ρ form a representation: $\rho(1) = 1$ and $\rho(gg') = \rho(g)\rho(g')$. For every $g \in \mathcal{P}$, $U(g)\mathcal{D} \subset \mathcal{D}$ and $U(g)\Omega = \Omega$. For every g, ℓ, f and $\psi \in \mathcal{D}$,

$$U(g)\Phi(\ell, f)\psi = \Phi(\rho(g)\ell, g \cdot f)U(g)\psi\,. \tag{2.20}$$

W3. If $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} f'$, then, for every $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$,

$$\Phi(\ell, f)\Phi(\ell', f')\psi = \Phi(\ell', f')\Phi(\ell, f)\psi\,. \tag{2.21}$$

The finite products of smeared fields in (2.18) are called field words. Their values on the vacuum span the domain itself; the empty word gives the vacuum. We use the Poincaré transformations themselves. In (2.17), $p \cdot b$ is the Euclidean dot product; the forward-cone condition uses the Minkowski form m from Definition 2.14.

We collect the continuum objects in a hierarchy: fields and their correlations, the component states and their average, the continuation used to recover real-time field expectations, and the coefficients for local product expansions.

Definition 2.17 (Hierarchy)**.** A hierarchy over the Wilson family of Definition 2.7 consists of the following data.

H1. A relativistic datum of Definition 2.15.

H2. For every $d \in \mathbb{N}$, a finite-dimensional complex normed space F_d and a complex linear map $\iota_d : F_d \to \mathcal{L}$. Set $\mathbb{F} = \bigsqcup_{d\in\mathbb{N}} F_d$, $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$.

H3. A map $\pi : \mathcal{T} \to \mathbb{F}$, assigning a continuum field to each observable index (Definition 2.6).

H4. A compact metric space X, called the phase space, with a Borel measure ν and a selected point $\xi_* \in X$.

H5. For every $n \in \mathbb{N}$ and $s \in X \sqcup \{\mathrm{mix}\}$, a correlation function $S^s(a; x) \in \mathbb{C}$ on $\mathbb{F}^n \times (\mathbb{R}^4)^n$. The extra symbol mix distinguishes the average state supplied by the lattice limit from the phases $\xi \in X$.

H6. For every $n \in \mathbb{N}$, a continuation $\mathsf{F}_n(a; z, \vec{x}) \in \mathbb{C}$ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$. It extends ordered Euclidean-time correlations to complex times, from which real-time field expectations are obtained as boundary values.

H7. For every $j \in \mathbb{N}$, a finite index set Ch_j naming the fields retained at degree j, and fields $O_{j,\kappa} \in F_j$ for those indices. For every finite index set I, $a \in \mathbb{F}^I$ and $\kappa \in \mathrm{Ch}_j$, there is a coefficient function $c_{j,\kappa}(a; x, o)$ on $(\mathbb{R}^4)^I \times \mathbb{R}^4$. Here $x = (x_i)_{i\in I}$ is the list of insertion positions and o is the centre at which the cluster is replaced by $O_{j,\kappa}$.

H8. A distinguished degree-four field $\Theta \in F_4$, used in the trace two-point test below.

The maps in H2–H3 relate three different objects: a lattice observable index v, the resulting continuum field (d, A), and its quantum field label ℓ:

$$v \in \mathcal{T} \overset{\pi}{\mapsto} (d, A) \in \mathbb{F} \overset{\iota}{\mapsto} \ell = \iota_d A \in \mathcal{L}\,. \tag{2.22}$$

The field label specifies the operator family $f \mapsto \Phi(\ell, f)$ (Definition 2.15). In our construction, v is the degree-and-polynomial index from Definition 6.4; it identifies which classical local quantity the lattice insertion represents.

The degree is an index used in the product estimates below. The definition imposes no inclusions between different F_d, no prescribed curvature interpretation and no compatibility of the ι_d across degrees. The functions S^s are defined even where points coincide, but the conclusions constrain them only at separated points.

2.4 Reconstruction and phases

The mixed state is the limit of lattice moments and averages the component correlations over X. The selected point ξ_* picks one component; only that component's Euclidean correlations are continued to the relativistic fields. Thus the mixed state and the selected state have different roles.

Definition 2.18 (Reconstruction properties)**.** A hierarchy of Definition 2.17 has the reconstruction properties if the following hold. Here $s \in X \sqcup \{\mathrm{mix}\}$, $n \in \mathbb{N}$ and $a \in \mathbb{F}^n$.

R1. Its relativistic datum satisfies W0–W3.

R2. For every nonzero $\ell \in \mathcal{L}$, there exists $f \in \mathcal{S}$ with $\Phi(\ell, f) \neq 0$. This is faithfulness: no nonzero field label represents the zero field.

R3. Each ι_d is injective.

R4. Some $a \in \mathbb{F}$ and $f \in \mathcal{S}$ satisfy $\Phi(\iota(a), f) \neq 0$.

R5. The map $a \mapsto \iota(a)$ from $\mathbb{F}$ onto $\mathcal{L}$ is surjective.

R6. The map $\pi : \mathcal{T} \to \mathbb{F}$ is surjective.

R7. The measure ν has total mass one.

R8. Every nonempty open subset of X has positive ν-measure; this is full support.

R9. For every state s, order n, list $a \in \mathbb{F}^n$, $x \in \mathrm{Conf}_n$ and permutation σ,

$$S^s((a_{\sigma(i)})_i; (x_{\sigma(i)})_i) = S^s(a; x)\,. \tag{2.23}$$

R10. For every s, n, a, the function $S^s(a; \cdot)$ is smooth on Conf_n.

R11. For every n, a and $x \in \mathrm{Conf}_n$, the function $\xi \mapsto S^\xi(a; x)$ is continuous.

R12. For every n, a and $x \in \mathrm{Conf}_n$,

$$\int_X S^\xi(a; x)\, \nu(d\xi) = S^{\mathrm{mix}}(a; x)\,. \tag{2.24}$$

R13. For every n, a and fixed real spatial configuration $\vec{x}$, the function $z \mapsto \mathsf{F}_n(a; z, \vec{x})$ is holomorphic on

$$\mathcal{Z}_n = \{z \in \mathbb{C}^n : \mathrm{Im}\, z_1 < \cdots < \mathrm{Im}\, z_n\}\,. \tag{2.25}$$

R14. For every n, a and real configuration x with $t(x_1) < \cdots < t(x_n)$,

$$\mathsf{F}_n(a; (it(x_i))_i, (\vec{x}_i)_i) = S^{\xi *}(a; x)\,. \tag{2.26}$$

R15. For every n, a, real heights $h_1 < \cdots < h_n$ and tests $f_1, \dots, f_n \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int_{(\mathbb{R}^4)^n} \mathsf{F}_n(a; (t(x_i) + irh_i)_i, (\vec{x}_i)_i) \prod_{i=1}^{n} f_i(x_i)\, dx \\ = \langle \Omega, \Phi(\iota(a_1), f_1) \cdots \Phi(\iota(a_n), f_n)\Omega \rangle\,. \tag{2.27}$$

The scalar on the right of (2.27) is the vacuum expectation of the indicated product of smeared fields. In (2.27), the integral is assigned the value zero if its integrand is not integrable; this convention makes the boundary-value limit a defined statement without asserting positive-height integrability. The integrals in the lattice limit and (2.24) are ordinary absolutely convergent integrals, by compact support and smoothness in the first case and by phase continuity and compactness in the second.

2.5 Products and the trace test

The product expansion groups nearby insertions into a cluster and replaces them by a finite sum of fields at the cluster centre. The other fixed insertions, kept a positive distance from that centre, are called spectators. One choice of coefficients must work with every list of such fields and in every state. The threshold on the cluster radius, however, may depend on the fixed spectator fields.

Definition 2.19 (Cluster geometry)**.** For a hierarchy (Definition 2.17) and finite index sets I, J, write $S^s(a, x; b, y)$ for the correlation on the concatenated lists, enumerated in one fixed order on $I \sqcup J$. For $r > 0$, $M \geq 0$ and $\eta > 0$, call (x, o) an (r, M, η)-cluster if

$$|x_i - o| \leq Mr \quad (i \in I), \qquad |x_i - x_{i'}| \geq \eta r \quad (i \neq i')\,. \tag{2.28}$$

For $c > 0$, call y c-clear of o if

$$|o - y_j| \geq c \quad (j \in J), \qquad |y_j - y_{j'}| \geq c \quad (j \neq j')\,. \tag{2.29}$$

Set $w(a) = \sum_{i \in I} \deg a_i$.

Definition 2.20 (Operator product expansion)**.** For a hierarchy (Definition 2.17), the operator product expansion (OPE) property consists of the following estimates on clusters and distant points as in Definition 2.19.

O1. For every finite I, $a \in \mathbb{F}^I$, $M \geq 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\kappa \in \mathrm{Ch}_j$, there exists $\delta > 0$ such that, for every $0 < r < \delta$ and every (r, M, η)-cluster,

$$|c_{j,\kappa}(a; x, o)| \leq r^{j - w(a) - \varepsilon}\,. \tag{2.30}$$

O2. For every finite I, J, $a \in \mathbb{F}^I$, $b \in \mathbb{F}^J$, $M \geq 0$, $\eta, c, \varepsilon > 0$ and $D \in \mathbb{N}$, there exists $\delta > 0$ such that, for every $0 < r < \delta$, every (r, M, η)-cluster, every c-clear spectator configuration and every $s \in X \sqcup \{\mathrm{mix}\}$,

$$\left| S^s(a, x; b, y) - \sum_{j=0}^{D} \sum_{\kappa \in \mathrm{Ch}_j} c_{j,\kappa}(a; x, o) S^s\big((j, O_{j,\kappa}), o; b, y\big) \right| \leq r^{D+1-w(a)-\varepsilon}\,. \tag{2.31}$$

The coefficient functions are those already chosen in H7; the same coefficients work for all spectators, states and truncation degrees. The bounds are written with constant 1: all dependence on the fixed data is absorbed by choosing δ smaller. For small enough r, $Mr < c$ ensures that all points in the tested correlations are separated. The sum stops at degree D because the assertion controls only a finite approximation; it does not claim convergence of an infinite series or uniqueness of the coefficients.

Definition 2.21 (The trace two-point function)**.** For the field $\Theta \in F_4$ of Definition 2.17 and $r \in \mathbb{R}$, set

$$G_\Theta(r) = S^{\xi_*}\Big((4,\Theta),(4,\Theta);-\frac{r}{2}e_4,\frac{r}{2}e_4\Big)\,. \tag{2.32}$$

Free-field two-point kernels are built from the energy $E_{\mathsf{m}}(p)$ associated with mass m and spatial momentum p. In the next definition, q specifies a power of this energy and h is a spatial-momentum factor of homogeneity degree j; the comparison class uses finitely many such kernels, their products and finite sums. We use these kernels only at positive separations $r > 0$. It specifies exactly the comparisons excluded by the theorem.

Definition 2.22 (Finite Wick profiles)**.** A mode here means one kernel specified by a mass $\mathsf{m} \geq 0$, integers $j, q \in \mathbb{N}$, and a continuous real function h on $\mathbb{R}^3$ satisfying $h(\lambda p) = \lambda^j h(p)$ for every $\lambda > 0$. Set

$$E_{\mathsf{m}}(p) = \sqrt{\mathsf{m}^2 + |p|^2}, \qquad K_{\mathsf{m},q,h}(r) = \int_{\mathbb{R}^3} \frac{E_{\mathsf{m}}(p)^q h(p)}{2E_{\mathsf{m}}(p)} e^{-rE_{\mathsf{m}}(p)}\,dp\,. \tag{2.33}$$

The integrand can be assigned value zero at a zero denominator, a set of measure zero. A finite Wick profile is

$$W(r) = \sum_{\alpha=1}^{A} c_\alpha \prod_{\beta=1}^{n_\alpha} K_{\mathsf{m}_{\alpha\beta},q_{\alpha\beta},h_{\alpha\beta}}(r), \qquad c_\alpha \in \mathbb{C}, \quad A, n_\alpha \in \mathbb{N}\,. \tag{2.34}$$

Empty products are allowed.

The trace field is intended to come from the stress tensor: a collection of fields whose integrals express how translations and rotations act. Its Euclidean trace is the sum of its diagonal fields. The construction subtracts the vacuum expectation of this trace; "centred" means this subtraction makes every vacuum one-point expectation zero. The hierarchy below still specifies only the resulting degree-four field, not a stress tensor.

Definition 2.23 (Trace test)**.** The hierarchy of Definition 2.17 passes the trace test if the following hold, with G_Θ as in Definition 2.21 and Wick profiles as in Definition 2.22.

T1. For every $f \in \mathcal{S}$, $\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0$.

T2. The number $A_G = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4}$ is positive.

T3. The following limit holds in $\mathbb{C}$ over all positive radii:

$$\lim_{r\downarrow 0} r^8 \left(\log\frac{1}{r}\right)^2 G_\Theta(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4}\,. \tag{2.35}$$

T4. For every finite Wick profile W and every $\delta > 0$, there exists $r \in (0,\delta)$ with $G_\Theta(r) \neq W(r)$.

The public trace test concerns the chosen degree-four field and the time axis; the construction identifies it with the centred trace just described, as used in Proposition 5.5.

3 The main theorem

The theorem ties the continuum correlations to the original Wilson measures. The lattice family is chosen once; the later choice of a volume subsequence determines one continuum hierarchy with all four conclusions below.

Theorem A. *For every gauge group $(m, G, \mathfrak{g})$ with $\mathfrak{g}$ simple (Definition 2.1), there exists a Wilson family $\mathcal{W}$ (Definition 2.7) satisfying the regularity conditions Reg1–Reg4 of Definition 2.8 such that, for every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $q : \mathbb{N} \to \mathbb{N}$ and a hierarchy over $\mathcal{W}$ (Definition 2.17) with the following properties.*

P1. *For every $n \in \mathbb{N}$, $v \in \mathcal{T}^n$ and separated test f (Definition 2.10), the test is supported inside volume $p(q(k))$ eventually in k (Definition 2.11), and there exist $R_j \in \mathbb{C}$ such that*

$$\begin{aligned} \lim_{N\to\infty} C_{N,j}(v; f) &= R_j \qquad (f \text{ supported in volume } j), \\ \lim_{k\to\infty} R_{p(q(k))} &= \int f(x) S^{\mathrm{mix}}(\pi(v_1), \dots, \pi(v_n); x)\, dx\,. \end{aligned} \tag{3.1}$$

Here $C_{N,j}$ is the Wilson moment of Definition 2.9; these are the ordered limits of Definition 2.12.

P2. *The datum reconstructed from the selected point ξ_* satisfies the Wightman conditions W0–W3 of Definition 2.16. The hierarchy has all reconstruction properties R1–R15 of Definition 2.18; in particular, its continuation has the selected phase's Euclidean correlations as values at ordered imaginary times. The mixed correlations are the phase average*

$$S^{\mathrm{mix}}(a; x) = \int_X S^{\xi}(a; x)\, \nu(d\xi) \qquad (x \in \mathrm{Conf}_n)\,.$$

P3. *The single coefficient system in H7 satisfies both OPE estimates of Definition 2.20.*

P4. *The field $\Theta \in F_4$ has zero vacuum expectation as in T1, and its selected-phase two-point function (Definition 2.21) satisfies*

$$\lim_{r\downarrow 0} r^8 \left(\log \frac{1}{r}\right)^2 G_\Theta(r) = \frac{3 \dim_{\mathbb{R}} \mathfrak{g}}{\pi^4} > 0\,.$$

For every finite Wick profile W (Definition 2.22) and every $\delta > 0$, there exists $r \in (0, \delta)$ with $G_\Theta(r) \neq W(r)$. These are the trace conditions T1–T4 of Definition 2.23.

The lattice data are chosen before p. The refinement, field spaces, phases, selected relativistic theory, product coefficients and trace field may depend on p. The same refinement works for all orders of correlation, observable indices and separated tests. The lattice limit in P1 is the mixture; the relativistic theory in P2 is attached to ξ_*.

The theorem does not choose a unique infinite-volume limit, show that different lattice choices give the same limit, or take the spacing and volume limits simultaneously. It does not prove a mass gap, meaning a positive lower bound on nonzero energies above the vacuum. It does not show that widely separated groups of observations become independent in the mixed state, that different phases agree, or that any one phase is itself the limit of lattice states. The correlations are defined only when insertion points are distinct; the theorem does not define their smeared extension at coincident points. The comparison excludes only the finite profiles in Definition 2.22; it does not exclude every polynomial combination of free fields or theories built from a continuum of free-field masses. No Euclidean covariance or reflection-positivity assertion is included among P1–P4; those properties are established internally in the construction.

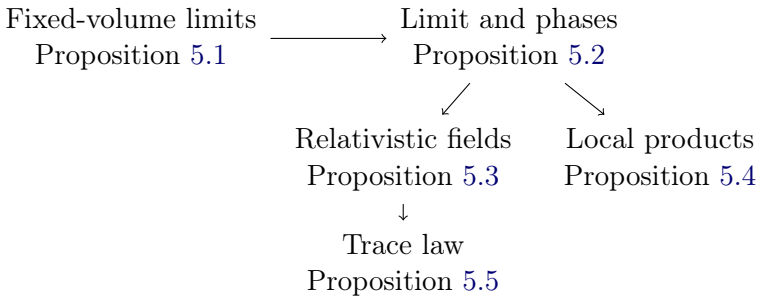
4 Strategy

The diagram records which proposition depends on which earlier conclusion.

From small spacings to a fixed box. Refining the lattice adds variables describing shorter distances. We integrate groups of these variables while retaining the variables on the coarser lattice; this produces a density for the coarse variables with the same total probability integral. After each integration, we compare the coefficient of the quadratic energy at one fixed reference spacing. The bare coupling is chosen so that this reference coefficient stays fixed. The insertion coefficients are changed at the same time, using the reference normalisation of Definition 6.11, so the same labelled observable is followed through the refinements. Proposition 5.1 gives one such family and the limit at each fixed volume. Its local estimates can then be used in boxes of any size.

From boxes to phases. Bounds that do not grow with box size allow a subsequence on which all correlations converge; this is the compactness step in Proposition 5.2. The limit may still average several component states. The positive-time reflection estimate in Section 8 supplies the Hilbert-space construction used in the later field and phase arguments. Long-time averages isolate the zero-energy part of this inner-product space. Once rotations are established, the resulting operators commute; the phase argument represents them as functions on a compact space X. Integrating those functions against a probability measure gives (2.24). This decomposition is a further step after compactness.

Three conclusions from one hierarchy. Proposition 5.3 uses the component at ξ_*; its ordered Euclidean-time correlations extend to complex times and give the relativistic field expectations at the boundary. Proposition 5.4 replaces a nearby product by a finite sum of fields at its centre, with an error bound inside correlations that also contain separated insertions. The coefficients are selected once and work for every phase and for the average. Proposition 5.5 uses the degree-four field identified with the centred stress trace; its logarithmic two-point limit rules out the finite comparison profiles in Definition 2.22. The dependence of these five results is summarised by



For each gauge group, Proposition 5.1 chooses one family; for each prescribed increasing volume sequence, Proposition 5.2 chooses one subsequence and one hierarchy used by all five propositions. The arrows record this order; compactness alone does not supply the later estimates.

5 The principal propositions

The sketches below give the main mathematical steps; the detailed estimates are developed in Part II.

Removing the lattice spacing. At a fixed physical volume, increasing the number of sites must not change what the model looks like at a fixed larger distance. The coupling and the normalisation of each observable are chosen to achieve this compatibility. The first proposition gives both one such choice and convergence along its entire spacing sequence.

Proposition 5.1 (Ultraviolet construction)**.** *For every gauge group with simple $\mathfrak{g}$ (Definition 2.1), there exists a regular Wilson family $\mathcal{W}$ (Definitions 2.7 and 2.8) such that, for every $p, n \in \mathbb{N}$ and $v \in \mathcal{T}^n$, there exists a smooth kernel $K_{p,v}$ on the periodically separated set of Definition 2.13 with*

$$\lim_{N\to\infty} C_{N,p}(v; f) = \int f(x) K_{p,v}(x)\, dx \tag{5.1}$$

for every separated test f supported inside volume p (Definitions 2.9, 2.10 and 2.11).

Idea of the proof.

Step 1. Integrate the fine link variables while holding the coarse link variables fixed. If ρ_j is the density before this step and ρ_{j+1} the resulting density, the integration preserves the total integral:

$$\int \rho_{j+1}(U')\, dU' = \int \rho_j(U)\, dU.$$

Step 2. Split the integral according to whether each plaquette matrix is close to the identity, using the distance and threshold of (6.3) and (7.8). Near the identity, the Wilson energy is compared with its quadratic approximation. This term is the squared size of the infinitesimal transport around a small square (the curvature). Away from the identity, an energy bound controls the contribution. Both regions remain in the integral.

Step 3. Repeating the integrations changes the coefficient of this quadratic curvature term. Choose the bare coupling to compensate for that change, with the logarithmic growth in Reg1.

Step 4. Change the insertion coefficients along with the density, using the same reference level in Definition 6.11. In (6.23), P_d turns a classical polynomial into a polynomial of nearby links, $S_{\theta_N^{-1}}$ rescales its number of factors, $\tau_{N,p,d}^{-1}$ carries the chosen reference coefficients back to the fine level, and a_N^{-d} supplies the physical scaling. This keeps the reference coefficients fixed when cutoffs are compared.

Step 5. Compare two cutoffs by stopping both at the same intermediate spacing. If $C_{N,p}^{(j)}$ denotes the moment after the integrations stop at spacing j, the comparison has the form

$$|C_{N,p} - C_{M,p}| \le |C_{N,p} - C_{N,p}^{(j)}| + |C_{N,p}^{(j)} - C_{M,p}^{(j)}| + |C_{M,p}^{(j)} - C_{M,p}|.$$

Make the two fine-spacing errors small by choosing j, then let N, M increase at that fixed j. This is the intended Cauchy argument for the full sequence.

Unresolved. The paper still needs bounds for both parts of the plaquette split, a proof of the change in the quadratic energy coefficient after each integration, and a fixed-spacing Cauchy estimate whose remaining fine-spacing error is uniform in the cutoff. These estimates are needed to prove the fixed-volume limit in the proposition.

Enlarging the volume and separating phases. The estimates for a fixed collection of observables must remain bounded when the surrounding box grows. They give compactness, but the limit can retain several possible vacuum states. The next proposition includes the separate argument that resolves this average into phases.

Proposition 5.2 (Continuum limit and phases)**.** *For the jointly chosen family of Proposition 5.1 and every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $q : \mathbb{N} \to \mathbb{N}$ and hierarchy data (Definition 2.17) satisfying the limit P1 and the phase and regularity conditions R7–R12 of Definition 2.18.*

Idea of the proof.

Step 1. Divide the box into reflected cells C as in Definition 8.2. For a positive-time observable F_C in a cell, $\Theta_C F_C$ is its reflected, complex-conjugated copy; $\|F_C\|_{\mathcal{C}}$ is the one-cell bound of (8.5). Reflection positivity and the volume-independent bound (8.6) give

$$\left|\left\langle \prod_C \Theta_C F_C \right\rangle\right| \le \prod_C \|F_C\|_{\mathcal{C}}.$$

Thus the joint bound does not grow with the surrounding box.

Step 2. Integrating the insertion along one coordinate slice bounds the change when its point moves by h:

$$|S^s(a; x_1, \dots, x_i + he_\mu, \dots, x_n) - S^s(a; x_1, \dots, x_i, \dots, x_n)| \le C|h|.$$

All other insertion labels and positions are fixed in this estimate, and the segment from x_i to $x_i + he_\mu$ stays away from the other points. Uniform boundedness and this continuity estimate give a common convergent subsequence for the countable polynomial-basis lists. For a smooth compactly supported test f, summation by parts moves the finite difference from the correlation to f; its limit is the distributional derivative, represented by the correlation with the corresponding higher-degree insertion.

Step 3. The full fixed-volume limits from Proposition 5.1 identify the inner limits. Only the outer volume refinement remains, in the order of (2.13).

Step 4. The reflection inner product completes to a Hilbert space. Its nonnegative energy operator H generates positive-time evolution e^{-tH}; P_0 denotes the orthogonal projection onto $\ker H$. The spectral theorem gives

$$e^{-tH} \xrightarrow[t\to\infty]{} P_0.$$

Inserting observations between long evolutions gives operators on this subspace. The required rotation argument must show that each such pair satisfies $AB = BA$; that identity is still unproved, as recorded in the gap below.

Step 5. If these bounded operators commute, their joint spectral representation assigns them coordinate functions on their compact joint-value space X. The vacuum vector induces a probability measure ν there, so the mixed correlation is the average of the component correlations:

$$S^{\mathrm{mix}}(a; x) = \int_X S^{\xi}(a; x)\, \nu(d\xi).$$

Unresolved. The proof still needs volume-independent bounds for this chosen family's local observables and their derivative representatives. It must also show that a stress-tensor integral supported arbitrarily far away contributes zero in the limit, and prove the displayed commutation of the long-time operators. Compactness gives none of these estimates.

Recovering relativistic fields. In Euclidean time a state of energy $E \geq 0$ is multiplied by e^{-tE}. Continuing t to imaginary values changes this factor into unitary time evolution, which is why complex-time correlations can give relativistic vacuum expectations. The estimates must remain controlled as the imaginary-time separations tend to zero, when the resulting field expectations are tested against smooth functions.

Proposition 5.3 (Relativistic reconstruction)**.** *For the jointly chosen hierarchy of Proposition 5.2, the selected theory satisfies R1–R6 and the continuation satisfies R13–R15 of Definition 2.18, with the same fields and phase as in Definition 2.17.*

Idea of the proof.

Step 1. Test each insertion with a smooth function supported in a positive-time interval, with the intervals separated and ordered. Shrinking their widths gives limits of the ordered products as distributions, meaning after integration against smooth tests.

Step 2. Use the positive-time reflected form described in the construction guide for the field words generated by (2.18). Identify vectors of zero norm and complete; a further smeared field acts by adding it to the left of a word.

Step 3. Once the required Euclidean rotation identities are proved, continue them to the corresponding spacetime transformations. The nonnegative Euclidean energy gives the forward-cone momentum condition. When two supports are spacelike separated, the two analytic orderings agree, so the fields commute there.

Step 4. Identify two field labels if every tested correlation involving their difference is zero, as in Proposition 8.13. This quotient gives faithful labels; the constant lattice observable gives a nonzero field.

Unresolved. The reconstruction proof still needs the Euclidean rotation identities, uniform bounds as the imaginary-time separations vanish, and a proof that the two analytic orderings have the same boundary values at spacelike separation. It must also verify that the null-correlation quotient preserves the stated field domains and symmetries.

Replacing a cluster by a single field. A distant observation cannot resolve every detail of several nearby insertions. The operator product expansion expresses their effect through a finite list of fields at the cluster centre. The coefficients must be independent of which distant observations are subsequently made.

Proposition 5.4 (Common operator product expansion)**.** *The fields and coefficients in H7 of the jointly chosen hierarchy satisfy O1–O2 of Definition 2.20, for the clusters of Definition 2.19.*

Idea of the proof.

Step 1. Fix the nearby insertion positions and move the cluster's centre to the origin by translation. Compare its correlations with those of the finite list of fields $O_{j,\kappa}$ placed at that centre. The

coefficients are selected before specifying any distant insertions, so the same list is tested against every later choice.

Step 2. Keep degrees at most D. The first omitted degree controls the error, with an arbitrarily small loss to absorb logarithmic factors:

$$\left| S^s(a,x;b,y) - \sum_{j=0}^{D} \sum_{\kappa\in \mathrm{Ch}_j} c_{j,\kappa}(a;x,o) S^s\big((j,O_{j,\kappa}),o;b,y\big) \right| \le r^{D+1-w(a)-\varepsilon},$$

the estimate of (2.31).

Step 3. Choose the coefficients using a fixed collection of distant-field correlations, before choosing the spectators or state. To make that choice work for every later correlation, the proof must bound the remainder as an operator on the Hilbert space; taking its expectation in any phase or in the average then preserves the bound.

Unresolved. The product argument still needs to prove the coefficient bound (2.30), the finite-sum remainder bound (2.31), and the operator bound that makes one coefficient choice valid for every spectator list and state.

Detecting the logarithmic correction. The construction supplies the stress tensor described after Definition 2.23 and subtracts the selected vacuum expectation from the sum of its diagonal fields. At short distances it compares the two-point function of this centred trace with the free Maxwell curvature kernels of Section 9. The resulting logarithmic factor differs from the small-distance behaviour of the finite profiles in Definition 2.22.

Proposition 5.5 (Trace law and finite-profile exclusion)**.** *The field $\Theta \in F_4$ of the jointly chosen hierarchy satisfies T1–T4 of Definition 2.23 in the theory of Proposition 5.3, with the two-point function of Definition 2.21.*

Idea of the proof.

Step 1. Compare each normalised trace insertion at short distance with the corresponding free Maxwell curvature expression of Section 9. Its coefficient contains one factor of the inverse running coupling; the two insertions therefore contribute two inverse-coupling factors.

Step 2. The logarithmic growth required by Reg1 turns those two factors into the short-distance law

$$\lim_{r\downarrow 0} r^8 \left(\log\frac{1}{r}\right)^2 G_\Theta(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4},$$

as in (2.35); the coefficient of the coupling's logarithmic growth and the trace normalisation determine the amplitude.

Step 3. First the comparison is proved on a sequence of radii. Positivity of energy gives a positive measure σ and the Laplace representation

$$G_\Theta(r) = \int_{[0,\infty)} e^{-rE}\, d\sigma(E), \qquad d\sigma \ge 0,$$

where σ is the positive spectral measure supplied by the spectral theorem for the selected trace two-point function and the nonnegative energy operator H from Proposition 5.2. Thus G_Θ decreases for $r > 0$. Bounds at neighbouring radii in the sequence then squeeze every intermediate value and give the full right limit.

Step 4. Each finite profile in Definition 2.22 is a finite sum of products of the displayed momentum integrals. Its small-radius expansion has a different power-and-logarithm form from the positive inverse-logarithmic law in (2.35); this proves the stated exclusion once those expansions are established.

Unresolved. The trace proof still needs the normalisation of the stress trace, the comparison on a sequence dense enough to squeeze all positive radii, and the small-radius expansions of the kernels in (2.33). It must show that every finite profile has an asymptotic incompatible with (2.35).

Deduction of Theorem A from the principal propositions. Fix the gauge group and choose the family of Proposition 5.1, including its insertions and positive regularity slope. For a prescribed strictly increasing p, take the common q and hierarchy of Proposition 5.2. This gives P1 and R7–R12 on that refinement. Proposition 5.3 supplies the remaining reconstruction properties, hence P2. Propositions 5.4 and 5.5 give P3 and P4 for those same objects. □

Part II
The construction

Part I states what the limiting theory must satisfy. Here we explain how to construct it from the Wilson measure of Definition 2.4. The construction first removes the spacing at each fixed volume, then takes a subsequence as the volume grows. The later sections explain how those limiting correlations define fields, symmetries and phases.

Following an observable across scales. Section 6 starts with the finite probability law and its exact symmetries. It describes a classical local observable by a gauge-invariant polynomial in the Taylor coefficients of a smooth connection (Definition 6.4); the connection specifies infinitesimal parallel transport, and these coefficients record its behaviour near one point. A polynomial in nearby link matrices represents that classical polynomial on the lattice. Integrating out fine links can change its coefficients, so Definition 2.6 gives it a fixed index and (6.23) fixes the coefficients at a reference level. This lets us follow one observable through different lattice spacings.

Section 7 constructs the link polynomials and organises products of insertions for integration. It records each contribution together with the lattice region on which it depends; terms where a plaquette is far from the identity are kept and tracked by their sites. This bookkeeping prepares the estimates but does not prove them.

The later ultraviolet sections perform these integrations one spacing at a time. They average over finer link variables to produce a density for the remaining coarse links, preserving the original integral. The estimates must control both this density and each insertion after many such steps. When every plaquette matrix in a region is close to the identity, the Wilson energy is compared with its quadratic approximation. When a plaquette is farther away, its contribution is bounded using the energy instead of being discarded. Contributions are grouped by the sites on which they depend, which is what allows bounds for one local observable to be independent of the surrounding box. The change in the coefficient of the quadratic curvature term is included in the coupling assigned to the original lattice measure.

These arguments must provide two kinds of estimates. At a fixed box size, they must show that correlations approach a limit along the full sequence of spacings, as in Proposition 5.1. Uniform bounds on local observables and on their finite differences as an insertion point moves must also hold independently of box size. The latter bounds give a convergent subsequence as the box grows; the fixed-volume limits identify its inner limits with those of the full spacing sequence. For the moments of Definition 2.9, the intended passage is the ordered limit (2.13):

$$C_{N,p}(v;f) \longrightarrow R_p \qquad (N\to\infty,\ p \text{ fixed}),$$
$$R_{p(q(k))} \longrightarrow \int f(x) S^{\mathrm{mix}}(\pi(v_1),\dots,\pi(v_n);x)\,dx \qquad (k\to\infty).$$

Here f is a separated test of Definition 2.10; the inner limit is required at every volume containing its support (Definition 2.11). The family is chosen before the prescribed volume sequence p, and the subsequence q is the same for every observable list and test.

From correlation functions to fields. Section 8 applies reflection positivity: for positive-time products F, G, reflect F across time zero and take its complex conjugate, writing the result as ϑF. The resulting positive form is $S^{\mathrm{mix}}((\vartheta F)G) \geq 0$; the finite-lattice inequality is (8.8), and its repeated-cell version is (8.9). Repeated reflection bounds a joint correlation by one-cell bounds,

which do not grow with the box. These bounds give one subsequence for all correlations. To identify derivatives, the proof compares a finite difference of an insertion's position with a higher-degree observable. Two observables are identified as the same continuum field when every tested limiting correlation of their difference is zero; removing this null subspace gives the field spaces of the limit. Smoothness therefore needs estimates for these comparisons, not just compactness. Finally the fixed-volume theorem removes the temporary spacing subsequence and gives the ordered limit of Definition 2.12.

Section 9 compares the limiting fields at short distances with the explicit curvature fields of the free Maxwell theory. These are the field strengths of a free electromagnetic field; finite tests distinguish the field classes, and comparison maps transfer the stress-tensor estimates to the interacting theory. Constructing these maps requires short-distance bounds and a separate dimension bound. The comparison does not make the interacting theory free. For the symmetry argument, the stress-tensor integral identities are a separate required input to establish continuous rotations on this space; they are not proved in the sections presented here.

Reading the construction. When reading an estimate, identify the quantities held fixed, the limit being taken, and whether its bound depends on volume. This is especially important when a fixed-volume spacing limit is used inside the later volume subsequence.

6 Lattice fields

The field at a point will be represented by observables on finitely many nearby links. There are two choices to make: which local quantity an observable represents, and how to change its coefficients as the lattice spacing decreases. This section specifies these choices without yet integrating out any scales.

We begin with the exact symmetries of the Wilson measure in Subsection 6.1. Subsection 6.2 describes classical local observables by polynomials in Taylor coefficients and chooses finite bases for them. Subsection 6.3 states how a polynomial in link matrices must reproduce these classical observables. Subsection 6.4 then fixes the coefficients of the lattice insertions at a reference scale. Section 7 constructs the local polynomials and introduces sums of finite-region terms used in the integration argument.

We use the lattice, plaquettes and measure of Definitions 2.2, 2.3 and 2.4. The gauge group is as in Definition 2.1; simplicity and connectedness are not needed in this section.

6.1 The Wilson measure

The action depends on the transport around each elementary square, or plaquette, as explained in Definition 2.3. Conjugating that transport or reversing its direction leaves the action unchanged. This observation gives exact symmetries of the probability measure, before any limit is taken.

Definition 6.1 (Walk transport)**.** A walk w records a starting site x_0 and a finite ordered list of steps $(\sigma_j, \mu_j)_{j=1}^{r}$, with $\sigma_j \in \{-1, 1\}$ and $1 \le \mu_j \le 4$. Set $x_j = x_{j-1} + \sigma_j e_{\mu_j}$. For the links of Definition 2.2, its transport is the ordered product

$$U(w) = W_1 \cdots W_r, \qquad W_j = \begin{cases} U(x_{j-1}, \mu_j), & \sigma_j = 1, \\ U(x_j, \mu_j)^{-1}, & \sigma_j = -1. \end{cases} \tag{6.1}$$

The empty walk has transport 1.

For the walk of Definition 6.1, the start and endpoint are $x = x_0$ and $z = x_r$. The intermediate gauge matrices cancel in this product. The gauge action of Definition 2.5 gives

$$U^\gamma(w) = \gamma(x)U(w)\gamma(z)^{-1}\,. \tag{6.2}$$

For the estimates below, abbreviate the single-plaquette action and define its associated distance from the identity by

$$s(u) = 1 - \frac{1}{m}\operatorname{Re}\operatorname{tr} u, \qquad \chi(u) = \frac{\|u-1\|_{\mathrm{F}}}{\sqrt{m}} \qquad (u \in G), \tag{6.3}$$

where the Frobenius norm is given by $\|M\|_{\mathrm{F}}^2 = \operatorname{Re}\operatorname{tr}(M^*M)$ and M^* is the conjugate transpose. For $u, v \in G$, unitarity gives

$$\chi(u)^2 = 2s(u), \qquad 0 \le s(u) \le 2, \qquad \chi(uv) \le \chi(u) + \chi(v)\,. \tag{6.4}$$

Indeed, $\|u-1\|_{\mathrm{F}}^2 = 2m - 2\operatorname{Re}\operatorname{tr} u$ and $\|u-1\|_{\mathrm{F}} \le 2\sqrt{m}$. For the last inequality, write $uv - 1 = (u-1) + u(v-1)$ and use invariance of the Frobenius norm under unitary multiplication. Both s and χ are continuous and invariant under conjugation and inversion.

The coordinate symmetries we use are the permutations and reversals of the four lattice axes. They act on link configurations by reading the transport along the image of each positive link.

Definition 6.2 (Coordinate symmetries)**.** Let H_4 be the group of signed permutation matrices, with one nonzero entry ± 1 in each row and column. For $A(y) = hy + z$, with $h \in H_4$ and $z \in \Lambda_\ell$, the pullback A^*U reads the transport along the image of each positive link. If $he_\mu = e_\nu$, then $(A^*U)(x,\mu) = U(Ax,\nu)$; if $he_\mu = -e_\nu$, then $(A^*U)(x,\mu) = U(Ax - e_\nu, \nu)^{-1}$, since the image link is traversed in the reverse direction. The lattice and link conventions are those of Definition 2.2.

Write $\mathcal{U}_\ell$ for the configuration space of Definition 2.2, $C(\mathcal{U}_\ell;\mathbb{C})$ for its continuous complex functions, and $B_b(\mathcal{U}_\ell;\mathbb{C})$ for its bounded Borel complex functions. Here $L^1(\mu_{\ell,\beta})$ denotes the integrable functions for the Wilson measure of Definition 2.4. The following proposition justifies ordinary expectations of local polynomials and the changes of variables used throughout the construction.

Proposition 6.3 (The lattice law)**.** *For $\ell \ge 1$ and $\beta \in \mathbb{R}$, the measure of Definition 2.4 satisfies*

$$\mu_{\ell,\beta}(\mathcal{U}_\ell) = 1, \qquad C(\mathcal{U}_\ell;\mathbb{C}) \cup B_b(\mathcal{U}_\ell;\mathbb{C}) \subseteq L^1(\mu_{\ell,\beta})\,. \tag{6.5}$$

For every gauge transformation γ of Definition 2.5, coordinate symmetry A of Definition 6.2, and $F \in L^1(\mu_{\ell,\beta})$,

$$\int F(U^\gamma)\,d\mu_{\ell,\beta}(U) = \int F(U)\,d\mu_{\ell,\beta}(U) = \int F(A^*U)\,d\mu_{\ell,\beta}(U)\,. \tag{6.6}$$

Proof. There are three checks: the density is finite and positive, each change of variables preserves product Haar measure, and each preserves the action. Together they preserve the probability law. Write η_ℓ for the product Haar measure in (2.4).

1. *Normalisation and integrability.* The closed group $G \subset U(m)$ and its finite configuration space are compact. There are six plaquettes per site, each with energy in $[0,2]$. Thus

$$0 \le E_\ell \le 12\ell^4, \qquad e^{-12|\beta|\ell^4} \le e^{-\beta E_\ell(U)} \le e^{12|\beta|\ell^4}\,. \tag{6.7}$$

Integrating against probability Haar gives a finite positive normalising constant $Z_{\ell,\beta}$ as in (2.4). A continuous function on the configuration space is bounded, so both classes in (6.5) are integrable.

2. *Reference measure.* Haar invariance allows a fixed matrix to be multiplied on either side of a link, and allows its orientation to be reversed. Let h be the probability Haar measure of Definition 2.4; T_*h denotes its image under a map T. For $R_g(u) = ug$, $\mathrm{inv}(u) = u^{-1}$ and every integrable $f : G \to \mathbb{C}$, the identities are

$$(R_g)_*h = h, \qquad \mathrm{inv}_*h = h, \qquad \int_G f(aub)\, h(du) = \int_G f(u)\, h(du) \quad (a, b \in G)\,. \tag{6.8}$$

Indeed, $(R_g)_*h$ is left invariant with mass one, so Haar uniqueness identifies it with h. The image under inversion is then left invariant by right invariance of h, again with mass one. Applying these identities independently to each link gives, for the gauge map $\Gamma_\gamma : U \mapsto U^\gamma$,

$$(\Gamma_\gamma)_*\eta_\ell = \eta_\ell, \qquad (A^*)_*\eta_\ell = \eta_\ell\,. \tag{6.9}$$

For the second identity, the coordinate symmetry permutes the positive links and inverts those whose image orientation is negative.

3. *Action.* By (6.2), a gauge transformation conjugates a plaquette transport at its initial vertex. A coordinate symmetry sends each elementary square to one elementary square, possibly with reversed orientation. Write $he_\mu = \sigma_\mu e_{\pi(\mu)}$, with $\sigma_\mu \in \{-1, 1\}$. Write $\mathbf{1}_{\{P\}}$ for 1 when the condition P holds and 0 otherwise. The base site of the image square when both directions are written positively is

$$hy + z - \mathbf{1}_{\{\sigma_\mu = -1\}} e_{\pi(\mu)} - \mathbf{1}_{\{\sigma_\nu = -1\}} e_{\pi(\nu)},$$

and its pair of directions is the increasing rearrangement of $\{\pi(\mu), \pi(\nu)\}$. This gives a bijection of the plaquettes, even when $\ell = 1$ or 2. Since conjugation and inversion preserve s, we have

$$E_\ell(U^\gamma) = E_\ell(U), \qquad E_\ell(A^*U) = E_\ell(U)\,. \tag{6.10}$$

Thus the transformed density agrees pointwise with the original density:

$$e^{-\beta E_\ell(TU)} = e^{-\beta E_\ell(U)}, \qquad T \in \{\Gamma_\gamma, A^*\}.$$

For either $T = \Gamma_\gamma$ or $T = A^*$, equations (6.9) and (6.10) now give

$$\int F(TU)\, d\mu_{\ell,\beta}(U) = \frac{1}{Z_{\ell,\beta}} \int F(TU) e^{-\beta E_\ell(TU)}\, d\eta_\ell(U) = \int F(U)\, d\mu_{\ell,\beta}(U)\,.$$

This proves (6.6) for every integrable F.

□

6.2 Classical polynomials

We now specify the classical quantities that the lattice insertions of Definition 2.6 will represent. A classical connection is a smooth $\mathfrak{g}$-valued one-form $A = \sum_\mu A_\mu(z)\, dz^\mu$ near the origin. We describe it in radial gauge, meaning that its component along the vector from the origin to z vanishes: $\sum_\mu z^\mu A_\mu(z) = 0$. Its Taylor coefficients then satisfy a linear constraint, degree by degree. The constant coefficient vanishes: a constant one-form with zero contraction against every radial vector must be zero. A polynomial in finitely many such coefficients describes a classical local observable; invariance under simultaneous conjugation removes the choice of gauge at the origin.

Definition 6.4 (Gauge-invariant polynomials)**.** For $\mathfrak{g}$ of Definition 2.1 and $j \in \mathbb{N}$, let $\mathcal{A}_j$ be the real space of one-forms A_j whose components $A_{j,\mu}(z)$ are homogeneous polynomials of total degree $j+1$, with values in $\mathfrak{g}$, satisfying

$$\sum_{\mu=1}^{4} z^\mu A_{j,\mu}(z) = 0 \qquad (z \in \mathbb{R}^4)\,. \tag{6.11}$$

The list of Taylor coefficients, called a real jet, is a sequence $J = (A_j)_{j\geq 0}$ with $A_j \in \mathcal{A}_j$. The convention is

$$A_{j,\mu}(z) = \frac{1}{(j+1)!} D^{j+1} A_\mu(0)[z,\dots,z],$$

so the formal Taylor series of each component in radial gauge is $\sum_{j\geq 0} A_{j,\mu}(z)$. Choose real coordinates on each $\mathcal{A}_j$, assigning weight $j+2$ to each coordinate. The weight of a monomial is the sum of the weights of its factors, counted with multiplicity. For $d \in \mathbb{N}$, let V_d be the space of complex polynomials in these coordinates, of total weight at most d, invariant under simultaneous conjugation $\mathrm{Ad}(g)X = gXg^{-1}$ by $g \in G$.

The displayed convention specifies the Taylor coefficients, including their factorials; for example, $A_{0,\mu}(z) = DA_\mu(0)[z]$. The weight $j+2$ records the scaling of a link integral: the polynomial one-form contributes $j+1$ powers of length and integration contributes one more. Weights add under multiplication. We also count the number of factors in a monomial, its ordinary polynomial degree; the two gradings will serve different purposes.

Definition 6.5 (Actions and gradings)**.** Let $O(4) = \{h \in M_4(\mathbb{R}) : h^{\mathsf{T}} h = 1\}$ be the group of real orthogonal matrices, where h^{T} is the transpose. An element $h \in O(4)$ acts on the one-forms and invariant polynomials of Definition 6.4 by

$$(T_h A_j)_\mu(z) = \sum_\nu h_{\mu\nu} A_{j,\nu}(h^{-1}z), \qquad (h \cdot v)(J) = v(T_{h^{-1}} J)\,. \tag{6.12}$$

Coefficient conjugation is denoted by $v \mapsto \bar{v}$; a real polynomial has $\bar{v} = v$. Let $V_d^{w,f}$ be the subspace homogeneous of weight w and factor number f. Every monomial of such a polynomial has weight w and ordinary polynomial degree f.

The action (6.12) preserves (6.11) and commutes with conjugation by G.

The weight bound leaves only finitely many coordinates and monomials. We can therefore choose one finite basis at each degree, retaining the basis elements already chosen at lower degrees. Real homogeneous bases make conjugation and the two gradings explicit.

Lemma 6.6 (Finite polynomial spaces)**.** *For the spaces of Definition 6.4, evaluation on real jets is injective, and*

$$\dim_{\mathbb{C}} V_d < \infty, \qquad 1 \in V_d \subseteq V_{d+1}, \qquad V_d = \bigoplus_{\substack{0\leq w\leq d\\ 0\leq f\leq w}} V_d^{w,f}\,. \tag{6.13}$$

Each summand is preserved by (6.12) *and coefficient conjugation. There are nested real bases, homogeneous in both gradings and containing* 1*.*

Proof. The basis claim follows from three finite-dimensional facts: only finitely many monomials occur, the invariant condition splits by the two grades, and each real homogeneous piece has a real basis.

1. *Finitely many monomials.*

   Only coordinates with $j + 2 \le d$ can occur in a monomial of weight at most d. There are finitely many such coordinates, and the exponent of each is bounded by d. Thus finitely many monomials span the polynomials under consideration, and their invariant subspace is finite dimensional.

2. *Evaluation and homogeneous pieces.* A complex polynomial vanishing on every real jet vanishes identically, by the one-variable polynomial identity theorem applied successively in these finitely many real coordinates.

   The simultaneous matrix action $A_j \mapsto \mathrm{Ad}(g)A_j$, where $\mathrm{Ad}(g)X = gXg^{-1}$, acts linearly within each $\mathcal{A}_j$ and therefore preserves both grades. Writing $v = \sum_{w,f} v^{w,f}$, comparison of the homogeneous terms gives

   $$v(\mathrm{Ad}(g)J) = v(J) \Longrightarrow \quad v^{w,f}(\mathrm{Ad}(g)J) = v^{w,f}(J) \qquad (g \in G)\,. \tag{6.14}$$

   Thus the homogeneous parts of an invariant polynomial are themselves invariant. Every coordinate has weight at least two; in particular the factor number cannot exceed the weight, giving the index range in (6.13). The same linearity argument applies to (6.12), and these real linear changes commute with coefficient conjugation. Each invariant homogeneous component is the complex span of its real polynomials, by decomposition into real and imaginary coefficient parts.

3. *Nested real bases.* Choose a real basis in each component, beginning with $\{1\}$ for the constant component, and retain these choices as d increases.

□

Definition 6.7 (Chosen polynomial bases)**.** Fix the nested bases of Lemma 6.6. For $d \in \mathbb{N}$, write

$$(b_{d,i})_{0 \le i < \dim_{\mathbb{C}} V_d}, \qquad b_{d,0} = 1, \qquad b_{d,i} \in V_d^{w_{d,i}, f_{d,i}},$$

with the two gradings of Definition 6.5.

We identify each polynomial with its coefficient vector when applying a linear map to V_d. The continuum spaces F_d of Definition 2.17 will arise after taking limits of lattice correlations and quotienting by polynomials whose limiting correlations with every allowed test vanish. The spaces V_d contain the classical observable polynomials before these relations are imposed.

6.3 Local representatives

A polynomial in Taylor coefficients is not yet a lattice observable. We seek a polynomial in nearby link matrices whose Taylor expansion, when tested on a classical connection, recovers the prescribed classical observable polynomial. The linear map assigning such a link polynomial to each $v \in V_d$ is the local representative map: it assigns to each classical polynomial an observable on finitely many links, with the required Taylor coefficients through degree d.

On $\mathbb{Z}^4$, the gauge action of Definition 2.5 and the coordinate pullbacks of Definition 6.2 use the same formulas as on a torus. Polynomial functions on a finite set of links can then be evaluated on any periodic lattice.

Definition 6.8 (Local polynomial functions)**.** For a finite set $\Sigma \subset \mathbb{Z}^4$, let $\mathcal{P}(\Sigma)$ be the polynomial functions of matrix entries and conjugate entries of links with both endpoints in Σ. For $P \in \mathcal{P}(\Sigma)$ and $o \in \Lambda_\ell$, write $P(o;U)$ for its value after substituting

$$U(x,\mu) \longmapsto U(o + [x]_\ell, \mu)\,. \tag{6.15}$$

Here $[x]_\ell$ is reduction of $x \in \mathbb{Z}^4$ modulo ℓ; the link conventions are those of Definition 2.2.

This implements the finite-site locality requirement Reg4 of Definition 2.8. The substitution is meaningful even when distinct links of $\mathbb{Z}^4$ become identical after reduction modulo ℓ.

To test the classical normalisation, retain the Taylor coefficients that can occur in V_d and form their polynomial connection. Its transport along one edge will be the test link.

Definition 6.9 (Truncated connections and test links)**.** For $d \in \mathbb{N}$, a real jet $J = (A_j)$ of Definition 6.4 and $\varepsilon \in \mathbb{R}$, set

$$\mathcal{J}_d = \bigoplus_{j+2\le d} \mathcal{A}_j, \qquad A^J_\mu(z) = \sum_{j+2\le d} A_{j,\mu}(z), \qquad (\mathrm{sc}_\varepsilon J)_j = \varepsilon^{j+2} A_j\,. \tag{6.16}$$

The finite tuple $(A_j)_{j+2\le d} \in \mathcal{J}_d$ determines A^J; its dependence on d is suppressed in this notation. A finite tuple is viewed as a jet by setting its higher coefficients to zero. For $x \in \mathbb{Z}^4$ and $1 \le \mu \le 4$, define the test link $U_{J,\varepsilon}(x,\mu)$ at spacing ε by the endpoint of

$$\begin{aligned} B(t) &= \varepsilon A^J_\mu\big(\varepsilon(x + te_\mu)\big), & u'(t) &= u(t)B(t), \\ u(0) &= 1, & U_{J,\varepsilon}(x,\mu) &= u(1)\,. \end{aligned} \tag{6.17}$$

Here $t \in [0,1]$ parametrises the edge $x + te_\mu$ and $B(t)$ is the matrix coefficient of its transport equation.

We justify existence, uniqueness and analytic dependence of these links before using them. Successive integration gives the ordered series

$$\begin{aligned} u(t) &= 1 + \sum_{r\ge 1} \int_{0<s_1<\cdots<s_r<t} B(s_1)\cdots B(s_r)\, ds_1 \cdots ds_r, \\ \left\| \int_{0<s_1<\cdots<s_r<t} B(s_1)\cdots B(s_r)\, ds_1\cdots ds_r \right\|_{\mathrm{F}} &\le \frac{M^r}{r!} \end{aligned} \tag{6.18}$$

for $r \ge 1$ and $0 \le t \le 1$ when $\|B(s)\|_{\mathrm{F}} \le M$. It proves the four claims as follows.

1. For a fixed real jet, B is polynomial in ε and the factorial bound holds uniformly on a small complex ε-neighbourhood of zero. The series converges locally uniformly there, solves the equation, and its endpoint is real analytic in ε near zero.

2. The integral equation has at most one solution: iterating the equation for the difference of two solutions gives the same factorial bound, which tends to zero.

3. For real ε, step-function approximations to B have solutions that are products of matrices $\exp(\Delta t\, B(t_j)) \in G$.

4. The series bound proves these products converge to u. Since G is closed, the endpoint link in (6.17) belongs to G.

The local representative map must preserve the exact lattice symmetries and the constant observable. Its classical normalisation requires equality of Taylor coefficients through the degree being represented. These are the conditions in the next definition.

Definition 6.10 (Normalised local representative map)**.** A normalised local representative map in degree $d \in \mathbb{N}$ is a finite set $\Sigma_d \subset \mathbb{Z}^4$ and a complex linear map $P_d : V_d \to \mathcal{P}(\Sigma_d)$, with V_d from Definition 6.4 and polynomial functions as in Definition 6.8, such that

$$0 \in \Sigma_d, \qquad h\Sigma_d = \Sigma_d \quad (h \in H_4), \tag{6.19}$$

and, for all $v \in V_d$, link configurations U on $\mathbb{Z}^4$, $\gamma : \mathbb{Z}^4 \to G$ and $h \in H_4$,

$$\begin{aligned} P_d[v](U^\gamma) &= P_d[v](U), & P_d[1] &= 1, \\ P_d[\bar{v}](U) &= \overline{P_d[v](U)}, & P_d[v](h^*U) &= P_d[h \cdot v](U)\,. \end{aligned} \tag{6.20}$$

For every real jet J and integer $0 \le k \le d$, it also satisfies

$$\frac{d^k}{d\varepsilon^k} P_d[v](U_{J,\varepsilon})\Big|_{\varepsilon=0} = \frac{d^k}{d\varepsilon^k} v(\mathrm{sc}_\varepsilon\, J)\Big|_{\varepsilon=0}\,. \tag{6.21}$$

The jets and their coordinate action are as in Definitions 6.4 and 6.5; H_4 and its pullbacks are from Definition 6.2, and the scaling and test links are from Definition 6.9.

The Taylor identity imposes no equality at nonzero spacing. No compatibility of P_d and P_{d+1} is required. Proposition 7.2 constructs these maps and proves derivative estimates used when link variables are integrated in groups to pass between lattice spacings.

6.4 Renormalised insertions

As shorter scales are integrated out, the coefficient of one classical polynomial can come to depend on several fine-lattice coefficients. We normalise the observable by its coefficients at one reference level, chosen independently of cutoff and volume; the coefficient transfer records this dependence from the fine lattice to that reference level. Its inverse determines which fine-lattice polynomial represents a prescribed polynomial at that reference level. This describes a particular choice of the insertions in Definition 2.6, not an additional condition on that definition.

Use a_N, N, p from Definition 2.7, the basis from Definition 6.7, and a local representative map from Definition 6.10. In the terminology of Definition 2.6, the observable index will be (d, v), where $v \in V_d$ is the classical polynomial and d its assigned scaling degree.

Definition 6.11 (Reference normalisation)**.** For each degree d, choose a reference level $r(d) \in \mathbb{N}$, independently of N, p. Choose any positive factor rescaling parameter $\theta_N > 0$ at each level N. It is used only in $S_{\theta_N^{-1}}$ below; small-plaquette conditions in the activity estimates use the separate cutoff t of (7.8). Define the linear factor scaling on the basis of Definition 6.7 by

$$S_t b_{d,i} = t^{f_{d,i}} b_{d,i} \qquad (t > 0)\,. \tag{6.22}$$

For an invertible complex linear map $\tau_{N,p,d} : V_d \to V_d$, the candidate insertion is

$$I_{N,p}\big((d, v), o; U\big) = \frac{1}{a_N^d} P_d\Big[S_{\theta_N^{-1}} \tau_{N,p,d}^{-1} v\Big]\,(o; U), \tag{6.23}$$

with placement (6.15).

Factor scaling multiplies a component with f factors by $\frac{1}{\theta_N^f}$; the prefactor $\frac{1}{a_N^d}$ supplies the physical degree normalisation.

For now, take as data complex linear maps $x_j : V_d \to V_d$, for $r(d) \le j \le N$, that are intended to send coefficients at the finest level N to coefficients after integrating down to level j. Their construction for the Wilson family remains open. Require $N \ge r(d)$. The conditions that make $\tau_{N,p,d}$ a reference transfer are

$$x_N(v) = v, \qquad x_{r(d)}(v) = \tau_{N,p,d} v, \qquad x_j(a_N^d 1) = a_j^d 1 \quad (r(d) \le j \le N)\,. \tag{6.24}$$

These equalities fix how coefficients are normalised between levels; they assert no pointwise identity between observables on different lattices. Reality and signed-permutation covariance require

$$\tau_{N,p,d}\bar{v} = \overline{\tau_{N,p,d} v}, \qquad \tau_{N,p,d}(h \cdot v) = h \cdot (\tau_{N,p,d} v) \quad (h \in H_4)\,. \tag{6.25}$$

The locality of the resulting insertion follows from its polynomial formula, even on a torus smaller than the chosen set of sites. The unit normalisation is different: it uses the reference-transfer conditions and therefore applies only beyond the reference level. The following proposition separates these two conclusions.

Proposition 6.12 (Locality of the insertions)**.** *Let P_d satisfy Definition 6.10, $\theta_N > 0$ and $\tau_{N,p,d} \in \mathrm{GL}_{\mathbb{C}}(V_d)$. For all N, p, the map $v \mapsto I_{N,p}\big((d,v),o;\cdot\big)$ in (6.23) is complex linear and satisfies Reg2–Reg4 of Definition 2.8, with finite set Σ_d. Moreover, where the translated configuration is defined by $U(\cdot + z)(x,\mu) = U(x+z,\mu)$,*

$$I_{N,p}\big((d,v),o;U(\cdot+z)\big) = I_{N,p}\big((d,v),o+z;U\big)\,. \tag{6.26}$$

If $N \ge r(d)$ and (6.24)–(6.25) hold, then

$$\begin{aligned} I_{N,p}\big((d,\bar{v}),o;U\big) &= \overline{I_{N,p}\big((d,v),o;U\big)}, \\ I_{N,p}\big((d,h\cdot v),A(o);U\big) &= I_{N,p}\big((d,v),o;A^*U\big), \\ I_{N,p}\big((d,a_{r(d)}^d 1),o;U\big) &= 1, \qquad x_{r(d)}(\tau_{N,p,d}^{-1} v) = v \end{aligned} \tag{6.27}$$

for $A(o) = ho + z$, $h \in H_4$, as in Definition 6.2.

Proof. The conclusions follow in three steps: the defining polynomial gives regularity and locality, the symmetries of its factors give covariance, and the transfer endpoint fixes the unit.

1. *Locality and continuity.*

   The coefficient changes in (6.23) are complex linear. A polynomial in link entries and their conjugates is continuous, and the local representative map is gauge invariant by (6.20); this proves Reg2 and Reg3. Placement (6.15) reads only links with both endpoints in $o + [\Sigma_d]_{L_pD_N}$, proving Reg4. Identifying vertices of $\mathbb{Z}^4$ after reduction modulo L_pD_N does not enlarge that set. Translating the placement proves (6.26).

2. *Reality and coordinate covariance.* Factor scaling commutes with coefficient conjugation and H_4, since both actions preserve factor number by Lemma 6.6. Together with (6.25) and (6.20), this gives the first two identities in (6.27).

3. *Reference unit.* For the unit, the endpoint conditions (6.24) give

$$\begin{aligned} \tau_{N,p,d}^{-1}(a_{r(d)}^d 1) &= a_N^d 1, \\ I_{N,p}\big((d, a_{r(d)}^d 1), o; U\big) &= \frac{1}{a_N^d} P_d\big[S_{\theta_N^{-1}}(a_N^d 1)\big](o; U) = \frac{a_N^d}{a_N^d} P_d[1](o; U) = 1\,. \end{aligned} \tag{6.28}$$

The remaining identity follows by applying the reference endpoint condition to $\tau_{N,p,d}^{-1} v$.

□

The disjoint index set $\mathcal{T} = \bigsqcup_d V_d$ retains the degree d even if v belongs to a smaller polynomial space. Proposition 6.3 also shows that each finite-volume product of these continuous observables is integrable. The running-coupling condition Reg1 and convergence of the moments remain later ultraviolet obligations.

Unresolved. The invertible coefficient transfers satisfying (6.24)–(6.25) must still be constructed for the same Wilson family. Proposition 6.12 assumes them; no transfer through scale integration is proved here.[1]

7 Local polynomials and activities

Section 6 specified which classical observable a lattice polynomial must reproduce at small spacing. We now construct those polynomials and introduce the form in which local interactions will be carried through the integration over scales. The common principle is to retain the region and the coefficients of every term: this allows multiplication and summation without losing the estimates on individual observables. An activity is a scalar function attached to a finite set of lattice sites, used to represent one term in the weight after some link variables have been integrated out.

Subsection 7.1 recovers finitely many Taylor coefficients from finitely many transported plaquettes. Subsection 7.2 turns this recovery into the local representative maps of Definition 6.10, with derivative bounds independent of the torus size. Subsection 7.3 records insertions as coefficients of auxiliary commuting variables whose squares vanish. Subsection 7.4 gives bounds for these activities and proves absolute convergence of their sums.

7.1 Recovering Taylor coefficients

The curvature of a connection A is the two-form with components

$$F_{\mu\nu}(A) = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

At first order in the size of A, a plaquette transport records the average of the linear part $\partial_\mu A_\nu - \partial_\nu A_\mu$ over its square. The commutator term is quadratic, so it does not enter this first variation. Different squares give different linear measurements of the Taylor coefficients. For any fixed degree, finitely many such measurements suffice; radial gauge makes recovery of the connection from this linear curvature possible.

Fix $d \in \mathbb{N}$. Use the spaces $\mathcal{A}_j, V_d$ of Definition 6.4 and the finite tuples $J \in \mathcal{J}_d$ of Definition 6.9. Such a tuple specifies the polynomial connection A^J and the test links $U_{J,\varepsilon}$ defined there. On $\mathfrak{g}$ use the norm $|X|_{\mathfrak{g}} = \frac{\|X\|_{\mathrm{F}}}{\sqrt{m}}$, and let $P_{\mathfrak{g}}$ be real Frobenius orthogonal projection onto $\mathfrak{g}$. The following

[1] Formal correspondence at revision `3456e5e`: Transfer conditions: `YM/Source/Insertion.lean`, lines 89–114; same-family application: `YM/FullVolume/Cross.lean`, lines 30–71.

matrix-valued function maps a plaquette matrix to the Lie algebra; its derivative at the identity is the identity map, so it retains the linear curvature term just described. Here $u \in G$, and χ is the distance from the identity in (6.3):

$$\mathcal{F}(u) = P_{\mathfrak{g}}\Big(\frac{u - u^{-1}}{2}\Big), \qquad |\mathcal{F}(u)|_{\mathfrak{g}} \le \chi(u), \qquad \frac{d}{ds}\mathcal{F}(e^{sX})\Big|_{s=0} = X \quad (X \in \mathfrak{g})\,. \tag{7.1}$$

Projection commutes with conjugation by G, since conjugation is orthogonal and preserves $\mathfrak{g}$. The bound follows by writing $u - u^{-1} = (u-1) - (u^{-1}-1)$ and using (6.4). The derivative follows from $e^{sX} - e^{-sX} = 2sX + O(s^3)$. Also $\mathcal{F}(u^{-1}) = -\mathcal{F}(u)$.

For a plaquette p based at $x \in \mathbb{Z}^4$ and a walk ω from 0 to x, transport its curvature back to the origin:

$$Q_{\omega,p}(U) = \mathrm{Ad}\big(U(\omega)\big)\mathcal{F}(U_p), \qquad Q_{\omega,p}(U^\gamma) = \mathrm{Ad}\big(\gamma(0)\big)Q_{\omega,p}(U)\,. \tag{7.2}$$

Here $U(\omega)$ is the ordered walk transport of Definition 6.1, U_p is the plaquette matrix of Definition 2.3, and $\mathrm{Ad}(g)X = gXg^{-1}$ as in Definition 6.4. The second identity uses the gauge transformation of Definition 2.5 and (6.2). All these quantities now transform at the same site, so invariant polynomials in them will be gauge invariant. They read only links in the walk and plaquette, with both endpoints included.

The next lemma supplies a local analytic recovery map. It uses several plaquettes because their combined measurements have enough independent components to recover all Taylor coefficients through degree d.

Lemma 7.1 (Finite sampling)**.** *There is a finite family $(\omega_a, p_a)_{a=1}^{M_d}$ for which*

$$\mathcal{Q}_d(J) = \big(Q_{\omega_a,p_a}(U_{J,1})\big)_{a=1}^{M_d}, \qquad \mathcal{Q}_d : \mathcal{J}_d \longrightarrow \mathfrak{g}^{M_d}, \tag{7.3}$$

has injective derivative at 0. There is a real analytic map $R_d : \mathfrak{g}^{M_d} \to \mathcal{J}_d$, defined near 0, satisfying

$$R_d(0) = 0, \qquad R_d\mathcal{Q}_d = \mathrm{id}, \qquad R_d(\mathrm{Ad}(g)q) = \mathrm{Ad}(g)R_d(q) \quad (g \in G)\,. \tag{7.4}$$

The last two equalities hold near the origin, for each fixed $g \in G$; the test links are from Definition 6.9, and $\mathrm{Ad}(g)$ acts on every component of a tuple.

Proof. The recovery has three steps: all square measurements together detect every jet, finite dimensionality reduces them to finitely many measurements, and the inverse function theorem recovers the jet from their nonlinear values. The two properties to establish are

$$\mathcal{Q}_d(J) = \big(Q_{\omega_a,p_a}(U_{J,1})\big)_{a=1}^{M_d}, \qquad R_d\mathcal{Q}_d(J) = J \ \text{ near } 0.$$

1. *All square measurements detect the jet.*

   Recall that $A_\mu^J = \sum_{j+2\le d} A_{j,\mu}$ is the truncated connection of Definition 6.9. Vary its size by $\lambda \in \mathbb{R}$ in the transport equation (6.17), keeping the edge fixed. At $\lambda = 0$ the solution is 1, so its first variation is the integral along that edge:

   $$\frac{d}{d\lambda}U_{\lambda J,1}(x,\mu)\Big|_{\lambda=0} = \int_0^1 A_\mu^J(x + te_\mu)\,dt\,.$$

   Since $\mathcal{F}(1) = 0$, differentiating the conjugating transport in (7.2) contributes zero. Consequently, for the unit square $p = (x;\mu,\nu)$,

   $$D\big(Q_{\omega,p}(U_{J,1})\big)_{J=0}[J] = \int_0^1\int_0^1 (\partial_\mu A_\nu^J - \partial_\nu A_\mu^J)(x + se_\mu + te_\nu)\,ds\,dt\,. \tag{7.5}$$

This is independent of ω.

Suppose all these integrals vanish for integer x. Each is a polynomial in x; a polynomial vanishing on $\mathbb{Z}^4$ vanishes everywhere, by successive one-variable polynomial identities. For the highest homogeneous term A_j, the highest part of that integral is $\partial_\mu A_{j,\nu} - \partial_\nu A_{j,\mu}$. It must therefore vanish. Differentiating the radial identity (6.11) and using homogeneity gives

$$(j+2)A_{j,\mu}(z) = \sum_\nu z^\nu (\partial_\nu A_{j,\mu} - \partial_\mu A_{j,\nu})(z)\,. \tag{7.6}$$

Thus $A_j = 0$, and descending in degree proves $J = 0$.

2. *Retain finitely many measurements.* The kernels of the linear measurements have zero intersection. While their intersection is nonzero, another measurement decreases its dimension, so finitely many suffice. For $d < 2$ the source space is zero and the assertion is immediate.

3. *Invert the finite nonlinear map.* The map $\mathcal{Q}_d$ is analytic near zero by the transport series (6.18). Set $B = D\mathcal{Q}_d(0)$. Conjugating the connection by a constant $g \in G$ conjugates its test links, by uniqueness of (6.17). Equation (7.2) and differentiation therefore give

$$\mathcal{Q}_d(\mathrm{Ad}(g)J) = \mathrm{Ad}(g)\mathcal{Q}_d(J), \qquad B\,\mathrm{Ad}(g) = \mathrm{Ad}(g)B\,.$$

The adjoint action on $\mathfrak{g}^{M_d}$ is orthogonal and preserves the range $\operatorname{ran} B = \{BJ : J \in \mathcal{J}_d\}$ of B in both directions. Let Π be the orthogonal projection onto this range; it commutes with that action. Here B^{-1} denotes the inverse from $\operatorname{ran} B$ to $\mathcal{J}_d$. The linear map $C = B^{-1}\Pi$ satisfies

$$CB = \mathrm{id}, \qquad C\,\mathrm{Ad}(g) = \mathrm{Ad}(g)C, \qquad D(C\mathcal{Q}_d)(0) = \mathrm{id}\,.$$

Since $\mathcal{Q}_d(0) = 0$, the analytic inverse function theorem gives a local inverse g_d of $C\mathcal{Q}_d$ fixing 0. Then $R_d = g_d C$ has the first two properties in (7.4). The last identity in (7.4) follows from uniqueness of the local inverse, for each fixed $g \in G$.

□

7.2 Constructing the local polynomials

The recovery map is analytic only near the zero connection. Its Taylor polynomial, however, can be evaluated on every link configuration. Composing this polynomial with the classical observable polynomials $v \in V_d$ of Definition 6.4 gives the required local representatives. A finite average restores the signed coordinate symmetries exactly.

We measure derivatives by varying each link matrix within G. The permitted size of a variation grows with its distance from the insertion site, so we first specify that distance. On the lattice Λ_ℓ of Definition 2.2, periodic maximum distance is

$$d_\ell(x,y) = \min_{z \in \ell\mathbb{Z}^4} \|\widetilde{x} - \widetilde{y} + z\|_\infty\,, \tag{7.7}$$

where tildes denote arbitrary integer representatives and $\|z\|_\infty = \max_\mu |z^\mu|$. For $0 < t \le 1$ and a set of sites $X \subseteq \Lambda_\ell$, define the bad sites and the configurations good on X by

$$\begin{aligned} \mathcal{B}_t(U) &= \{x : \text{some plaquette } p \text{ containing } x \text{ has } \chi(U_p) > t\}, \\ \mathcal{G}_t(X) &= \{U : X \cap \mathcal{B}_t(U) = \varnothing\}\,. \end{aligned} \tag{7.8}$$

Every plaquette touching X is tested, and equality $\chi(U_p) = t$ belongs to the good set. The bad-site indicators are Borel and gauge invariant, by continuity and conjugation invariance of χ.

Fix an insertion site $o \in \Lambda_\ell$ and integers $k, q \in \mathbb{N}$. Here k is the largest derivative order and q controls the permitted growth with distance. A direction A assigns an element of $\mathfrak{g}$ to each positive link e. The path $U(e) \mapsto e^{sA(e)}U(e)$ changes each matrix by multiplication on the left and remains in G for real s (Definition 2.1). Differentiate along this path and impose the bound

$$D_A F(U) = \frac{d}{ds} F\big((e^{sA(e)}U(e))_e\big)\Big|_{s=0}, \qquad |A(e)|_{\mathfrak{g}} \le t\big(1 + d_\ell(e^-, o)\big)^q . \tag{7.9}$$

Here e^- is the initial site of the link. Write $\mathrm{Dir}_{t,q}(o)$ for the directions satisfying the bound in (7.9), and define

$$\|F\|_{t,k,q;o,X} = \sup_{\substack{U \in \mathcal{G}_t(X),\ 0 \le r \le k \\ A_1,\dots,A_r \in \mathrm{Dir}_{t,q}(o)}} |D_{A_1} \cdots D_{A_r} F(U)| . \tag{7.10}$$

For $r = 0$, the expression under the supremum is $|F(U)|$. This is a seminorm on functions with these derivatives; no value outside $\mathcal{G}_t(X)$ is measured.

The small factor in the following bound counts the number of classical factors, independently of their weights. The local representative maps are chosen before the derivative order, the threshold and the torus.

Proposition 7.2 (Normalised local representative maps)**.** *For the bases $b_{d,i}$ and factor numbers $f_{d,i}$ of Definition 6.7, one can choose local representative maps P_d satisfying Definition 6.10, simultaneously for all d. Each $P_d[b_{d,i}]$ is a polynomial in real coordinates of finitely many quantities (7.2), with every monomial containing at least $f_{d,i}$ factors. For every $k, q \in \mathbb{N}$ with $k \ge 6$, $q \ge 2$, there is $A_{d,k,q} \ge 0$ such that*

$$\big\|P_d[b_{d,i}](o;\cdot)\big\|_{t,k,q;o,\,o+[\Sigma_d]_\ell} \le A_{d,k,q} t^{f_{d,i}} \qquad (\ell \ge 1,\ o \in \Lambda_\ell,\ 0 < t \le 1) . \tag{7.11}$$

Proof. The construction has three parts: approximate the inverse to the required Taylor order, average over signed coordinate changes, and estimate each product of transported plaquettes on the good set. The two key controls are

$$\frac{d^r}{d\varepsilon^r} R_d^{[d]}(\mathcal{Q}_d(\mathrm{sc}_\varepsilon J))\Big|_{\varepsilon=0} = \frac{d^r}{d\varepsilon^r} \mathrm{sc}_\varepsilon J\Big|_{\varepsilon=0} \quad (0 \le r \le d), \qquad \|P_d[b_{d,i}]\|_{t,k,q;o,\,o+[\Sigma_d]_\ell} \le A_{d,k,q} t^{f_{d,i}} .$$

By (7.4), the composition $R_d^{[d]}\mathcal{Q}_d(J)$ has the same Taylor coefficients as J through order d at $J = 0$.

1. *Match the Taylor coefficients and retain factors.*

   Let $R_d^{[d]}$ be the ordinary Taylor polynomial at 0 of the recovery map R_d in Lemma 7.1, through degree d. Apply it to the finite tuple of transported plaquettes chosen in that lemma and set

   $$P_d^0[v](U) = v\Big(R_d^{[d]}\big((Q_{\omega_a,p_a}(U))_a\big)\Big) . \tag{7.12}$$

   The polynomial $R_d^{[d]}$ has zero constant term, real coefficients, and commutes with conjugation by G, by differentiating (7.4). Hence P_d^0 is gauge invariant, respects coefficient conjugation, and sends 1 to 1. Inverses of unitary matrices are conjugate transposes, so (7.12) is a polynomial function of link entries and their conjugates. Substituting polynomials with zero constant term into a classical observable polynomial with $f_{d,i}$ factors leaves at least that many

transported-curvature factors in every monomial. Only the recovery map R_d is replaced by its degree- d Taylor polynomial. After composition with v, we keep the full resulting polynomial in the link variables, so it is defined on every link configuration.

For the scaling of jets in Definition 6.9, homogeneity in the transport equation and uniqueness give

$$U_{J,\varepsilon} = U_{\mathrm{sc}_\varepsilon J,1}\,. \tag{7.13}$$

The compositions $R_d^{[d]}\mathcal{Q}_d$ and $R_d\mathcal{Q}_d$ have equal Taylor coefficients through degree d at zero. Composing with $\mathrm{sc}_\varepsilon J$ and then v proves (6.21) for P_d^0.

2. *Restore the exact signed symmetries.* Average over the group of Definition 6.2:

$$P_d[v](U) = \frac{1}{|H_4|}\sum_{h\in H_4} P_d^0[h^{-1}\cdot v](h^*U)\,. \tag{7.14}$$

Changing the summation index gives the signed coordinate identity in (6.20); the unit and reality identities pass to the average as well. For the Taylor identity, the test links satisfy

$$h^*U_{J,\varepsilon} = U_{T_{h^{-1}}J,\varepsilon} \qquad (h\in H_4)\,. \tag{7.15}$$

This follows by substitution in (6.17) and uniqueness; on a reversed link the solution is $u(1)^{-1}u(1-t)$. In each summand of (7.14), the inverse action on polynomials from Definition 6.5 cancels the change in the test jet, preserving (6.21). A transformed transported plaquette is another transported plaquette, with a possible sign from reversal of its circulation. Since the action on polynomials preserves factor number, the lower factor bound survives averaging. Enclose all walks and plaquettes occurring in this finite sum in one centred integer cube Σ_d. This gives every condition in Definition 6.10.

3. *Bound derivatives on the good set.* For the bound, each transported curvature Q has value at most t on the good set. All links occurring in its fixed walks and plaquettes have maximum distance at most a fixed R from the origin on $\mathbb{Z}^4$. After periodic placement their distance is still at most R, so the allowed direction there has size at most $t(1+R)^q$. Each differentiation of a link product introduces one such direction. For $1\le r\le k$, this gives

$$|Q(U)|_{\mathfrak{g}} \le t, \qquad |D_{A_1}\cdots D_{A_r}Q(U)|_{\mathfrak{g}} \le C_{r,q}t^r \le C_{r,q}t\,. \tag{7.16}$$

The constant includes the fixed matrix projection and finitely many path factors. Leibniz's rule therefore gives Ct^s for every derivative through order k of a monomial with s factors. Sum its finitely many coefficients and use $s\ge f_{d,i}$ and $t\le 1$ to obtain (7.11). Repeated links on a small torus affect only the finite differentiation multiplicities, so the constant is independent of ℓ.

□

These polynomial functions are smooth along the link variations (7.9) at every configuration; the estimate applies on the closed good set, including its boundary. The derivatives are taken on the polynomial itself, before multiplying by any indicator of the good set. When regular terms are defined in Subsection 7.4, their derivatives are likewise taken before multiplication by the good-set indicator.

7.3 Products of insertions

To record one observable in a weighted product, introduce a commuting formal variable ζ with $\zeta^2 = 0$. The coefficient of ζ then records exactly one insertion. Several such variables record finite lists of insertions, and restrictions on allowed lists can be imposed in the same algebra.

Let V be a finite set indexing the formal variables, and let $\mathcal{R} \subseteq 2^V$ be the collection of allowed sets of insertions. Assume that $\mathcal{R}$ is closed under taking subsets and contains $\varnothing$ and every singleton. For $I \in \mathcal{R}$, write $\zeta_I = \prod_{v \in I} \zeta_v$. Define the algebra and its weighted coefficient norm by

$$\begin{aligned} \mathcal{A}_{\mathcal{R}} &= \bigoplus_{I \in \mathcal{R}} \mathbb{C}\zeta_I, \qquad a = \sum_{I \in \mathcal{R}} a_I \zeta_I, \quad 1 = \zeta_\varnothing, \\ \zeta_I \zeta_J &= \begin{cases} \zeta_{I \cup J}, & I \cap J = \varnothing,\ I \cup J \in \mathcal{R}, \\ 0, & \text{otherwise}, \end{cases} \\ \|a\|_\omega &= \sum_{I \in \mathcal{R}} \omega^I |a_I|, \qquad \omega^I = \prod_{v \in I} \omega_v, \quad \omega_v > 0\,. \end{aligned} \tag{7.17}$$

The variables commute; their squares, and products excluded by $\mathcal{R}$, are zero. For three factors, either both orders of multiplication give zero, or both give the monomial indexed by the union of their pairwise disjoint variable sets; hence multiplication is associative.

The weights multiply exactly when disjoint insertions are combined. Thus the estimates below have no extra factor depending on the number of formal variables.

Proposition 7.3 (Square-zero products)**.** *For $a, b \in \mathcal{A}_{\mathcal{R}}$ of (7.17),*

$$\|1\|_\omega = 1, \qquad \|ab\|_\omega \le \|a\|_\omega \|b\|_\omega, \qquad \left\| \sum_{r=0}^{|V|} \frac{a^r}{r!} \right\|_\omega \le e^{\|a\|_\omega}\,. \tag{7.18}$$

If $a_\varnothing = b_\varnothing = 0$, then $a^{|V|+1} = 0$ and the finite exponential $\operatorname{Exp}(a) = \sum_{r=0}^{|V|} \frac{a^r}{r!}$ satisfies

$$\operatorname{Exp}(a+b) = \operatorname{Exp}(a)\operatorname{Exp}(b), \qquad \operatorname{Exp}(-a)\operatorname{Exp}(a) = 1\,. \tag{7.19}$$

Proof. The norm estimate follows because only disjoint variable sets multiply nontrivially, and their weights multiply. The exponential identities follow from nilpotence and the finite binomial formula. We prove these as follows.

1. *Product bound.* A nonzero coefficient in ab comes from disjoint sets I, J, with $\omega^{I \cup J} = \omega^I \omega^J$; hence
$$\|ab\|_\omega \le \sum_{\substack{I,J \in \mathcal{R} \\ I \cap J = \varnothing,\ I \cup J \in \mathcal{R}}} \omega^I \omega^J |a_I||b_J| \le \left(\sum_I \omega^I |a_I| \right) \left(\sum_J \omega^J |b_J| \right)\,.$$

2. *Norm of the finite exponential.* Since $\|1\|_\omega = 1$, submultiplicativity and the triangle inequality give
$$\|\operatorname{Exp}(a)\|_\omega \le \sum_{r=0}^{|V|} \frac{\|a\|_\omega^r}{r!} \le e^{\|a\|_\omega}.$$

3. *Exponential identities.* If the constant coefficient vanishes, every factor of a uses a nonempty set of variables. A product of more than $|V|$ such factors is zero. Thus $a^{|V|+1} = b^{|V|+1} = 0$, and the binomial formula yields
$$\operatorname{Exp}(a+b) = \operatorname{Exp}(a)\operatorname{Exp}(b), \qquad \operatorname{Exp}(-a)\operatorname{Exp}(a) = 1.$$

□

The same norm justifies countable summation: if $\|a_j\|_\omega \le c_j$ and $\sum_j c_j < \infty$, the partial sums are Cauchy and $\|\sum_j a_j\|_\omega \le \sum_j c_j$. Completeness follows from finite dimensionality. For one variable, let ρ be a scalar weight and I an observable, viewed as functions of the link configuration. Multiplication by $1+\zeta I$ records the term containing one copy of I while leaving the weight attached. For a finite index set $\mathcal{I}$, let K_j be scalar factors and H_j the factors with an insertion, for $j \in \mathcal{I}$. The insertion rule is then

$$[\zeta]\,\rho(1+\zeta I) = \rho I, \qquad [\zeta]\prod_{j\in\mathcal{I}}(K_j+\zeta H_j) = \sum_{j\in\mathcal{I}} H_j \prod_{i\in\mathcal{I}\setminus\{j\}} K_i\,. \tag{7.20}$$

Here $[\zeta]$ means the coefficient of ζ. For any measure μ for which these functions are integrable, linearity of integration gives the corresponding identity

$$\int [\zeta]\rho(1+\zeta I)\,d\mu = \int \rho I\,d\mu = [\zeta]\int \rho(1+\zeta I)\,d\mu.$$

For the finite product in (7.20), the same fact is the displayed identity

$$\int [\zeta]\prod_{j\in\mathcal{I}}(K_j+\zeta H_j)\,d\mu = \sum_{j\in\mathcal{I}}\int H_j \prod_{i\in\mathcal{I}\setminus\{j\}} K_i\,d\mu = [\zeta]\int \prod_{j\in\mathcal{I}}(K_j+\zeta H_j)\,d\mu,$$

provided the terms are integrable. Thus the coefficient of an insertion in an expectation is obtained by integrating the product of that observable and its weight.

7.4 Local interactions

When links at one lattice spacing are integrated out, the resulting function of the remaining links is the effective weight. We seek to write it as a sum of scalar terms, each assigned to a finite set of sites; different indices may refer to the same set. For a region X, the good-set condition $\mathcal{G}_t(X)$ from (7.8) means that every plaquette touching X has $\chi(U_p) \le t$. Activities controlled by derivatives on this good set are the regular terms. Terms supported on configurations with large plaquette distances retain the sites where $\chi(U_p) > t$; the formula below calls these bad sites.

Use the lattice of Definition 2.2, the threshold $0 < t \le 1$, the derivative order $k \ge 6$ and distance exponent $q \ge 2$ from (7.10). The support weight counts the balls in a smallest connected cover: each ball contributes a factor e^α, so a larger or more separated region receives a larger weight.

Definition 7.4 (Connected ball covers)**.** Fix an integer radius $B \ge 1$ and $\alpha > 0$. A ball of radius B about a is $\mathbb{B}_{\ell,B}(a) = \{x \in \Lambda_\ell : d_\ell(x,a) \le B\}$. A set of sites is connected if any two of its sites can be joined within the set by a finite sequence of steps of maximum distance at most one. For $X \subseteq \Lambda_\ell$, define

$$\mathfrak{c}_B(X) = \min_{\substack{D\subseteq\Lambda_\ell \\ X\subseteq\bigcup_{a\in D}\mathbb{B}_{\ell,B}(a) \\ \bigcup_{a\in D}\mathbb{B}_{\ell,B}(a)\ \text{connected}}} |D|\,. \tag{7.21}$$

Thus $\mathfrak{c}_B(\varnothing) = 0$; the support weight is $e^{\alpha\mathfrak{c}_B(X)}$.

The minimum exists: taking every site as a centre covers the torus, which is connected by coordinate steps.

We call the activities controlled by derivatives on good configurations the regular terms. Fix finite sets of sites $X_i \subseteq \Lambda_\ell$, sites $o_i \in X_i$, and real Borel gauge-invariant functions E_i supported on the links inside X_i in Definition 2.6. They have the derivatives measured by (7.9) on the good set $\mathcal{G}_t(X_i)$ of (7.8). Their contribution is $\mathbf{1}_{\mathcal{G}_t(X_i)}E_i$. For the other terms, fix finite site sets $Y_j \subseteq \Lambda_\ell$. Choose a nonempty set $I_j \subseteq Y_j$ of sites that the term requires to be bad, and an optional extra condition $\mathcal{E}_j$ on configurations; if no extra condition is imposed, take $\mathcal{E}_j = \mathcal{U}_\ell$. Require $\mathcal{E}_j$ to be Borel and gauge invariant. For each j, choose a real Borel gauge-invariant function K_j that vanishes unless all sites of I_j and no other sites of Y_j are bad and $\mathcal{E}_j$ holds:

$$K_j(U) \neq 0 \implies Y_j \cap \mathcal{B}_t(U) = I_j \quad \text{and} \quad U \in \mathcal{E}_j\,. \tag{7.22}$$

Here the internal links of Y_j are those with both endpoints in Y_j, as in the link-support convention of Definition 2.6. Both K_j and $\mathcal{E}_j$ depend only on those links and the bad-site indicators in Y_j. The fixed sets and extra conditions are retained when an insertion changes the coefficients.

Evaluating the bad-site indicators of (7.8), either for a regular term or for the conditions (7.22), may require extra links. For a set of sites Y, write $\mathrm{vertices}(p)$ for the four corner sites of a plaquette p. All such links have endpoints in

$$\widehat{Y} = Y \cup \bigcup_{\mathrm{vertices}(p)\cap Y \neq \varnothing} \mathrm{vertices}(p)\,. \tag{7.23}$$

The two activity bounds count terms differently: regular terms are summed at their chosen site, and the other terms at every site in their support. Precisely, bounds $\delta, w \geq 0$ mean that there exist nonnegative majorants e_i, k_j (upper bounds for the individual activities) with

$$\begin{aligned} &\|E_i\|_{t,k,q;o_i,X_i} \leq e_i, \qquad \sup_U |K_j(U)| \leq k_j, \\ &\sum_{i:o_i=x} e^{\alpha \mathfrak{c}_B(X_i)} e_i \leq \delta, \qquad \sum_{j:x\in Y_j} e^{\alpha \mathfrak{c}_B(Y_j)} k_j \leq w \quad (x \in \Lambda_\ell)\,. \end{aligned} \tag{7.24}$$

The last two sums are convergent series of nonnegative terms. Indices range over $\mathbb{N}$ and zero functions may be added.

Let $c \in \mathbb{R}$ be a chosen constant per site; it contributes the overall factor $e^{c|\Lambda_\ell|}$ to the weight. Let S be the part of the action kept separate from the activities, assumed bounded, real, Borel and gauge invariant; write $\|S\|_\infty = \sup_U |S(U)|$. For a finite selection $\mathcal{I} \subset \mathbb{N}$ of indices, multiply the factors K_j. The sum $\mathcal{Z}_K$ retains only disjoint regions whose specified bad sites cover the entire measured bad set; E_t sums the regular contributions:

$$\begin{aligned} E_t(U) &= \sum_i \mathbf{1}_{\mathcal{G}_t(X_i)}(U) E_i(U), \\ \mathcal{Z}_K(U) &= \sum_{\substack{\mathcal{I}\subset\mathbb{N} \text{ finite} \\ Y_j\ (j\in\mathcal{I}) \text{ pairwise disjoint} \\ \bigcup_{j\in\mathcal{I}} I_j = \mathcal{B}_t(U)}} \prod_{j\in\mathcal{I}} K_j(U), \\ \rho(U) &= \exp\bigl(c|\Lambda_\ell| - S(U) - E_t(U)\bigr)\mathcal{Z}_K(U)\,. \end{aligned} \tag{7.25}$$

The empty selection contributes 1 precisely when $\mathcal{B}_t(U) = \varnothing$. The sum requires the regions Y_j to be disjoint, but their inspected sets $\widehat{Y}_j$ may overlap: those larger sets only record the extra links needed to decide which sites are bad. The function ρ is an unnormalised scalar factor intended to be integrated against product Haar; positivity and probability normalisation are not asserted here.

These formulas are useful only if their infinite sums converge absolutely. The per-site bounds give this convergence at each finite volume. To identify the represented weight with one obtained from the Wilson measure, a later step must:

1. integrate links at one spacing;
2. collect the resulting terms by finite site region; and
3. verify the prescribed values and Taylor terms of total degree two at the identity, together with the activity bounds.

Here the identity configuration is $U(x,\mu) = 1$ for every positive link. Nearby links are written $U(x,\mu) = \exp(X_{x,\mu})$ with $X_{x,\mu} \in \mathfrak{g}$ near zero, so the degree-two terms are the homogeneous Taylor terms of total degree two in these coordinates. This construction is not supplied here.

Proposition 7.5 (Convergence of the activity sums)**.** *Under* (7.22)–(7.24)*, both series in* (7.25) *converge absolutely and uniformly in* U*. They are Borel and gauge invariant, and*

$$\begin{aligned} |E_t(U)| &\le |\Lambda_\ell|\delta, & |\mathcal{Z}_K(U)| &\le e^{|\Lambda_\ell| w}, \\ |\rho(U)| &\le \exp\bigl(c|\Lambda_\ell| + \|S\|_\infty + |\Lambda_\ell|(\delta + w)\bigr)\,. \end{aligned} \tag{7.26}$$

In particular ρ *is integrable against the product Haar measure* η_ℓ *of Definition 2.4. If* $\mathcal{B}_t(U) = \varnothing$*, then* $\mathcal{Z}_K(U) = 1$*.*

Proof. At fixed finite volume, the per-site hypotheses give summable bounds for individual terms; the sum over compatible selections is then bounded by an unrestricted product. The argument has three steps.

1. *Sum individual activities.* Since each regular term has one chosen site and every Y_j is nonempty, the weights in (7.24) are at least one and give

$$\sum_i e_i \le |\Lambda_\ell|\delta, \qquad \sum_j k_j \le |\Lambda_\ell| w\,. \tag{7.27}$$

   The regular term $\mathbf{1}_{\mathcal{G}_t(X_i)} E_i$ is bounded by e_i everywhere, so its series converges uniformly and absolutely.

2. *Sum the selections.* The disjoint-region sum is dominated by all finite selections, whose configuration-independent majorant satisfies

$$\sum_{\mathcal{I}\subset\mathbb{N}\ \text{finite}} \prod_{j\in\mathcal{I}} k_j = \prod_{j\ge 0}(1 + k_j) \le \exp\left(\sum_j k_j\right)\,. \tag{7.28}$$

   The equality follows by expanding finite products and taking increasing limits; the inequality uses $1 + s \le e^s$ for $s \ge 0$. This proves absolute uniform convergence and the first two bounds in (7.26).

3. *Pass the properties to the sum.* All indicators in (7.25) are Borel and gauge invariant, as are the activities, so the limits have these properties. The exponent in ρ is bounded using $|S(U)| \le \|S\|_\infty$ and $|E_t(U)| \le |\Lambda_\ell|\delta$; thus

$$|\rho(U)| \le e^{c|\Lambda_\ell| + \|S\|_\infty + |\Lambda_\ell|(\delta + w)}.$$

   The product of the normalised Haar measures in Definition 2.4, equation (2.4), is a probability measure, so this bound gives integrability. If there are no bad sites, a nonempty selection cannot have empty union of its nonempty I_j; only the empty selection remains.

□

The estimates here allow the local bounds δ, w to be independent of volume, while their global sums grow with $|\Lambda_\ell|$. They justify summation and integration of supplied activities.

8 Infinite volume

At finite spacing and volume, an insertion is a function of finitely many link matrices, and a correlation is the expectation of a product of insertions. This section lets the physical box grow and constructs limits of those expectations on $\mathbb{R}^4$, for the single Wilson family chosen in Proposition 5.1. For example, an insertion measuring one plaquette is copied to many reflected cells; positivity of a reflected pair bounds the expectation of that repeated product. This gives bounds uniform in the box size.

The argument has five steps. Subsection 8.1 proves the reflection bounds. Subsection 8.2 turns those bounds into control of correlations under small changes of insertion points, then extracts limits. Subsection 8.3 identifies coefficients that no separated correlation can detect and represents derivatives by fixed coefficients. Subsection 8.4 makes the resulting correlations smooth and proves their symmetries and positivity. Subsection 8.5 returns from an auxiliary spacing subsequence to the full fixed-volume limit. The later construction uses this same limiting table. Three ultraviolet inputs remain separate: the centred root bound (8.6), the polynomial comparisons in Subsection 8.3, and full-spacing fixed-volume convergence in Subsection 8.5. Rotation and phase arguments do not supply them.

8.1 Lattice estimates

The polynomial coefficient $v \in V_d$ of Definition 6.4 specifies the local quantity measured at a lattice point; for example, the constant polynomial belongs to every degree space, while curvature products require higher degrees. The pair (d, v) is the observable index of Definition 2.6, and its finite-lattice representative is the insertion (6.23). We retain d because the same polynomial can be included in several degree spaces, while the degree determines which estimates apply.

Definition 8.1 (Coefficient conventions)**.** For the Wilson measures of Definitions 2.7 and 2.4, write

$$\langle F\rangle_{N,p} = \int F \, d\mu_{N,p}, \qquad \langle F\rangle_{\beta} = \int F \, d\mu_{\ell,\beta} \tag{8.1}$$

when the current torus has ℓ sites per side. Choose norms on V_d using the fixed bases of Lemma 6.6, and equip linear maps with the induced operator norms. For the signed permutation group H_4 of Definition 6.2, write

$$\rho_d(h)v = h \cdot v, \qquad u_d = a^d_{r(d)}1, \qquad v \in V_d,\ h \in H_4.$$

The action is (6.12); $r(d)$ is the reference level of Definition 6.11, and u_d is the physical unit polynomial of Proposition 6.12. A list $v = (v_1, \dots, v_n)$ includes specified degrees d_i with $v_i \in V_{d_i}$. We suppress the pairs (d_i, v_i) in insertion and moment notation. For a coefficient or field list, $v_{i\to w}$ means replacement of its ith entry by w. The symbol $\partial_{i,\mu}$ means differentiation in coordinate μ of the ith physical insertion point.

On physical coordinate spaces, $|\cdot|$ denotes the Euclidean norm and $|\cdot|_\infty$ the maximum of the absolute coordinate values. The sites and links are those of Definition 2.2, with physical positions as in Definition 2.9.

The spacing sequence is built in blocks: one step of length a_j contains ℓ_j steps of length a_{j+1}. The odd integer ℓ_j is the block factor. Its oddness puts the boundary of a centred coarse cell halfway between fine-lattice sites, making it a bond reflection plane. The following conditions are additional requirements on the particular family constructed in Proposition 5.1; they do not follow from Definition 2.8:

$$a_0 = 1, \qquad a_{j+1} = \frac{a_j}{\ell_j}, \qquad \ell_j \in 2\mathbb{N}+1, \quad \ell_j \geq 3, \qquad L_p \in 4\mathbb{N}, \qquad \beta_{N,p} \geq 0\,. \tag{8.2}$$

The ratio of successive scales is exactly the number of finer lattice steps in a coarse step. We also require these block factors to grow slowly relative to the inverse spacing:

$$\frac{\log \ell_j}{\log a_j^{-1}} \longrightarrow 0 \qquad (j \longrightarrow \infty)\,. \tag{8.3}$$

The quotient at $j = 0$ plays no role. The reference level $r(d)$ is the scale at which degree d is normalised (Definition 6.11). Beyond that scale, Proposition 6.12 supplies translation and conjugation identities, invariance under signed coordinate permutations, and a fixed unit insertion. The slow-growth condition (8.3) is used later to turn bounds at discrete cell sizes into a power bound at every physical separation.

Definition 8.2 (Lattice reflections and cells)**.** For a coordinate unit vector e_μ, reflection in the physical plane $x^\mu = c$ is

$$r_{\mu,c}x = x + 2(c - x^\mu)e_\mu, \qquad \begin{cases} c \in a_N\mathbb{Z}, & \text{site reflection,} \\ c \in a_N\big(\mathbb{Z} + \frac{1}{2}\big), & \text{bond reflection.} \end{cases} \tag{8.4}$$

The site plane passes through sites; the bond plane bisects the links between neighbouring site layers. On the torus, coordinates are reduced modulo L_p. For a link from x to $x + a_N e_\nu$, the pullback r^*U uses the reflected endpoints and the coordinate action of Definition 6.2. A reflected link may point opposite to its original orientation; since a link matrix is assigned to an oriented edge, reversing that edge replaces U by U^{-1}. Set $\Theta_r F = \overline{F \circ r^*}$.

A cell tiling $\mathcal{C}$ divides the torus into congruent cubes, with the same even number along each axis and one common reflection type. Starting with a base cube, successive reflections copy its observable into every cell. An observable is interior if it is bounded, Borel measurable, gauge invariant, and depends only on links with both endpoints strictly inside the cell or half-torus. Choose one cell as the base cell, where the observable is initially given. Write $\Theta_C F$ for the reflected copy in cell C, conjugated after an odd number of reflections. The resulting expectation has one copy in each cell; its geometric mean is the chessboard root:

$$\|F\|_{\mathcal{C}} = \left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F \right\rangle_{N,p} \right|^{\frac{1}{|\mathcal{C}|}}. \tag{8.5}$$

The centred tiling $\mathcal{C}_j(y)$ has cell side a_j and base cell centred at the physical point $\widehat{y}$ of Definition 2.9:

$$\big(\mathcal{C}_j(y)\big)_0 = \widehat{y} + \Big[-\frac{a_j}{2}, \frac{a_j}{2}\Big]^4 \pmod{L_p}.$$

Its odd side ratio a_j/a_N places the boundaries on bond planes for $N \geq j$.

The root measures the cost of repeating one observation throughout the box. The ultraviolet estimate bounds its growth as the cell shrinks, uniformly in volume. "Transformed" coefficients are the signed-coordinate and conjugate images in Proposition 6.12; enlarging R bounds this finite list. For every $d \in \mathbb{N}$ and $R, \varepsilon > 0$, there exists $A \geq 0$ such that, for every $j \geq r(d)$, eventually in N, uniformly in p, y and $\|v\|_{V_d} \leq R$, the insertion is interior and

$$\|I_{N,p}(v, y)\|_{\mathcal{C}_j(y)} \leq A a_j^{-d-\varepsilon}. \tag{8.6}$$

The cutoff threshold may depend on j. [2]

Unresolved. For the family chosen in Proposition 5.1, prove the scale relations (8.2), slow growth (8.3), the repeated-cell estimate (8.6), and the insertion identities of Proposition 6.12. All four inputs are required for this same family; they do not follow from regularity in Definition 2.8.[3]

Definition 8.3 (Ordered tails)**.** For chosen spacing and volume refinements N_j, p_k, an assertion $B(N, p)$ holds eventually on ordered tails if

$$\exists k_0\ \forall k \geq k_0\ \exists j_0(k)\ \forall j \geq j_0(k), \qquad B(N_j, p_k). \tag{8.7}$$

A coefficient or error tends to zero on these tails if its norm is eventually less than every positive tolerance in this order. Before a refinement is selected, use $N_j = j$ and $p_k = k$. A refining ultrafilter is a maximal proper filter on index pairs containing every such tail. It is a compactness device: bounded vectors in finite dimensions have one limit along it, and the tail condition preserves the order "volume first, then spacing." It does not change the numerical limits specified by the tails.

Thus the spacing threshold may depend on volume, in the order specified by Definition 2.12.

Reflection positivity pairs an observable with its reflected complex conjugate and gives a nonnegative expectation. The positive-kernel formula below is the finite-matrix version of this fact; it will also be used to control the movement of insertion points.

Definition 8.4 (Positive kernels)**.** A complex kernel K on a set E is positive if, for every finite list $X_i \in E$ and $c_i \in \mathbb{C}$,

$$\sum_{i,j} \bar{c}_i c_j K(X_i, X_j) \geq 0.$$

For a measured configuration space (E, μ), the same kernel acts on functions by $(Kf)(X) = \int_E K(X, Y) f(Y)\, d\mu(Y)$ whenever the integral exists. Its matrix positivity makes the corresponding integral quadratic form nonnegative, first for finite sums and then by approximation.

Lemma 8.5 (Reflection estimates)**.** *For the Wilson measure of Definition 2.4 on an even torus, $\beta \geq 0$, and bounded Borel gauge-invariant observables depending only on links whose endpoints lie strictly inside the indicated half-torus or cells,*

$$\langle \Theta_r F\, F \rangle_\beta \geq 0 \qquad \textit{(r a site or bond reflection)}, \tag{8.8}$$

$$\left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F_C \right\rangle_\beta \right| \leq \prod_{C \in \mathcal{C}} \|F_C\|_{\mathcal{C}}, \tag{8.9}$$

$$\|F + G\|_{\mathcal{C}} \leq \|F\|_{\mathcal{C}} + \|G\|_{\mathcal{C}}.$$

[2] Formal correspondence at revision `3456e5e`: `Extraction/Center.lean`: the uniform centred estimate and its projection to each extraction.

[3] Formal correspondence at revision `3456e5e`: `MergeProof/Gardens.lean` and `Source/Construction.lean`.

Proof. The proof first establishes positivity across one reflection plane, then repeats reflections to bound products over all cells, and finally removes a small regularising term.

Step 1. For bond reflection, the factors crossing the plane form a positive matrix kernel on boundary-slice configurations; its positive quadratic form is derived below from the trace inner product. For site reflection, the two halves separate directly. Integrating their weights against this kernel gives $\langle \Theta_r F\, F\rangle_\beta \geq 0$.

Step 2. Fix a finite list of observables. An assignment $i = (i_C)$ chooses one list member for each cell; let $\mathcal{Z}(i)$ be the absolute reflected expectation of that choice. A fold copies either half of the assignment across the reflection plane, giving choices i^+, i^-. Reflected Cauchy–Schwarz gives $\mathcal{Z}(i)^2 \leq \mathcal{Z}(i^+)\mathcal{Z}(i^-)$. The maximisation argument below shows that a largest value is attained when every cell has the same observable.

Step 3. Divide each cell observable by its chessboard root plus η. A constant assignment then has absolute expectation at most one; the maximisation from Step 2 bounds every assignment by one. Let $\eta \downarrow 0$.

Positivity. For the reflection plane under consideration, call an unoriented link $\{x, x + a_N e_\mu\}$ a crossing link when its endpoints lie on opposite sides of the plane. Such a link joins the two halves of the torus, so it is the only kind of link whose matrix is shared across the reflection plane. For bond reflection the crossing-link interaction can be written as a kernel of the two boundary-slice configurations; for site reflection no such link crosses the plane. For matrices $U, V \in G$, the real inner product $\operatorname{Re}\operatorname{tr}(UV^*)$ is a positive kernel on G in Definition 8.4: on a finite list, its matrix is the Gram matrix of those matrices for the real trace inner product. A plaquette weight is the exponential of a sum of these trace pairings. Entrywise products of positive matrices and the exponential series preserve positivity, so the Wilson interaction with crossing links set to the identity gives a continuous positive kernel $K_0(X, Y)$ of the two boundary-slice configurations X, Y. The factors belonging to links and plaquettes on the boundary slice are divided equally between the two halves; multiplying the two half-weights restores the original Wilson density. A vertexwise gauge transformation of boundary data is $g \cdot X$ (Definition 2.5). Independent Haar averages on the two copies restore the crossing-link interaction by replacing each point mass with its orbit distribution:

$$K(X, Y) = \iint K_0(g \cdot X, h \cdot Y)\, dg\, dh.$$

For positivity, let μ_X be the Haar distribution of $g \cdot X$ and put $\nu = \sum_i c_i \mu_{X_i}$. Then

$$\sum_{i,j} \bar{c}_i c_j K(X_i, X_j) = \iint K_0(X, Y)\, d\bar{\nu}(X)\, d\nu(Y) \geq 0. \tag{8.10}$$

Approximate the finite measures by finite sums and apply positivity of K_0. The integrand itself need not be nonnegative for independent g, h.

Split the variables into boundary links u and links inside the two open halves, v, w. Their first and last slices give X_-, X_+; the remaining links belong to one half. The boundary kernel after integrating crossing links is

$$P_u(v, w) = \begin{cases} 1, & \text{site reflection,} \\ K\big(X_-(w), X_-(v)\big) K\big(X_+(v), X_+(w)\big), & \text{bond reflection,} \end{cases}$$

where X_-, X_+ read the two boundary slices of a half-torus. It is positive: in the bond case it is a product of two positive kernels, one with exchanged arguments. Let Z be the Wilson normalising constant of (2.4), and $A_u(v)$ the square root of the weight of links and plaquettes in one open half,

including half of every boundary factor. The two square roots restore each full factor. Complete u, v arbitrarily to a full configuration $U_{u,v}$ and set $f_u(v) = A_u(v)F(U_{u,v})$. Interior support makes this independent of unused links. With normalised product Haar measures on these data, the density factors into the two half-weights and the boundary kernel. Integrating the halves first gives

$$\langle \Theta_r F\, F\rangle_\beta = \frac{1}{Z}\int \left[\iint P_u(v,w) f_u(v)\overline{f_u(w)}\, dv\, dw\right] du \geq 0. \tag{8.11}$$

All functions are measurable and bounded on compact configuration spaces, so the changes of integration order are valid.[4]

Repeated reflections. Choose the finite list of base-cell observables $F_1, \dots, F_s$ from the proof outline. With the assignment notation introduced there, write

$$\mathcal{Z}(i) = \left| \left\langle \prod_{C\in\mathcal{C}} \Theta_C F_{i_C} \right\rangle_\beta \right|.$$

A fold keeps the assignment on one half-torus and copies it onto the other half by reflection. Let i^+, i^- be the two choices. Cauchy–Schwarz for the reflected form gives

$$\mathcal{Z}(i)^2 \leq \mathcal{Z}(i^+)\mathcal{Z}(i^-). \tag{8.12}$$

Thus a maximising assignment stays maximising under either fold:

$$M = \max_i \mathcal{Z}(i), \quad \mathcal{Z}(i) = M \quad \Longrightarrow \quad M^2 \leq \mathcal{Z}(i^+)\mathcal{Z}(i^-) \leq M^2.$$

If $M = 0$, every assignment is maximising; otherwise both factors equal M.

It remains to show that some assignment attaining the largest value is constant, meaning that the same observable occupies every cell. Fix a coordinate with $2q$ cells. A band of width w is a consecutive group of w cells in the coordinate circle; it is constant there if one observable occupies every cell in the band. Every assignment has such a band of width one. Folding at the first boundary extends the band by copying its first value to the preceding cell; once it has width q, reflecting the half-circle fills the whole coordinate circle. With t denoting the band's initial cell, the two useful steps are

$$(t, w) \longmapsto (t-1, w+1) \quad (1 \leq w < q), \qquad (t, q) \longmapsto \text{the whole coordinate circle.}$$

The second step reflects a constant half-circle onto its complement. These folds preserve constancy in coordinates already treated. Doing this in all four coordinates gives a constant maximising assignment.

Normalisation. For the observables in (8.9), set

$$\widetilde{F}_C = \frac{F_C}{\|F_C\|_{\mathcal{C}} + \eta}, \qquad \eta > 0.$$

Every constant assignment from this finite list has expectation of absolute value at most 1, because its root is exactly the denominator in (8.5). The maximising argument therefore gives

$$\left| \left\langle \prod_C \Theta_C \widetilde{F}_C \right\rangle_\beta \right| \leq 1, \qquad \left| \left\langle \prod_C \Theta_C F_C \right\rangle_\beta \right| \leq \prod_C \big(\|F_C\|_{\mathcal{C}} + \eta\big).$$

[4] Formal correspondence at revision `3456e5e`: `YM/Slice/Positive.lean`, `averaged` and `step_positive`; `YM/Slice/Gate.lean`, `wall`, `links` and `portPos`; `YM/Slice/Reflection.lean`, `readers` and `link_positive`: the exact boundary form.

Letting $\eta \downarrow 0$ proves (8.9). Finally, expand the reflected product of $F + G$ and apply that estimate term by term:

$$\|F + G\|_{\mathcal{C}}^{|\mathcal{C}|} \le \big(\|F\|_{\mathcal{C}} + \|G\|_{\mathcal{C}}\big)^{|\mathcal{C}|}.$$

Taking the root proves subadditivity.[5] □

8.2 Bounds and compactness

The next three estimates turn point separation into compactness.

Step 1. Choose cell sizes with $a_m < r$ and $6na_k < a_m$; the centred root bound then gives a uniform bound when the insertion points are pairwise at least r apart.

Step 2. In one coordinate direction, the insertions occupy slabs and the empty slices between successive slabs are gaps; let Y_i count the slices in gap i. Moving a slab by h steps changes two adjacent counts from (Y_i, Y_j) to $(Y_i + h, Y_j - h)$. The transfer kernel between slice configurations has nonnegative quadratic form; this positivity bounds the change in correlation by a constant times the physical displacement.

Step 3. Local boundedness and this movement estimate give uniform Cauchy convergence on compact sets; a diagonal choice over the countable polynomial bases and compact exhaustions gives one subsequence for all correlations.

Definition 8.6 (Coordinate separation)**.** For the configuration space of Definition 2.10, put

$$\mathrm{Gen}_n = \{x \in \mathrm{Conf}_n : x_i^\mu \neq x_j^\mu \text{ for } i \neq j,\ \mu = 1, \dots, 4\}. \tag{8.13}$$

At fixed period L_p, let $\mathrm{Gen}_n^{(p)}$ be the same set with inequalities taken modulo $L_p\mathbb{Z}$. These are the coordinate-separated configurations, in ordinary space and on the periodic lattice respectively.

This temporary restriction leaves an empty coordinate slice between each pair of points in every coordinate direction, so each point can be moved in any coordinate without meeting another insertion. The sets are open and their complements have measure zero; the periodic set lies in Definition 2.13. Smoothness in Subsection 8.4 restores configurations with equal coordinate projections.

To compare different spacings on a common domain, extend each lattice correlation to a step function on physical space.

Definition 8.7 (Lattice extensions)**.** With $\ell_{N,p} = \frac{L_p}{a_N}$ as in (2.7), round each coordinate up to the next lattice point and reduce it modulo the period:

$$y_{N,p}(x) = \big[\lceil a_N^{-1} x \rceil\big] \in \Lambda^n_{\ell_{N,p}}, \qquad M^{\mathrm{step}}_{N,p}(v; x) = M_{N,p}(v; y_{N,p}(x)). \tag{8.14}$$

Brackets denote the periodic class, the ceiling acts componentwise, and $M_{N,p}$ is the moment of Definition 2.9. The ceiling assigns each point to the upper corner of its lattice cube, fixing one representative and moving each scalar coordinate by less than one lattice spacing; hence the product configuration moves by at most $4na_N$ in Euclidean norm, modulo L_p.

Proposition 8.8 (Auxiliary extraction)**.** *Assume* (8.2)*,* (8.3) *and* (8.6) *and the insertion identities of Proposition 6.12. Every pair of strictly increasing spacing and volume sequences has refinements N_j, p_k and multilinear kernels P_p, S_0. Here P_p is the fixed-volume limit after refining the spacing, and S_0 is the subsequent large-volume limit. The same choices work for every insertion list.*

[5] Formal correspondence at revision `3456e5e`: `Cells/Flood.lean`: extension of a band by one cell and successive coordinates; `Cells/Chess.lean`: maximisation, normalisation and subadditivity.

(i) *The moment functions converge uniformly on each compact subset of the sets in Definition 8.6,*

$$\begin{aligned} M^{\mathrm{step}}_{N_j,p}(v;\cdot) &\longrightarrow P_p(v;\cdot) \in C(\mathrm{Gen}_n^{(p)}) \quad (p \in \mathbb{N}), \\ P_{p_k}(v;\cdot) &\longrightarrow S_0(v;\cdot) \in C(\mathrm{Gen}_n). \end{aligned} \tag{8.15}$$

(ii) *For every separated test f of Definition 2.10, its support eventually lies inside volume p_k (Definition 2.11). The product with the limiting kernel is integrable, and the smeared moments have two limits; $R_p(v;f)$ denotes the fixed-volume value:*

$$\begin{aligned} &\lim_{j\to\infty} C_{N_j,p}(v;f) = R_p(v;f) \qquad\qquad (f \text{ supported inside volume } p), \\ &\lim_{k\to\infty} R_{p_k}(v;f) = \int f(x) S_0(v;x)\,dx. \end{aligned} \tag{8.16}$$

Proof. We now prove these estimates in order.

Step 1. The centred roots give local bounds on separated configurations; the cell choices satisfy $a_m < r$ and $6na_k < a_m$.

Step 2. Slice positivity controls movements by $|M^{\mathrm{step}}_{N,p}(v;x) - M^{\mathrm{step}}_{N,p}(v;y)| \le C_K|x-y| + o(1)$.

Step 3. A finite-net argument gives locally uniform limits, first at each volume and then along one common volume refinement.

Step 4. Dominated convergence of the step-function smearings gives the two ordered limits for every separated test.

Local bounds. The centred estimate (8.6) controls an observable repeated in cells centred on its insertion site. It also controls repetitions in translated cells, provided the insertion stays away from their boundaries. At aligned smaller-cell centres, apply (8.9) and put 1 in the unused cells. Between these centres, let $R(z)$ be the chessboard root as the insertion moves by z lattice steps along one axis. A fold gives the discrete convexity inequality

$$R(z)^2 \le R(z-1)R(z+1) \quad\Longrightarrow\quad R(z) \le \max\{R(z_-), R(z_+)\} \qquad (z_- \le z \le z_+).$$

Here z_-, z_+ are consecutive aligned centres; the inequality says an intermediate placement cannot have a larger root than both endpoint placements. Applying this in each direction bounds every translated root by the largest centred root of the reflected coefficients. It suffices that the insertion site is at least $3a_k$ from each boundary and its supporting links lie within distance a_k of that site.[6]

Now suppose n insertion sites have pairwise physical torus sup-distance at least $r > 0$. Choose a cell size $a_m < r$ to put different sites in different cells, and then a smaller size a_k for the boundary margins. For each insertion, translations that put its support too close to a cell boundary occupy at most three strips of width $2a_k$ in one period a_m. Across n insertions their total length is at most $6na_k$. Consequently

$$m \le k, \quad a_m < r, \quad 6na_k < a_m \quad\Longrightarrow\quad \left| \left\langle \prod_{i=1}^{n} I_{N,p}(v_i, y_i) \right\rangle_{N,p} \right| \le A^n \tag{8.17}$$

for sufficiently large N, where A bounds the centred roots at level k for the finitely many reflected coefficients. Since $6na_k < a_m$, some translation avoids all forbidden intervals. Use it to put the insertions strictly inside different cells, then apply the translated-root bound and (8.9). The finite

[6] Formal correspondence at revision `3456e5e`: `YM/Extraction/Around.lean`: translated cells; `YM/Cells/Slide.lean`: discrete convexity.

supports fit inside their cells once N is sufficiently large. Thus fixed separated compact sets have local bounds, uniform in volume on the ordered tails of Definition 8.3.

Moving an insertion. For at least two insertions, group links into slices perpendicular to an axis. The positive transfer kernel K acts across each gap, and the trace identifies the first and last slice around the torus. If the torus has M slices and each slab has w slices, the Wilson factorisation gives

$$\langle \prod_i I_i \rangle = \frac{\operatorname{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})}{Z}, \qquad Z = \operatorname{Tr} K^M > 0, \qquad nw + \sum_i Y_i = M.$$

Here $I_i = I_{N,p}(v_i, y_i)$, and the trace is the cyclic integral over slice configurations. The slice kernel and inserted kernels are continuous, since the original insertions are continuous by Reg2. The adjoint kernel is $B_i^*(X, Y) = \overline{B_i(Y, X)}$. Let A_i bound the chessboard root of the ith insertion. For $M = 2qw$, place alternating reflected copies of the insertion and its adjoint in the cells. This repeated configuration is the trace of $(B_i^* B_i)^q$, so (8.9) gives

$$\left(\frac{\operatorname{Tr}\big((B_i^* B_i)^q\big)}{Z} \right)^{\frac{1}{2q}} \le A_i,$$

The period condition in (8.2) makes q an integer when the slabs have twice a centred cell's width.

Moving one slab by h steps changes the two adjacent distances by $+h$ and $-h$. Approximate the cyclic integrals by finite weighted sums with nonnegative weights. For the resulting positive matrix K, define $K^z = \exp(z \log K)$ on its positive eigenspaces and set it equal to zero on its kernel. Its matrix entries are entire in z, and it agrees with the ordinary powers at positive integer exponents. For the two adjacent interval indices i, j, set

$$T(z) = \operatorname{Tr}\big(B_1 K^{Y_1(z)} \cdots B_n K^{Y_n(z)}\big), \qquad Y_l(z) = \begin{cases} Y_i + z, & l = i, \\ Y_j - z, & l = j, \\ Y_l, & l \ne i, j. \end{cases}$$

For a finite matrix, set $|B| = (B^* B)^{\frac{1}{2}}$ and $\|B\|_s = (\operatorname{Tr} |B|^s)^{\frac{1}{s}}$; this is the ℓ^s norm of its singular values. Choose $b > 0$ with all $Y_l > 2b$. For $|z| \le 2b$, imaginary powers have operator norm at most one. Hölder's inequality for singular values (the Schatten norm inequality) gives

$$|T(z)| \le \prod_l \|B_l\|_{2q} \, Z^{1 - \frac{n}{2q}}, \qquad \frac{n}{2q} + \frac{\sum_l \operatorname{Re} Y_l(z)}{M} = 1.$$

The analytic function T is uniformly bounded on these discs; Cauchy's estimate therefore bounds its derivative along the real interval. For $0 \le h \le b$, this gives

$$\frac{|T(h) - T(0)|}{Z} \le \frac{h}{b} \prod_l \left(\frac{\operatorname{Tr} |B_l|^{2q}}{Z} \right)^{\frac{1}{2q}} \le \frac{h}{b} \prod_l A_l.$$

Pass to the limit in the finite weighted sums. The integer endpoint exponents stay positive, so the choice on the zero eigenspace changes neither endpoint.[7]

On a compact coordinate-separated set, choose fixed slabs leaving a positive physical interval between successive insertions in each coordinate direction. In lattice units, each such interval

[7] Formal correspondence at revision `3456e5e`: `YM/Complex/Gaps.lean`: finite trace difference; `YM/Slice/Calendar.lean`: passage to kernels; `YM/Extraction/Difference.lean`: insertion movements.

contains at least b empty slices for sufficiently fine N. The integer b can then be chosen so that $a_N b$ stays bounded below by a positive physical length, independently of N. The preceding estimate costs at most a constant times the physical displacement $a_N h$. Move the $4n$ coordinates in turn and include the rounding error of Definition 8.7. On each compact neighbourhood K in $\mathrm{Gen}_n^{(p)}$, this gives, for sufficiently close $x, y \in K$,

$$|M_{N,p}^{\mathrm{step}}(v; x) - M_{N,p}^{\mathrm{step}}(v; y)| \le C_K |x - y| + o(1), \qquad N \longrightarrow \infty, \tag{8.18}$$

uniformly in x, y. For fixed $K \subset \mathrm{Gen}_n$, the same constants work independently of sufficiently large volumes on ordered tails. For one insertion use translation invariance; for no insertions the moment is 1.

Common subsequences. Fix a coefficient basis tuple and a compact set K. Local boundedness gives a subsequence converging at every point of a countable dense subset. Choose a finite δ-net E_δ from that subset. Then (8.18) gives

$$\sup_K |M_j - M_l| \le 2C_K \delta + o_j(1) + o_l(1) + \max_{z \in E_\delta} |M_j(z) - M_l(z)|.$$

Here M_j denotes the lattice extension along the selected subsequence; the movement estimate transfers convergence from the net to all of K. First let $j, l \to \infty$ and then $\delta \downarrow 0$. The sequence is uniformly Cauchy on K, and its limit is continuous. This is the compactness mechanism of Arzelà–Ascoli, even though the approximating step functions need not be continuous.

Take the diagonal over all fixed volumes, numbers of insertions, degree lists, basis tuples and compact exhaustions. There are countably many such choices, so a diagonal subsequence gives every P_p; the same argument gives one volume subsequence and S_0. Bounds and movement estimates pass to the limits, and multilinearity extends them from the bases. Set the limits to zero off their coordinate-separated domains; the omitted configurations form a measure-zero set.

Smeared limits. Write the weighted lattice sum as the integral of its step function. A compact neighbourhood of $\operatorname{supp} f$ stays separated and avoids periodic collisions once the volume contains it; local bounds dominate, and rounding changes f by $o(1)$. Dominated convergence first in N_j and then in p_k gives both limits in (8.16). Finally $L_{p_k} \to \infty$ ensures eventual containment of the support.[8] □

8.3 Equation relations

On coordinate-separated points, integration by parts represents each derivative by a higher-degree polynomial. The ultraviolet comparisons identify these representatives modulo coefficients invisible to all correlations; the independent rank bound controls their size.

We now state these additional ultraviolet inputs precisely. Along a further common refinement, preserving Proposition 8.8, assume linear projections $P_{d,N,p}, P_d : V_d \to V_d$ with

$$P_{d,N,p}^2 = P_{d,N,p}, \qquad P_{d,N,p} \longrightarrow P_d, \qquad P_d^2 = P_d \quad \text{in operator norm.} \tag{8.19}$$

All limits and eventual assertions in this subsection use the ordered tails of Definition 8.3. The inverse coefficient transfer $\tau_{N,p,d}^{-1}$ of (6.24) converts a coefficient list at the reference scale back to the fine-cutoff coordinates; its norm measures the size of that list at the fine cutoff. We allow families satisfying

$$\forall \varepsilon > 0, \qquad \|\tau_{N,p,d}^{-1} w_{N,p}\|_{V_d} \le a_N^{-\varepsilon} \quad \text{eventually.} \tag{8.20}$$

[8] Formal correspondence at revision `3456e5e`: `YM/Extraction/Lipschitz.lean`: common subsequences; `YM/Extraction/Integrate.lean`: dominated convergence; `YM/Extraction/Held.lean`: their combination.

The ultraviolet estimates must imply (8.20) for bounded coefficient families. They do so by bounding the growth at every intervening scale. The product of these bounds has logarithm $o(\log a_N^{-1})$, so it grows more slowly than every fixed power of the inverse spacing.

Definition 8.9 (Cell errors)**.** An error $E_{N,p}(y)$ is a lattice observable with its configuration argument suppressed. It is small in reflected cells if one rate $\eta_{N,p} \to 0$ bounds its repeated-cell root at a suitable scale for every fixed number of insertions. For every integer $q \geq 0$ and lower level m, there exists $k \geq m$ such that eventually, for every translated bond cell of side a_k and every insertion site with integer coordinates z measured from the cell's first site in each direction and satisfying

$$M \leq z^\mu, \quad z^\mu + M + 1 \leq \frac{a_k}{a_N}, \quad M = \left\lfloor \frac{a_k}{8(q+1)a_N} \right\rfloor \quad (\mu = 1, \dots, 4), \tag{8.21}$$

the error is a bounded Borel gauge-invariant observable depending only on links whose endpoints lie strictly inside that cell, as in Definition 8.2, and its chessboard root (8.5) is at most $\eta_{N,p}$. The integer coordinates z count lattice steps from the cell's first site; the inequalities keep the support inside the cell by margin M, as needed for the chessboard estimate. A degree-e representative is an insertion $I_{N,p}(w_{N,p}, y)$ with $w_{N,p} \in V_e$; it is a higher-degree polynomial that stands for the original insertion up to the small cell error.

At every degree we require the following three comparisons:

(C1) Each fixed degree-d insertion has a degree-e representative satisfying (8.20) for $d \leq e$, differing by an error small in reflected cells.

(C2) The forward difference $\frac{I_{N,p}(v,y+e_\mu) - I_{N,p}(v,y)}{a_N}$ has such a representative of degree $d+1$.

(C3) A family satisfying (8.20) in $\ker P_{d,N,p}$ differs by such an error from bounded measurable observables whose correlations with every fixed list of the other insertions vanish exactly, eventually at each positive separation.

Boundedness in C3 is only at each cutoff and site, which suffices for integration against the finite Wilson law.

Fix a list, a slot i, the other coefficients and a separated test f (Definition 2.10). Varying $w \in V_d$ in that slot gives linear functionals

$$L_{N,p}(w) = C_{N,p}(v_{i\to w}; f), \qquad L(w) = \int f(x) S_0(v_{i\to w}; x)\, dx, \qquad N_d = \bigcap_L \ker L. \tag{8.22}$$

Intersect over all lists, slots, other polynomials and tests. Thus N_d is exactly the set of coefficients undetected by every separated smear. In the following rank input and comparisons, eventual assertions retain Definition 8.3:

$$\dim_{\mathbb{C}} N_d \leq \dim_{\mathbb{C}} \ker P_d\,. \tag{8.23}$$

The projectors and this inequality are obtained on a further common refinement, retaining all previous limits.

Unresolved. For the chosen family, prove the three polynomial comparisons above, the coefficient-growth bound (8.20), and the finite-dimensional detector estimate (8.23). In particular, show that the dimension of the undetectable coefficient space is no greater than the dimension removed by the

projection. This dimension estimate is a separate input; it does not follow from the repeated-cell bound.[9]

The next lemma replaces cutoff-dependent polynomials by fixed representatives for inclusions and derivatives, valid in every correlation.

Lemma 8.10 (Fixed representatives)**.** *Assume the hypotheses of Proposition 8.8, the projections (8.19), the coefficient bound (8.20) for bounded families, the error condition (8.21), the rank bound (8.23), and comparisons C1–C3. Then $N_d = \ker P_d$, and there are complex-linear maps*

$$\kappa_{d,e} : V_d \longrightarrow V_e \quad (d \le e), \qquad \delta_{d,\mu} : V_d \longrightarrow V_{d+1}$$

such that, for every separated test of Definition 2.10 and functionals (8.22),

$$\begin{aligned} \int f S_0(v_{i\to\kappa_{d,e}v_i}) &= \int f S_0(v), \\ \int f S_0(v_{i\to\delta_{d,\mu}v_i}) &= -\int (\partial_{i,\mu} f) S_0(v). \end{aligned} \tag{8.24}$$

Proof. The proof has two steps.

Step 1. Identify the invisible directions using $\ker P_d \subset N_d$ and $\dim N_d \le \dim \ker P_d$, hence $N_d = \ker P_d$.

Step 2. Divide each projected representative by $1 + \|P_{e,N,p} w_{N,p}\|$; compactness gives a nonzero normalised limit, which can be rescaled to represent every target correlation at once.

An error small in reflected cells has vanishing separated smearings. Choose its cell below the fixed separation and a finer centred scale for the other insertions; the margin factor $8(q+1)$ avoids the boundary strips for any fixed number of slots. The translated-cell estimate and (8.9) bound the mixed moment by $C\eta_{N,p}$. The weighted count of sites meeting $\operatorname{supp} f$ is bounded by the volume of a fixed neighbourhood, independently of N, p. Thus its smear tends to zero.

Since V_d is finite dimensional, convergence of $L_{N,p}$ on a basis gives operator-norm convergence. For fixed v, the family $v - P_{d,N,p} v$ is in the projection kernel, so comparison C3 gives $L(v - P_d v) = 0$. The dimension input closes the identification:

$$\ker P_d \subset N_d, \qquad \dim_{\mathbb{C}} N_d \le \dim_{\mathbb{C}} \ker P_d \quad \Longrightarrow \quad N_d = \ker P_d.$$

Let $\Delta_{i,\mu}$ be the forward lattice difference in slot i and direction μ, and let $e_{i,\mu}$ be that coordinate vector in the point list. Summation by parts on the torus gives

$$a_N^{4n} \sum_y f(\widehat{y}) \Delta_{i,\mu} M_{N,p}(v; y) = a_N^{4n} \sum_y \frac{f(\widehat{y} - a_N e_{i,\mu}) - f(\widehat{y})}{a_N} M_{N,p}(v; y)\,.$$

Eventually the support stays inside the fundamental box. The backward quotient converges uniformly to $-\partial_{i,\mu} f$; local bounds and the bounded weighted site count make the error vanish. Comparisons C1–C2 therefore give coefficient families $w_{N,p}$ with the required target limits for every choice of the other insertions and test at once.

Write $p_{N,p} = P_{e,N,p} w_{N,p}$, where e is the representing degree. Comparison C3 applied to the normalised remainder gives

$$L_{N,p}\left(\frac{w_{N,p} - p_{N,p}}{1 + \|p_{N,p}\|_{V_e}} \right) \longrightarrow 0\,.$$

[9] Formal correspondence at revision `3456e5e`: `MergeProof/Powder.lean` and `Next/RANK/Beam.lean`: same-family assembly and retained rank comparison.

Division by $1 + \|p_{N,p}\|_{V_e} \geq 1$ preserves (8.20), and the normalised projected vector is bounded. Take a cluster limit (c, z) of

$$\left(\frac{1}{1 + \|p_{N,p}\|_{V_e}}, \frac{p_{N,p}}{1 + \|p_{N,p}\|_{V_e}} \right)$$

in $[0,1] \times \overline{B}_{V_e}(0,1)$, where $\overline{B}_{V_e}(0,1)$ is the closed unit ball. Use an ultrafilter containing every ordered tail of Definition 8.3; it selects one common cluster limit for all finite-dimensional coordinates while preserving the required order of volume and spacing limits. If t_L is the target correlation or weak derivative in (8.24), the limits satisfy $P_e z = z$, $c + \|z\|_{V_e} = 1$ and $Lz = c\, t_L$. A vanishing c would contradict the normalisation:

$$c = 0 \implies z \in \operatorname{im} P_e \cap \ker P_e = \{0\} \implies c + \|z\|_{V_e} = 0 \neq 1.$$

Thus $\frac{z}{c}$ represents every target simultaneously. Choose representatives on each source basis and extend linearly. No further subsequence has been chosen for the uncountable collection of tests.[10] □

8.4 Smooth fields and positivity

Here a weak derivative is a derivative identified after integration against every smooth test, as in (8.24). Repeated use of Lemma 8.10 gives weak derivatives of every order and hence smooth representatives, including configurations where two points share a coordinate. Two polynomial coefficients define the same continuum field precisely when no separated correlation can distinguish them. Quotienting by these invisible coefficients gives the field spaces; their degree maps and derivatives are independent of representatives and enter Definition 2.17.

Definition 8.11 (Continuum fields)**.** For the null subspace N_d in (8.22), set

$$F_d = V_d / N_d, \qquad \pi_d v = [v], \qquad \|[v]\|_{F_d} = \inf_{w \in N_d} \|v + w\|_{V_d}. \tag{8.25}$$

These are normed finite-dimensional spaces: the quotient norm is the smallest norm of any polynomial coefficient representing the same field. The class $\pi_d u_d$ is denoted by $\mathbf{1}$ at degree d.

Definition 8.12 (Analytic conventions)**.** For a function on $\mathbb{R}^a$, D_x^s denotes its sth total derivative, viewed as an s-linear map and given its operator norm. For a multi-index $\alpha \in \mathbb{N}^a$, put $|\alpha| = \sum_l \alpha_l$ and $D^\alpha = \partial_1^{\alpha_1} \cdots \partial_a^{\alpha_a}$. For $u \in L^1(\mathbb{R}^a)$, use the Fourier transform

$$\widehat{u}(\xi) = \int_{\mathbb{R}^a} e^{-i\xi \cdot x} u(x)\, dx.$$

In applications to n insertion points, $a = 4n$.

Proposition 8.13 (Smooth correlations)**.** *Under the hypotheses of Lemma 8.10, there is a collection $S = (S_n)_{n \geq 0}$ of multilinear kernels on the polynomial lists of Definition 8.1, with quotient spaces as in Definition 8.11, such that:*

(i) *$S_n(v; \cdot) \in C^\infty(\mathrm{Conf}_n)$ on (2.11), $S_n|_{\mathrm{Gen}_n} = S_0$ on (8.13), and (8.16) holds with S_n.*

(ii) *$\ker \pi_d = N_d = \ker P_d$ is the null space of S: a coefficient lies in N_d exactly when every separated-point correlation containing it vanishes.*

[10] Formal correspondence at revision `3456e5e`: `Work/D01.lean`–`D04.lean`: coefficient costs, errors and differences; `Dash/Limits.lean`: normalised compactness; `Dash/Inert.lean`: fixed representatives.

(iii) *The maps $\kappa_{d,e}, \delta_{d,\mu}$ of Lemma 8.10 induce linear maps $j_{d,e} : F_d \hookrightarrow F_e$ and $\partial_\mu : F_d \to F_{d+1}$, with $j_{e,f} j_{d,e} = j_{d,f}$ and $j_{d,d} = 1$, satisfying*

$$\begin{aligned} S_n(\dots, j_{d,e} A_i, \dots; x) &= S_n(\dots, A_i, \dots; x), \\ S_n(\dots, \partial_\mu A_i, \dots; x) &= \partial_{i,\mu} S_n(\dots, A_i, \dots; x). \end{aligned} \tag{8.26}$$

The notation $S_n(A; x)$ denotes the induced correlation of quotient fields, with $A_i = \pi_{d_i} v_i$.

Proof. The weak identities give integrable kernels for every derivative. The following steps use cutoffs and Fourier inversion to obtain smooth representatives and show that they agree on overlaps.

Step 1. The weak identities (8.24) represent each coordinate derivative by another locally integrable correlation.

Step 2. For every compactly supported smooth cutoff χ, $\sup_\xi (1 + |\xi|)^b |\widehat{\chi S_0(v)}(\xi)| < \infty$ for all b, so Fourier inversion gives a smooth local representative.

Step 3. These representatives agree on overlaps; their pointwise identities descend to the quotient fields.

Equation (8.24) represents every weak coordinate derivative by another locally integrable kernel. Iteration gives derivatives of every order. For a smooth cutoff function χ supported in Conf_n, meaning a smooth function with compact support there, each derivative of $\chi S_0(v)$ is a finite sum of integrable compactly supported kernels. Integration by parts in the Fourier transform gives, for every integer $b \geq 0$,

$$\sup_\xi (1 + |\xi|)^b \left| \widehat{\chi S_0(v)}(\xi) \right| < \infty \,.$$

Fourier inversion supplies a smooth representative. Choose $\chi = 1$ near an arbitrary distinct-point configuration. The local representatives agree almost everywhere on overlaps, hence everywhere there by continuity. They glue to S and equal S_0 on the set (8.13). The same argument preserves multilinearity and smearings.[11]

The null-space description follows both ways: pointwise vanishing gives zero smearings, while smoothness and tests in small neighbourhoods recover pointwise vanishing from (8.22). The identities (8.24) then hold pointwise. A null input has zero correlation with every list of other fields. Its inclusion therefore also has zero correlations, and its derivative has zero correlations by differentiating the zero function. Thus κ and δ preserve the null subspaces and descend to F_d. If an included field is zero, its correlations vanish; the inclusion identity in (8.24) then says the original field also has zero correlations. This proves injectivity of $j_{d,e}$; applying the identities twice proves its composition law and $j_{d,d} = 1$. □

The ordered limit preserves the lattice symmetries and reflection positivity. The slow-growth scale condition converts fixed-cell bounds into distance bounds needed for the later Hilbert-space construction.

Fix a coordinate plane through the origin. A positive-side configuration is a list of distinct insertion points lying on one side, here $x_i^\mu > 0$. For $r_\mu = r_{\mu,0}$ of Definition 8.2, reflect the first list across this plane and pair it with the second list: For two such lists of lengths n and m, define the reflected correlation

$$G_S\big((v, x), (w, y)\big) = S_{n+m}(\rho(r_\mu)\bar{v}, w; r_\mu x, y). \tag{8.27}$$

[11] Formal correspondence at revision `3456e5e`: `Dash/Regular.lean`, `Dash/Twist.lean`, and `Org/Input.lean`: smooth representatives with the retained extraction.

Reflection acts on every point and coefficient in the first list. Thus each matrix entry pairs an observation with the reflected complex conjugate of another observation. For joint separated tests f_α of Definition 2.10 supported on positive configurations, put

$$G_{\alpha\beta}(f) = \iint \overline{f_\alpha(x)} f_\beta(y) G_S\big((v^\alpha, x), (v^\beta, y)\big)\, dx\, dy. \tag{8.28}$$

These are the Osterwalder–Schrader forms: their quadratic form is the expectation of an observation paired with its reflected conjugate, and positivity is the input for the later Hilbert-space construction. For a matrix G, the notation $G \succeq 0$ means $\sum_{\alpha,\beta} \bar{c}_\alpha c_\beta G_{\alpha\beta} \geq 0$ for every complex vector c. Derivative and norm notation is that of Definition 8.12; m in the estimate below is a derivative order, independently of the gauge matrix size.

Proposition 8.14 (Euclidean properties)**.** *The kernels of Proposition 8.13 satisfy the following identities and bounds.*

(i) *For the coefficient action and units of Proposition 6.12, every permutation σ, $h \in H_4$, $b \in \mathbb{R}^4$ and distinct configuration x of (2.11),*

$$\begin{aligned} S_\varnothing &= 1, & S_{n+1}(u_d, v; x_0, x) &= S_n(v; x), \\ S_n(v \circ \sigma; x \circ \sigma) &= S_n(v; x), & S_n(\bar{v}; x) &= \overline{S_n(v; x)}, \\ S_n(\rho(h)v; hx + b) &= S_n(v; x). \end{aligned} \tag{8.29}$$

(ii) *For every coordinate direction and finite collection of positive configurations, the matrices in (8.27)–(8.28) satisfy*

$$\Big[G_S\big((v^\alpha, x^\alpha), (v^\beta, x^\beta)\big)\Big]_{\alpha,\beta} \succeq 0, \qquad [G_{\alpha\beta}(f)]_{\alpha,\beta} \succeq 0. \tag{8.30}$$

(iii) *For every $n, m \in \mathbb{N}$, degree list (d_i) and coefficient list $v_i \in V_{d_i}$, there are $C \geq 0$ and $b \in \mathbb{N}$ such that*

$$\|D_x^m S_n(v; x)\| \leq C r^{-b} \quad (0 < r \leq 1,\ \forall i \neq j,\ |x_i - x_j|_\infty \geq r)\,. \tag{8.31}$$

The identities descend through π_d to the quotient fields: π_d forgets exactly those polynomial coefficients whose separated correlations all vanish.

Proof. The three steps pass the exact symmetries, reflected positivity and collision bounds to the limit.

Step 1. Exact lattice identities pass to the limit at grid points; density and continuity extend them from coordinate-separated configurations to all of Conf_n.

Step 2. Lattice reflection positivity gives a positive semidefinite matrix for any finite list of joint positive-side tests. Each entry converges as a separated smear, so the limiting matrix is positive semidefinite; concentrating smooth tests gives point positivity.

Step 3. If $a_m < r/2$ and $6na_k < a_m$, the two first-crossing bounds give $a_k \geq c' r^4$. The centred estimate at scale a_k then yields a power bound in r, and derivative identities give the stated derivative bound.

For the first step, at grid configurations y and lattice translations b, the exact identities include

$$\begin{aligned} M_{N,p}(v \circ \sigma; y \circ \sigma) &= M_{N,p}(v; y), \quad M_{N,p}(\bar{v}; y) = \overline{M_{N,p}(v; y)}, \\ M_{N,p}(\rho(h)v; hy + b) &= M_{N,p}(v; y) \qquad\qquad (h \in H_4,\ b \in \mathbb{Z}^4). \end{aligned}$$

The unit and empty-list identities pass in the same way. Equation (8.15) passes these formulas to the limit on Gen_n; density and smoothness extend them throughout Conf_n and to every real translation.

For the second step, fix finitely many joint tests supported in positive-side separated configurations and let $\mathcal{O}_{N,\alpha}$ denote their lattice smearings. Each insertion depends on finitely many links, whose physical distances from its insertion site tend to zero. Thus, for large p and then large N, these links lie strictly inside the positive half-torus, and Lemma 8.5 gives

$$\left[\langle \Theta_{r_\mu} \mathcal{O}_{N,\alpha}\, \mathcal{O}_{N,\beta}\rangle_{N,p}\right]_{\alpha,\beta} \succeq 0.$$

The two halves of each entry are internally separated and lie on opposite sides of the plane, so it is a separated smear. Pass each entry through the ordered limit (8.16); absolute convergence follows from smoothness on the compact supports, and the positive-semidefinite cone is closed. Therefore (8.30) holds for joint tests. Choose smooth nonnegative tests of integral one supported in shrinking neighbourhoods of each fixed positive configuration. Continuity makes their smeared matrix converge entrywise to the pointwise matrix, and positivity is preserved in this limit.

For the position-independent bound, (8.3) implies $a_{j+1} \geq c a_j^2$ for some $c > 0$, after adjusting finitely many levels. At the first scale with $a_j < \delta$, the previous scale is at least δ; using $a_j \geq c a_{j-1}^2$ gives $a_j \geq c\delta^2$ (with the finite initial levels absorbed into c). For $n \geq 1$ and fixed v, set $D = 1 + \max_i d_i$. Choose the first sufficiently large m with $a_m < \frac{r}{2}$, then the first $k \geq m$ with $a_k < \frac{a_m}{6(n+1)}$. Applying this first-crossing estimate first at $r/2$ and then at a fixed fraction of a_m gives $a_k \geq c' r^4$. Since $a_k \geq c' r^4$, the centred estimate (8.6) with $\varepsilon = 1$ contributes at most a fixed power of r^{-1} in each of the n cells. The separated-cell argument therefore yields

$$|S(v;x)| \leq C r^{-4nD} \qquad (0 < r \leq 1,\ \forall i \neq j,\ |x_i - x_j|_\infty \geq r)\,.$$

Pass first at coordinate-separated points, then use density while retaining separation $\frac{r}{2}$. Repeated derivative identities apply this bound to finitely many lists of higher-degree polynomials. A finite-dimensional norm comparison then bounds each Fréchet derivative norm. For no insertions use $S_\varnothing = 1$.[12] □

The null-space characterisation 8.13(ii) also gives the field symmetries. Inclusions commute with conjugation and H_4 and preserve unit classes. Derivatives commute with conjugation; the chain rule in $S(\rho(h)v; hx) = S(v;x)$ gives

$$\rho_{d+1}(h)\partial_\mu = \sum_\nu h_{\nu\mu}\partial_\nu \rho_d(h) \qquad (h \in H_4)\,.$$

These signed-coordinate identities and the derivative transformation are the discrete orthogonal symmetries already available here. Covariance under arbitrary continuous rotations requires a further argument.[13]

8.5 Removing the spacing refinement

Use the full fixed-volume convergence of Proposition 5.1. For a test supported inside volume p in Definition 2.11, write

$$\widehat{R}_p(v;f) = \lim_{N\to\infty} C_{N,p}(v;f)\,.$$

[12] Formal correspondence at revision `3456e5e`: `Extraction/Harvest.lean`: symmetry and positivity; `Work/D05.lean`: two-scale bound; `Work/B10.lean`: derivative bounds.

[13] Formal correspondence at revision `3456e5e`: `Work/D06.lean`: orientation of the derivative maps.

For every volume containing the test support, restriction to N_j and uniqueness in $\mathbb{C}$ give $\widehat{R}_p(v;f) = R_p(v;f)$, retaining the outer limit and smooth correlations. Start with the full spacing sequence and prescribed volume sequence p, and compose the volume refinements into one strictly increasing q. Then

$$\lim_{k\to\infty} \widehat{R}_{p(q(k))}(v;f) = \int f(x)S(v;x)\,dx\,. \tag{8.32}$$

At volumes not containing the support, assign arbitrary values as allowed by Definition 2.12. The ordered limit is therefore available for every coefficient list and separated test.[14]

Unresolved. Prove that each fixed-volume smeared correlation for the chosen Wilson family converges along the entire spacing sequence. This needs the fixed-level Cauchy estimates and the positional-limit construction. The uniqueness argument above uses this full-spacing theorem; it does not prove it.[15]

The smooth null space agrees with the separated-test null space, so the public projection is $\pi(d,v) = (d,\pi_d v)$ by Definition 2.17. Multiplying the separated-correlation identity by f transports (8.32) to Proposition 5.2; its phase clauses use the same table.

9 Free Maxwell fields

The limiting fields of Section 8 are described by their correlations, meaning the expectations of products of local polynomial measurements. For a short-distance comparison we use the free Maxwell field: the linear, non-interacting theory of a curvature two-form. It has one component for each basis direction in the Lie algebra of Definition 2.1; the index c below records that basis direction (often called colour). Its local observables are polynomials in the curvature and its derivatives. The two linear Maxwell equations say, respectively, that the curvature has zero cyclic derivative and zero divergence. In coordinates, the first says the signed sum around each three-index cycle vanishes; the second says each column has zero divergence.

Subsection 9.1 defines the Taylor coefficients of a Maxwell curvature and computes their free correlations by differentiating the curvature two-point function. Subsection 9.2 starts with positive pairings of reflected linear observations. To see the finite matching rule, use independent standard complex Gaussian coordinates z_a, each with density $\pi^{-1}e^{-|z_a|^2}$: they satisfy $\mathbb{E}[z_a\overline{z}_b] = \delta_{ab}$ and $\mathbb{E}[z_a z_b] = 0$. Thus a product has nonzero expectation only when each unbarred factor is paired with a barred factor of the same coordinate. The field correlations below are Wick ordered: pairings within one insertion are removed, leaving the matching rule between distinct insertions. This representation gives polynomial positivity, identifies coefficients that represent the same field, and supplies finitely many detecting tests at each degree. Subsection 9.3 states the additional same-family lattice estimate needed to compare free correlations with the already chosen continuum correlations. Subsection 9.4 makes the comparison maps agree across degrees and shows they are isomorphisms. Subsection 9.5 upgrades convergence of tested correlations to convergence of all derivatives in the insertion positions. The radii and field maps are chosen together. The further uniformity needed for a prescribed radius sequence is left to the later trace argument.

[14] Formal correspondence at revision `3456e5e`: `FullVolume/Finale/Program.lean`: removal of the spacing refinement; `HierarchyBridge.lean`: public ordered Wilson limit.

[15] Formal correspondence at revision `3456e5e`: `FullVolume/Fields/Filled.lean`: full-spacing fixed-volume convergence for the chosen run.

9.1 Curvature correlations

Use the gauge group of Definition 2.1, the classical polynomial spaces V_d of Definition 6.4, and their norms from Definition 8.1. The inner product on the Lie algebra and a corresponding orthonormal basis $E^1, \dots, E^{\dim \mathfrak{g}}$ are fixed by

$$\langle X, Y\rangle_{\mathfrak{g}} = -\frac{1}{m}\operatorname{Re}\operatorname{tr}(XY), \qquad \langle E^c, E^{c'}\rangle_{\mathfrak{g}} = \delta_{cc'}. \tag{9.1}$$

Here m is the defining matrix size, as in Definition 2.1. Thus c records which Lie-algebra component of the curvature is being measured; the normalisation fixes the trace convention in every later two-point function.[16]

Definition 9.1 (Maxwell Taylor coefficients)**.** The indices a, b, k range over the four coordinate directions. A Maxwell field is a smooth array $f = (f_{ab})$ of $\mathfrak{g}$-valued functions. Antisymmetry makes it a two-form: each component is attached to an oriented coordinate area, and reversing the area changes its sign. The cyclic equation is the zero exterior derivative condition, and the last equation is zero divergence. It satisfies

$$f_{ab} = -f_{ba}, \qquad \partial_a f_{bk} + \partial_b f_{ka} + \partial_k f_{ab} = 0, \qquad \sum_a \partial_a f_{ab} = 0. \tag{9.2}$$

For $j \in \mathbb{N}$, let $\mathcal{H}_j$ be the finite-dimensional real space of homogeneous polynomial solutions of degree j; these are the possible degree-j Taylor terms of a Maxwell curvature near the origin. A homogeneous polynomial Q_j is identified with its symmetric multilinear coefficient by evaluation on repeated arguments: its value at z is obtained by putting the same vector z in every Taylor slot. Our Taylor convention is

$$Q_j(z) = \frac{1}{j!} D^j f(0)[z, \dots, z].$$

Thus a Taylor tensor here includes the factorial denominator, so the sequence (Q_j) is exactly the formal Taylor expansion of the curvature. A sequence $Q = (Q_j)_{j\ge 0}$ with $Q_j \in \mathcal{H}_j$ is a formal Maxwell Taylor series; convergence of this series is not required. For a one-form A, use $z \cdot A = \sum_a z^a A_a$ and $(dA)_{ab} = \partial_a A_b - \partial_b A_a$. The radial potential is a polynomial one-form reconstructed from the curvature, with $dA = Q$. The condition $z \cdot A(z) = 0$ fixes the radial gauge; its explicit formula is

$$(\kappa_j Q)_a(z) = \frac{1}{j+2}\sum_b z^b Q_{ba}(z), \qquad z \cdot \kappa_j Q(z) = 0, \qquad d(\kappa_j Q) = Q. \tag{9.3}$$

For example, when Q_0 is a constant antisymmetric matrix, the formula gives $(\kappa_0 Q_0)_a(z) = \frac{1}{2}\sum_b z^b (Q_0)_{ba}$, a linear one-form whose curl is Q_0.

The last identity follows by differentiating the formula and using the cyclic equation in (9.2). Euler's identity, $z \cdot \nabla Q_j = j Q_j$, supplies the factor $j + 2$ in the denominator. Thus the coefficients may also be read in the radial coordinates of (6.11).[17]

For $z \in \mathbb{R}^4 \setminus \{0\}$, the scalar Green function Γ is the fundamental solution of the four-dimensional Laplacian. Its second derivatives are the two-point covariance: they give the free expectation of two

[16] Formal correspondence at revision `3456e5e`: `YM/Gauge/Colors.lean`, lines 43–83: physical colour coordinates and normalised trace.

[17] Formal correspondence at revision `3456e5e`: `YM/Maxwell/Shell.lean`: the constrained Taylor space and inverse radial curl.

linear curvature measurements at separation z. The indices a, b, e, f specify coordinate directions; c, c' are the colour indices of (9.1). Set

$$\begin{aligned} \Gamma(z) &= \frac{1}{4\pi^2|z|^2}, \qquad z \neq 0, \\ C^{cc'}_{ab,ef}(z) &= \delta_{cc'}\big(\delta_{be}\partial_a\partial_f\Gamma - \delta_{bf}\partial_a\partial_e\Gamma - \delta_{ae}\partial_b\partial_f\Gamma + \delta_{af}\partial_b\partial_e\Gamma\big)(z). \end{aligned} \tag{9.4}$$

The signs include differentiation at the second endpoint of a covariance depending on $x - y$. A linear probe of order j is a real linear functional on $\mathcal{H}_j$; it measures a specified linear combination of the degree-j Taylor coefficients. For probes $p \in \mathcal{H}_j^*$ and $q \in \mathcal{H}_\ell^*$, the dual spaces of these real tensor spaces, define

$$K_{p,q}(z) = \frac{(-1)^\ell}{j!\,\ell!}(p \otimes q)\big[D^{j+\ell}C(z)\big]. \tag{9.5}$$

The first j derivative slots belong to p and the remaining ℓ slots to q. Thus p and q select one Taylor coefficient at each of the two points; the factorials match the Taylor convention in (9.3). Colour and form indices are contracted by the two functionals themselves; no extra antisymmetry factor is inserted. One may extend these functionals to all tensors to evaluate the formula. The next lemma shows that the equations force the differentiated covariance into the Maxwell solution spaces, so the chosen extensions cannot affect the answer.

Lemma 9.2 (The contraction kernel)**.** *For $z \neq 0$, the kernel (9.4) satisfies (9.2) in both groups of tensor indices. Consequently (9.5) is well defined on $\mathcal{H}_j^* \times \mathcal{H}_\ell^*$, is real analytic, and satisfies*

$$K_{p,q}(z) = K_{q,p}(-z), \qquad K_{p,q}(\lambda z) = \lambda^{-j-\ell-4}K_{p,q}(z) \quad (\lambda > 0). \tag{9.6}$$

Proof. The verification has three steps. Here the curl is the antisymmetric derivative d defined above; the cyclic equation says its next exterior derivative vanishes, while harmonicity of Γ gives zero divergence away from the origin.

Step 1. Taking a curl at both endpoints gives the displayed covariance; antisymmetry and commuting derivatives give the cyclic equation.

Step 2. Harmonicity of Γ gives the divergence equation.

Step 3. Endpoint exchange and homogeneity give (9.6); the equations make the contraction independent of how each functional is extended.

For the second step, write $\Delta = \sum_a \partial_a^2$ for the Laplacian and suppress the colour factor:

$$\sum_a \partial_a C_{ab,ef} = \delta_{be}\partial_f\Delta\Gamma - \delta_{bf}\partial_e\Delta\Gamma = 0 \qquad (z \neq 0).$$

Antisymmetry is explicit in (9.4). Exchanging the endpoints proves the same equations in the other group of indices. Every Taylor tensor to which an extended functional is applied therefore lies in $\mathcal{H}_j$ or $\mathcal{H}_\ell$. An extension vanishing on that space contributes zero, proving independence. The exchange and scaling identities follow by differentiating the even, homogeneous function Γ; analyticity follows from its explicit formula.[18] □

[18] Formal correspondence at revision `3456e5e`: `YM/Maxwell/Curls.lean`: equations in both groups of tensor indices; `YM/Maxwell/Leg.lean`, lines 141–176: independence of the extensions.

Definition 9.3 (Free polynomial fields)**.** Choose coordinates on the spaces $\mathcal{H}_j$ of Definition 9.1 and give each coordinate weight $j+2$, its scaling degree as a curvature Taylor coefficient. Substitute the reconstructed potential (9.3) into a polynomial in the radial gauge fields. This gives the restriction map res_d on V_d; its kernel consists exactly of polynomials that vanish on every Maxwell solution. Write

$$\begin{aligned}(\mathrm{res}_d\, v)(Q) &= v\big((\kappa_j Q_j)_{j\ge 0}\big),\\ \mathcal{I}_d = \ker \mathrm{res}_d, &\qquad \mathcal{P}_d^0 = \mathrm{res}_d(V_d), \qquad F_d^0 = V_d/\mathcal{I}_d \simeq \mathcal{P}_d^0.\end{aligned} \tag{9.7}$$

For the orthogonal group $O(4)$ of Definition 6.5, write $\rho_d^0(h)$ for the action induced on these free polynomial spaces by pullback of the two-form and its Taylor tensors. Give each F_d^0 its fixed quotient norm from V_d; all vector and operator norms below use this norm on the free factors. The free unit is the constant class $\mathbf{1} = [1]$. The interacting unit is $\pi_d u_d$ in Definition 8.11; the comparison maps below will send the free unit to that class.

Wick ordering means subtracting the self-contractions at each insertion. In the resulting correlations, pairings join different insertion points only. The following finite sum implements this rule.

Definition 9.4 (Wick correlations)**.** For monomials $q_i = \prod_{e\in E_i} p_e$ in the linear probes defined above, regard each repeated factor as a separate occurrence. Let E be the union of these occurrence sets and let $i(e) = i$ identify the insertion containing e. A perfect matching pairs every occurrence with exactly one other occurrence. The disjoint-union convention keeps two copies of the same probe distinct when a monomial repeats it. Let $\mathfrak{M}_{\neq}(E)$ be the matchings that never pair two members of the same E_i. Set

$$W_n(q;x) = \sum_{P\in\mathfrak{M}_{\neq}(E)} \prod_{\{e,f\}\in P} K_{p_e,p_f}\big(x_{i(e)} - x_{i(f)}\big), \qquad x \in \mathrm{Conf}_n, \tag{9.8}$$

For $q = (q_1, \dots, q_n)$, extend this formula complex multilinearly to the polynomial fields of Definition 9.3.

The configuration space is that of Definition 2.10. The matching rule treats repeated occurrences separately and is unchanged by their enumeration. For example, a quadratic insertion has zero one-point expectation, whereas two quadratic insertions give

$$W_2(pq, st; x, y) = K_{p,s}(x-y)K_{q,t}(x-y) + K_{p,t}(x-y)K_{q,s}(x-y).$$

No value of a kernel at zero is used. Empty products give $W_0 = 1$.[19]

9.2 Positivity and finite observations

This subsection proves positivity for polynomial observations from the analytic result for linear probes.

Definition 9.5 (Reflected Maxwell observations)**.** Fix an axis μ and let $\theta = r_{\mu,0}$ be the reflection of Definition 8.2. A positive-side point has $x^\mu > 0$. Reflect the two-form indices and all Taylor arguments by

$$(\theta\cdot Q)_{ab}(z) = \sum_{c,d=1}^{4} \theta_{ac}\theta_{bd} Q_{cd}(\theta^{-1}z), \qquad (R_\theta p)(Q) = p(\theta^{-1}\cdot Q).$$

[19] Formal correspondence at revision `3456e5e`: `YM/Maxwell/Equation.lean`: restriction and equation kernel; `YM/Maxwell/Wick.lean`: matching correlations.

The colour component is unchanged. Extend R_θ multiplicatively to polynomials; on F_d^0 it is $\rho_d^0(\theta)$ of Definition 9.3. For a finite list of probes p_α at positive-side points x_α, their reflected Gram matrix, the matrix of reflected pairings, is

$$H_{\alpha\beta} = K_{p_\alpha, R_\theta p_\beta}\big(x_\alpha - \theta x_\beta\big). \tag{9.9}$$

The reflected point is separated from every positive-side point, even when two of the x_α coincide.

Lemma 9.6 (Reflected linear probes)**.** *The matrix* (9.9) *is positive semidefinite. If all $x_\alpha = x$ and the probes p_α are linearly independent as functionals on $\prod_j \mathcal{H}_j$, it is positive definite: every nonzero linear combination has strictly positive reflected norm.*

Unresolved. The analytic proof of Lemma 9.6 is still missing from this manuscript. It must prove nonnegativity for arbitrary positive-side probe lists and strictness at one point modulo exactly the Maxwell equations. In the strictness argument, the response is the pairing of a linear probe with the curvature field: if it vanishes on the reflected open set, analytic continuation makes it vanish at all separated points, and tests against homogeneous Maxwell solutions must then show that the probe is zero on $\mathcal{H}_j$. Positivity alone does not establish this strict assertion.[20]

Once strictness is established, it identifies the quotient (9.7) with the quotient by all separated null correlations, just as required for the interacting fields in 8.13(ii).

An inclusion keeps the same polynomial when the allowed degree is enlarged. A derivative moves the insertion point infinitesimally: the coefficient at degree $j+1$ contributes to the derivative at degree j, with the factor $j+1$ fixed by the Taylor factorials.

Definition 9.7 (Free field operations)**.** Write $j_{d,e}^0$ for the inclusion induced by $V_d \subset V_e$, for $d \le e$. For a formal Taylor series Q as in Definition 9.1, translating the observation point infinitesimally in direction μ shifts the Taylor coefficients as follows:

$$(\mathcal{D}_\mu Q)_j = (j+1)Q_{j+1}(e_\mu, \cdot, \dots, \cdot).$$

Here e_μ is the coordinate unit vector and the remaining arguments are the j symmetric Taylor arguments; form indices are retained. For a polynomial q, let $D_j q$ denote its ordinary finite-dimensional derivative in the Q_j variable and define

$$(\partial_\mu^0 q)(Q) = \sum_j D_j q(Q)\big[(\mathcal{D}_\mu Q)_j\big]. \tag{9.10}$$

The sum is finite, since q uses only finitely many Q_j.

The proof below shows that this operation maps $\mathcal{P}_d^0$ into $\mathcal{P}_{d+1}^0$ and differentiates its correlations with respect to the insertion point. In derivative bounds, D_x^s denotes the total Fréchet derivative with its operator norm, as in Definition 8.12.

Proposition 9.8 (Free fields)**.** *Grant Lemma 9.6. The correlations* (9.8) *descend faithfully to F_d^0 in* (9.7) *and have the following properties.*

(i) *They are normalised, unital and real analytic on* Conf_n*; they satisfy the symmetry identities* (8.29)*, with the full $O(4)$ action on coefficients, and both reflection-positivity conditions* (8.30)*.*

[20] Formal correspondence at revision `3456e5e`: `YM/Next/MAXWELL/Gram.lean`, lines 52–108; `YM/Next/MAXWELL/Strict.lean`, especially `level_detects`, and its imported energy estimates.

(ii) *The operations of Definition 9.7 give compatible inclusions* $j^0_{d,e} : F^0_d \hookrightarrow F^0_e$ *and derivatives* $\partial^0_\mu : F^0_d \to F^0_{d+1}$ *satisfying* (8.26) *and the conjugation and orientation laws of Section 8.4.*

(iii) *For every fixed field list and* $s \in \mathbb{N}$*, there are* $C \geq 0$ *and* $b \in \mathbb{N}$ *with*

$$\|D^s_x W_n(q;x)\| \leq C r^{-b} \quad \big(0 < r \leq 1,\ |x_i - x_j|_\infty \geq r \text{ for } i \neq j\big). \tag{9.11}$$

Proof. The proof has four steps. First we rewrite the reflection matrix as the ordinary inner product of polynomials with positive weights. Strict positivity then shows that only polynomials already zero in the Maxwell quotient are invisible. The remaining properties follow from the same matching formula: constants and symmetries pass directly, and differentiating a matching is the same as differentiating one insertion.

Step 1. A positive definite finite matrix has a square-root factorisation $H = L^{\mathsf{T}} L$. Use one independent standard complex Gaussian variable z_a for each coordinate and represent probe e by $\sum_a L_{ae} z_a$. The complex Gaussian moment rule pairs barred with unbarred coordinates; expanding products gives the polynomial inner product $\sum_\gamma \gamma!\, \overline{P_\gamma} Q_\gamma$.

Step 2. Strictness at one point makes this representation injective, so the only null coefficients are $\mathcal{I}_d$.

Step 3. The same finite matching formula gives positivity, normalization and symmetries.

Step 4. Differentiating the matching formula gives the field operations and the homogeneous-kernel bounds.

For a finite collection of products supported on the positive side, use the finite set of all their linear factors as the matrix indices. Within each product, sum over partial matchings between distinct insertion points, and replace each unpaired occurrence e by $\sum_a L_{ae} z_a$. This produces a polynomial $Q_\alpha(z)$ for each product α. Every full matching splits uniquely into a matching within each half and the pairs crossing between them. Write $P(z) = \sum_\gamma P_\gamma z^\gamma$, where γ is a multi-index of nonnegative integers and $z^\gamma = \prod_a z_a^{\gamma_a}$. The reflected Gram matrix is then

$$G_{\alpha\beta} = \langle Q_\alpha, Q_\beta \rangle_{\mathrm{F}}, \qquad \langle P, Q \rangle_{\mathrm{F}} = \sum_\gamma \gamma!\, \overline{P_\gamma} Q_\gamma, \qquad \gamma! = \prod_a \gamma_a!. \tag{9.12}$$

This is the finite-dimensional Gaussian factorisation: the reflected pairing is an ordinary polynomial inner product, and every monomial square has the positive weight $\gamma!$. Hence the reflected matrix is positive semidefinite. For a single insertion there are no internal pairs, so $Q(z) = q(L^{\mathsf{T}} z)$. For independent coordinate probes at one point, Lemma 9.6 makes L invertible and hence

$$q \neq 0 \quad \Longrightarrow \quad W_2\big(R_\theta \bar{q}, q; \theta x, x\big) = \sum_\gamma \gamma!\, |Q_\gamma|^2 > 0. \tag{9.13}$$

Thus the only separated null coefficients in V_d are $\mathcal{I}_d$. Integrating point Gram matrices against finite sums of positive-side joint tests gives smeared positivity. The integrals converge absolutely on the separated compact supports, by the explicit kernels.

The constant polynomial and the matching rule give the normalisation and unit identities. The kernel is invariant under translations and transforms as a two-form under $O(4)$; permuting occurrences proves permutation symmetry and conjugating coefficients proves reality. All these identities descend to the quotient.

For differentiation, use the infinitesimal translation of Definition 9.7. It raises weight by one, obeys the Leibniz rule, and preserves (9.2). The induced derivation preserves the images $\mathcal{P}_d^0$, and differentiating a matching sum in slot i gives

$$\partial_{i,\mu} W_n(q;x) = W_n(q_{i\to\partial_\mu^0 q_i};x).$$

Inclusions are induced by $V_d \subset V_e$; faithfulness makes them injective and compatible. Finally, each derivative of a contraction is homogeneous of a fixed negative degree and bounded on the unit sphere. The finite sums of products in (9.8) give (9.11), including the empty-list case.[21] □

Faithfulness means that every nonzero field is detected by some correlation. At each fixed degree, finitely many observations suffice: they turn convergence of correlations into information about the finite-dimensional coefficient maps.

Definition 9.9 (Tested observations)**.** For $v \in F_d^0$, a fixed list $b = (b_1, \dots, b_n)$ of free fields in the other insertion slots, and a separated test f of Definition 2.10, write

$$\ell(v) = \int f(x) W_{n+1}(v, b; x)\, dx. \tag{9.14}$$

Here (v, b) is the list obtained by placing v first.

Lemma 9.10 (Finite separating tests)**.** *Assume Lemma 9.6. For each d, finitely many functionals* (9.14) *and a constant $c_d > 0$ satisfy*

$$\max_{1\le a\le s_d} |\ell_a(v)| \ge c_d \|v\| \qquad (v \in F_d^0). \tag{9.15}$$

Proof. The proof has two steps. First, strict positivity says that no nonzero vector is missed by all possible measurements. Then choose one measurement that detects a vector in the remaining common kernel; each choice lowers its dimension, so the process stops after finitely many choices.

Step 1. By (9.13), every nonzero field has a correlation detected by some test, so the common kernel of all functionals (9.14) is zero.

Step 2. Choose each next functional nonzero on the current common kernel; its dimension drops at every step, leaving finitely many tests, and compactness of the unit sphere gives the positive constant in (9.15).

In detail, starting with F_d^0, choose a functional nonzero on the current common kernel until its dimension reaches zero:

$$E_0 = F_d^0, \qquad E_{a+1} = E_a \cap \ker \ell_{a+1}, \qquad E_a \neq 0 \implies \dim E_{a+1} < \dim E_a.$$

Only finitely many choices are needed. The maximum in (9.15) is then a continuous positive function on the compact unit sphere. Its minimum is the positive constant c_d.[22] □

[21] Formal correspondence at revision `3456e5e`: `YM/Next/MAXWELL/Fock.lean`, `YM/Next/MAXWELL/Studio.lean` and `YM/Next/MAXWELL/Core.lean`: polynomial factorisation and strictness; `YM/Next/MAXWELL/System.lean`, lines 29–114: quotient and derivative operations; `YM/Next/MAXWELL/Caps.lean`: analyticity and bounds.

[22] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Invert.lean`, lines 48–65; `YM/Work/A04.lean`, lines 14–36.

9.3 The comparison estimate

Return to the same limiting correlations S of Proposition 8.13 and the normed interacting fields F_d of Definition 8.11. A map from a free field to an interacting field is useful only if it compares their correlations. The needed estimate is uniform in the size of a joint test, including tests at distinct points with equal coordinate projections. For $R \geq 1$, put

$$K_{R,n} = \{x : |x| \leq R,\ |x_i - x_j|_\infty \geq R^{-1} \text{ for } i \neq j\}. \tag{9.16}$$

Every compact subset of Conf_n lies inside such a set for sufficiently large R.

Definition 9.11 (Finite-cutoff comparison)**.** The comparison data at finite lattice spacing have the following quantifiers. For every $D, N_*, R, \varepsilon, \delta$ with $D, N_* \in \mathbb{N}$, $R \geq 1$ and $\varepsilon, \delta > 0$, require a radius $0 < r \leq \delta$ and linear maps

$$A_{N,p,d} : F_d^0 \longrightarrow V_d \qquad (d \leq D). \tag{9.17}$$

These coefficient maps choose lattice polynomials representing free fields. Their operator norms are eventually bounded on the ordered tails of Definition 8.3; they commute with conjugation and signed permutations and preserve the degree-zero unit there. For tuples with specified degrees, abbreviate $A_{N,p}q = (A_{N,p,d_i}q_i)_i$. For a point list x, write $rx = (rx_1, \dots, rx_n)$. For every $n \leq N_*$, $d_i \leq D$, fixed tuple $q_i \in F_{d_i}^0$, separated test f supported in $K_{R,n}$, $M > 0$ with $\|f\|_\infty \leq M$, and $\eta > 0$, require eventually

$$\left| C_{N,p}\big(A_{N,p}q; f_r\big) - \int f(x) W_n(q; x)\, dx \right| \leq \varepsilon M \prod_i \|q_i\| + \eta, \qquad f_r(y) = r^{-4n} f\Big(\frac{y}{r}\Big). \tag{9.18}$$

The moments and tests are those of Definitions 2.9 and 2.10. The order is: fix accuracy, degree and point-separation window; then choose the radius and coefficient maps; only after that choose any fixed field tuple and test. This order is what makes one comparison map serve every such test. The cutoff threshold may depend on that tuple, test, M and η; no common cutoff threshold for all tests is required.

We also need the independent dimension bound

$$\dim_{\mathbb{C}} F_d \leq \dim_{\mathbb{C}} F_d^0 \qquad (d \in \mathbb{N}). \tag{9.19}$$

Unresolved. The ultraviolet sections must produce (9.17)–(9.18) and (9.19) for the same Wilson family and the same retained limits as in Section 8. The comparison includes bounded coefficient maps, symmetries, the degree-zero unit and all the displayed test quantifiers. Centred bounds alone do not provide it.[23]

For maps $U_d : F_d^0 \to F_d$, write $Uq = (U_{d_i} q_i)_i$, just as for the coefficient maps in Definition 9.11. First pass that comparison to the already constructed continuum theory. Bounded coefficient maps have a common cluster limit in finite dimensions. The extraction is used to estimate the same S, rather than to construct another limiting law. Equal coordinate projections are needed because the comparison tests are supported on configurations where distinct points may share one coordinate; those configurations belong to Conf_n even though the first compactness extraction used a smaller full-measure set.

[23] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Station.lean`: exact finite-cutoff comparison condition; `YM/Next/UV_STRESS/Granular.lean` and `YM/MergeProof/East/Plume.lean`: suppliers; `YM/Next/UV_STRESS/Invert.lean`, lines 104–116: the separate dimension input.

Lemma 9.12 (Passage to continuum fields)**.** *Assume* (9.17)–(9.18) *and the conclusions of Propositions 8.8, 8.13 and 8.14. There are linear maps* $U_d : F_d^0 \to F_d$, $d \le D$, *commuting with conjugation and signed permutations, with* $U_0\mathbf{1} = \mathbf{1}$, *such that*

$$\left| \int f(x) \big[S_n(Uq; rx) - W_n(q; x) \big] \, dx \right| \le \varepsilon M \prod_i \|q_i\| \tag{9.20}$$

for all the tuples and tests in (9.18).

Proof. There are two steps.

Step 1. The bounded finite-dimensional maps have a common cluster limit A_d on the ordered tails; set $U_d = \pi_d A_d$.

Step 2. Local bounds replace the varying coefficients by A_d in correlations, and the ordered limit turns the finite-cutoff estimate into (9.20) for the same S.

More precisely, use a refining ultrafilter as in Definition 8.3 and write its limits as A_d. Expansion in fixed polynomial bases and (8.17) give, for the fixed rescaled test,

$$|C_{N,p}(A_{N,p}q; f_r) - C_{N,p}(Aq; f_r)| \le C_{r,f,q} \sum_{d \le D} \|A_{N,p,d} - A_d\| \longrightarrow 0.$$

Put $U_d = \pi_d A_d$, using the quotient map of Proposition 8.13. The retained ordered limit and the change of variables $y = rx$ yield

$$C_{N,p}(A_{N,p}q; f_r) \longrightarrow \int f(x) S_n(Uq; rx) \, dx.$$

All symmetry and unit identities pass through the finite-dimensional limits. For each fixed test, take the limit in (9.18), then let $\eta \downarrow 0$. The same cluster maps work for every test because they were chosen before any test was fixed. For each fixed tuple and test, the local bound in (8.17) controls the coefficient replacement; the finite-cutoff inequality then passes to the limit along that already chosen common refinement.[24] □

9.4 Compatible field bases

We next choose comparisons at successively smaller radii. Compatibility means that if a free polynomial is included from degree d into degree e, its image is the same interacting field under either map. The same free field can appear in two degree spaces through an inclusion. The maps of Lemma 9.12 may initially send these two copies to different interacting fields, so we correct the higher-degree map to agree on the already included lower-degree part.

For a radius sequence r_k and maps $U_{k,d} : F_d^0 \to F_d$, write

$$E_k(q; x) = S_n(U_k q; r_k x) - W_n(q; x). \tag{9.21}$$

Definition 9.13 (Uniform tested convergence)**.** For every fixed field tuple, the convergence used below is

$$\sup_{\substack{f \text{ as in Definition 2.10} \\ \operatorname{supp} f \subset K,\ \|f\|_\infty \le 1}} \left| \int f(x) E_k(q; x) \, dx \right| \longrightarrow 0 \qquad (K \Subset \mathrm{Conf}_n), \tag{9.22}$$

Here $K \Subset \mathrm{Conf}_n$ means a compact subset of the separated configuration space.

[24] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Station.lean`, especially `zoom_integral` and `harvested`.

The supremum controls every test supported in the same compact region with the same size bound. This uniformity lets us use tests that vary with k in the derivative argument; convergence for one fixed test would not suffice.

Definition 9.14 (Moving tested observations)**.** For a finite list of tested observations ℓ_a on F_d^0 from Definition 9.9, let f_a be its test and b_a its list of m_a other fields. Given r_k and maps $U_{k,d}$, define

$$\ell_{a,k}(v) = \int f_a(x) S_{m_a+1}\big(U_k(v, b_a); r_k x\big)\, dx, \qquad B_k v = (\ell_{a,k}(v))_a, \quad Bv = (\ell_a(v))_a. \tag{9.23}$$

These are the same measurements, now made on the interacting fields at radius r_k.

Proposition 9.15 (Compatible comparisons)**.** *Assume the finite-cutoff input (9.17)–(9.18), the dimension bound (9.19), and the field operations and Euclidean properties of Propositions 8.8, 8.13, 8.14 and 9.8. There exist $r_k \downarrow 0$ strictly and complex-linear isomorphisms $U_{k,d} : F_d^0 \to F_d$ satisfying (9.22). For each fixed $d \le e$, eventually in k,*

$$\begin{aligned} U_{k,e} j_{d,e}^0 &= j_{d,e} U_{k,d}, & U_{k,d}\mathbf{1} &= \mathbf{1}, \\ U_{k,d}\bar{q} &= \overline{U_{k,d} q}, & U_{k,d}\rho_d^0(h) &= \rho_d(h) U_{k,d} \quad (h \in H_4). \end{aligned} \tag{9.24}$$

Proof. The construction has four steps.

Step 1. At stage k, require accuracy through degree and number of insertions k, and choose $r_k \downarrow 0$.

Step 2. Correct each map on the lower-degree image using a left inverse, so $\widetilde{U}_{k,d+1} j_d^0 = j_d \widetilde{U}_{k,d}$ while preserving conjugation and signed symmetries.

Step 3. Finite separating tests give $\|B_k v\|_\infty \ge (c_d/2)\|v\|$, hence each fixed-degree map is eventually injective.

Step 4. The independent dimension bound gives $\dim F_d^0 \le \dim F_d \le \dim F_d^0$, hence eventual isomorphisms degree by degree.

Apply Lemma 9.12 with

$$D_k = N_{*,k} = k, \qquad R_k = k+1, \qquad \varepsilon_k = \frac{1}{k+1}, \qquad r_{k+1} \le \min\Big(\frac{r_k}{2}, \frac{1}{k+2}\Big).$$

The comparison hypothesis permits the last condition by choosing the next radius bound first. Each compact separated set eventually fits inside $K_{R_k,n}$. For each fixed tuple, (9.20) therefore gives (9.22) for the initial maps.

Write $j_d^0 = j_{d,d+1}^0$ and $j_d = j_{d,d+1}$. Choose a linear left inverse $R_d : F_{d+1}^0 \to F_d^0$, so $R_d j_d^0 = 1$. It extracts from a higher-degree field the component lying in the copy of degree d. Thus $1 - j_d^0 R_d$ removes that old component before the new part is mapped. It may be chosen real and commuting with the H_4 action of Definition 8.1: average any left inverse first with its conjugate and then over H_4. Both averages remain left inverses because j_d^0 intertwines these symmetries. For $d < D_k$, define corrected maps recursively by

$$\widetilde{U}_{k,0} = U_{k,0}, \qquad \widetilde{U}_{k,d+1} = j_d \widetilde{U}_{k,d} R_d + U_{k,d+1}(1 - j_d^0 R_d). \tag{9.25}$$

Since $R_d j_d^0 = 1$, this gives $\widetilde{U}_{k,d+1} j_d^0 = j_d \widetilde{U}_{k,d}$. The symmetry identities pass through the recursion; the degree-zero unit identity propagates to every degree. For (9.22), expand a corrected correlation using

$$q = j_d^0 R_d q + (q - j_d^0 R_d q).$$

After finitely many expansions it is a sum of original comparisons with fixed free fields, possibly in lower degrees. The inclusion identities in both theories identify the corresponding sum of free correlations with $W_n(q;x)$. Thus the corrected maps retain (9.22).

Fix d and choose the finite functionals of Lemma 9.10. Use their maps $B_k : F_d^0 \to \mathbb{C}^{s_d}$ and B from Definition 9.14, with the corrected comparisons $\widetilde{U}_k$. Then (9.22), evaluated on a basis, gives operator-norm convergence to $B = (\ell_a)_a$. For large k, (9.15) implies

$$\|B_k v\|_\infty \geq \frac{c_d}{2}\|v\|, \qquad \widetilde{U}_{k,d} v = 0 \implies B_k v = 0 \implies v = 0.$$

Hence $\widetilde{U}_{k,d}$ is injective. Injectivity says the image has at least the free-space dimension. Together with the independent upper bound (9.19),

$$\dim F_d^0 \leq \dim F_d \leq \dim F_d^0,$$

so it is an isomorphism eventually at each fixed degree. Fill the finitely many exceptional early terms at each degree with any fixed isomorphism; the assertions are eventual. For a fixed tuple or a fixed pair of degrees these replacements have no effect on the eventual assertions, proving (9.24) and (9.22).[25] □

9.5 Derivatives of the comparison

The tested convergence (9.22) also determines how the derivatives of the comparison fields change with k. Integration by parts identifies their values against every separating functional; finite-dimensional invertibility then identifies the derivative maps themselves. This is the extra argument needed before using a smooth short-distance limit.

Proposition 9.16 (Smooth short-distance limits)**.** *Under the hypotheses of Proposition 9.15, for every $d \in \mathbb{N}$ and coordinate direction μ, the maps there satisfy, in operator norm,*

$$r_k U_{k,d+1}^{-1} \partial_\mu U_{k,d} \longrightarrow \partial_\mu^0. \tag{9.26}$$

For every fixed tuple q, compact $K \subset \mathrm{Conf}_n$ and $s \in \mathbb{N}$,

$$\sup_{x \in K} \|D_x^s E_k(q;x)\| \longrightarrow 0. \tag{9.27}$$

Proof. The proof has three steps.

Step 1. Integration by parts identifies the transported derivative through the finite tests: $B_k D_{k,d,\mu} q \to B\partial_\mu^0 q$.

Step 2. Repeated differentiation and finite-dimensional multilinearity give uniform tested convergence for every derivative list, hence $\|D^\alpha E_k\|_{L^1(K)} \to 0$.

Step 3. A compactly supported cutoff and Fourier inversion give $\|D^s(\chi E_k)\|_\infty \to 0$ on the target compact set.

[25] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Rates.lean`: diagonal choice of radii; `YM/Next/UV_STRESS/Arrange.lean`: equivariant retractions and exact degree correction; `YM/Next/UV_STRESS/Invert.lean`, especially `Stations.equivalences`: eventual injectivity and isomorphisms.

Put $D_{k,d,\mu} = r_k U_{k,d+1}^{-1} \partial_\mu U_{k,d}$. The factor r_k converts differentiation in the rescaled coordinate $r_k x$ back to differentiation in the original coordinate. The field derivative identity (8.26) and the chain rule give

$$\partial_{i,\mu} S_n(U_k q; r_k x) = S_n\big(U_k(q_{i\to D_{k,d_i,\mu} q_i}); r_k x\big).$$

Use the finite tested observations at degree $d+1$ from Lemma 9.10, with $\ell_{a,k}, f_a, b_a, m_a$ as in Definition 9.14. Integration by parts in the first insertion variable gives

$$\ell_{a,k}(D_{k,d,\mu} q) = -\int \partial_{1,\mu} f_a(x) S_{m_a+1}\big(U_k(q, b_a); r_k x\big)\, dx \longrightarrow \ell_a(\partial_\mu^0 q). \tag{9.28}$$

The maps B_k of Definition 9.14 converge in operator norm to the injective map $B = (\ell_a)_a$, as in the preceding proof. Choose a left inverse L of B, so $LB = 1$. Then $LB_k \to 1$, and for large k,

$$D_{k,d,\mu} q = (LB_k)^{-1} L\big(B_k D_{k,d,\mu} q\big) \longrightarrow \partial_\mu^0 q.$$

Convergence on a finite basis gives (9.26).

Repeated differentiation expresses each positional derivative in terms of higher-degree fields whose free coordinates converge, by (9.26). Multilinearity and expansion in finite bases extend (9.22) to these converging coordinate lists. For a coordinate multi-index α and compact $K \Subset \mathrm{Conf}_n$, this gives

$$\sup_{\substack{f \text{ as in Definition 2.10} \\ \operatorname{supp} f \subset K,\ \|f\|_\infty \le 1}} \left| \int f(x) D^\alpha E_k(q; x)\, dx \right| \longrightarrow 0.$$

Uniform control of integrals against all tests with sup norm at most one controls the integral of the absolute value: approximate the complex sign of each derivative by smooth tests on a slightly larger compact neighbourhood of K inside Conf_n. The dual formula for the integral of an absolute value therefore gives

$$\|D^\alpha E_k(q; \cdot)\|_{L^1(K)} \longrightarrow 0.$$

Choose a smooth cutoff function χ supported in that larger set and equal to one near K. The L1 convergence of every derivative makes the Fourier transform decay uniformly after integration by parts; Fourier inversion then bounds the derivatives pointwise. Writing $a = 4n$ for the dimension of the point configuration space and α for a coordinate multi-index, integration by parts in the Fourier transform and Fourier inversion give

$$\|D^s(\chi E_k)\|_\infty \le C_{\chi,s} \sum_{|\alpha| \le s+a+1} \|D^\alpha(\chi E_k)\|_{L^1} \longrightarrow 0.$$

This proves (9.27); the zero-dimensional case is just convergence of constants.[26] □

A simple nonabelian Lie algebra has no nonzero invariant linear functional: invariance forces $\lambda([X,Y]) = 0$, while its commutators span $\mathfrak{g}$. Apart from constants, each curvature coefficient has weight at least two, so a product of two has weight at least four. Below weight four, only one curvature coefficient or its first derivative can occur; gauge invariance would require an invariant linear functional on $\mathfrak{g}$.

[26] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Focus.lean`, especially `head_tilt`, `bend_limit` and `travels_jet`; `YM/Work/A04.lean`, `tendsto_of_fitted`: recovery from varying finite tests.

Corollary 9.17 (Degrees below four)**.** *If* $\mathfrak{g}$ *is simple in Definition 2.1, then, for* $d < 4$*,*

$$F_d^0 = \mathbb{C}\mathbf{1}, \qquad \dim_{\mathbb{C}} F_d^0 = 1. \tag{9.29}$$

Under (9.19)*, the interacting space also satisfies* $F_d = \mathbb{C}\mathbf{1}$*.*

Proof. There are two steps.

Step 1. Below weight four, only a single curvature or first-derivative coefficient could occur; gauge invariance would require a linear functional $\lambda : \mathfrak{g} \to \mathbb{C}$.

Step 2. Since $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, such a functional vanishes, leaving only constants; the unit survives in both quotients under the stated dimension bound.

Indeed, simplicity and noncommutativity imply $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, so

$$\lambda([X, Y]) = 0 \text{ for all } X, Y \in \mathfrak{g} \quad \Longrightarrow \quad \lambda = 0.$$

Thus V_d consists of constants, and $W_1(1) = 1$ shows that the constant survives the free quotient. The dimension bound and the nonzero interacting unit from (8.29) give the last assertion.[27] □

At each stage, Proposition 9.15 chooses a radius small enough for finitely many degree and test requirements. The construction does not ensure these radii form a subsequence of a sequence specified in advance. That stronger choice requires comparison estimates uniform over varying scale intervals, and is reserved for the trace section. The degree-four observable in Proposition 5.5 is the first nonconstant gauge-invariant field retained by this comparison. Identifying its normalisation requires further identities for differentiated correlations; the radii must also work uniformly across the prescribed scale intervals. Those later arguments use the field spaces and derivative limits established here.

References

[27] Formal correspondence at revision `3456e5e`: `YM/UVNext/FREE_STRESS_CERTIFICATE/Size.lean`, `just_one` and `dim_before`; `YM/Trace2/Transfer.lean`, `first_grade`.